

AT LEAST ONE OF THE 2-ADIC ZETA VALUES $\zeta_2(7)$, $\zeta_2(9)$ IS IRRATIONAL

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ABSTRACT. Let $\zeta_2(s) = L_2(s, \omega^{1-s})$ be the 2-adic zeta values of Kubota and Leopoldt. We prove that at least one of the two numbers $\zeta_2(7)$, $\zeta_2(9)$ is irrational. This sharpens the case $s = 3$ of a theorem of L. Lai, which says that at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$, $\zeta_2(13)$ is irrational. As a warm-up we give a shorter, independent proof that at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$ is irrational. It uses the member $(a, j) = (8, 3)$ of a family of rational functions whose member $(4, 1)$ is the construction of Lai, Sprang and Zudilin for $\zeta_2(5)$, together with Andrews' transformation and Lai's 2-adic dominant-term method. The proof for the pair uses Volkenborn integrals of a shifted, well-poised deformation of the member $(6, 3)$ of that family. Its main ingredients, new to our knowledge, are a count of the denominators that exploits zeros of the rational functions at half-integers, a saddle-point bound for the coefficients whose numerical part is certified by interval arithmetic, and a nonvanishing argument at the auxiliary prime $n - 1$, after Lai and Sprang. Both theorems are formalized in Lean 4 with Mathlib, without hypotheses and with only the standard axioms. The one input that is cited rather than formalized is the identification, due to Lai, Sprang and Zudilin, of the Volkenborn-integral definition of $\zeta_2(s)$ used in Lean with the Kubota–Leopoldt value. The mathematics, the proofs and the formalization were produced by AI agents under the direction of the author, and have not been refereed by a human expert.

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Date: 25 September 2026.

2020 Mathematics Subject Classification. Primary 11J72; Secondary 11S40, 11S80, 11M06, 33C20, 68V20.

Key words and phrases. p -adic zeta values, irrationality, Volkenborn integral, very-well-poised hypergeometric series, formal verification, Lean.

Disclosure of AI use. The mathematics of this paper (the constructions and the proofs), the Lean formalizations (apart from the vendored, human-written files for the prime number theorem [4]) and the text were produced by AI agents (Anthropic Claude models) under the direction of the author. The proofs were checked by adversarial AI verification and, in their formal version, by the Lean kernel. No human expert has refereed them. The formal theorem statements are machine-checked; the identification of the 2-adic zeta values used in Lean with the Kubota–Leopoldt values is cited from the literature and is not formalized. Details are in Section 7.

1. INTRODUCTION

1.1. **Background.** For a prime p and an integer $s \geq 2$ let

$$\zeta_p(s) := L_p(s, \omega^{1-s}),$$

where $L_p(s, \chi)$ is the Kubota–Leopoldt p -adic L -function and ω is the Teichmüller character (§2.3). The number $\zeta_p(s) \in \mathbb{Q}_p$ is the p -adic limit of $\zeta(k)$ as $k \rightarrow s$ p -adically through negative integers $k \equiv s \pmod{p-1}$, and $\zeta_p(s) = 0$ for even s . The odd values $\zeta_p(3), \zeta_p(5), \dots$ are p -adic analogues of the odd values of the Riemann zeta function, and they are even less understood: it is not even known that $\zeta_p(2k+1) \neq 0$ for all $k \geq 1$ and all primes p [22].

In the archimedean case, $\zeta(3)$ is irrational by Apéry’s theorem [3], infinitely many of the values $\zeta(2k+1)$ are irrational by the theorem of Rivoal and Ball–Rivoal [33, 5], and at least one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational by a theorem of Zudilin [38]. On the p -adic side, Calegari [7] proved that $\zeta_2(3)$ and $\zeta_3(3)$ are irrational, and Beukers [6] gave another proof. Calegari, Dimitrov and Tang proved that $\zeta_2(5)$ is irrational, with their arithmetic holonomy method [8, 9]. Lai, Sprang and Zudilin [23] gave a second proof by an explicit construction in the spirit of Apéry, which also bounds the irrationality measure: $\mu(\zeta_2(5)) \leq 16 \log 2 / (8 \log 2 - 5) = 20.342 \dots$. For primes $p \geq 5$ the irrationality of $\zeta_p(3)$ is described as open in [21].

Sprang [36] used Volkenborn integrals of rational functions to construct linear forms in 1 and p -adic zeta values, and proved a p -adic analogue of the Ball–Rivoal theorem. Lai and Sprang [22] showed that, for every prime p , at least $(c_p - \varepsilon)\sqrt{s/\log s}$ of the numbers $\zeta_p(3), \zeta_p(5), \dots, \zeta_p(s)$ are irrational, for every $\varepsilon > 0$ and every sufficiently large odd s , with a constant $c_p > 0$. The 2-adic analogue of Zudilin’s theorem is due to Lai [20, Theorem 1.1]: for every integer $s \geq 0$, at least one of the numbers $\zeta_2(j)$ with j odd and $s+3 \leq j \leq 3s+5$ is irrational. For $s=3$ this says that at least one of $\zeta_2(7), \zeta_2(9), \zeta_2(11), \zeta_2(13)$ is irrational. Lai, Lupu and Sprang [21] proved analogous statements for all primes $p \geq 5$; for instance, some $\zeta_p(i)$ with i odd in $[3, p + p/\log p + 5]$ is irrational.

1.2. **Results.** We call an element of $\mathbb{Q}_2$ *rational* if it lies in the image of $\mathbb{Q} \rightarrow \mathbb{Q}_2$.

Theorem A. *At least one of the two 2-adic numbers $\zeta_2(7), \zeta_2(9)$ is irrational.*

Theorem B. *At least one of the three 2-adic numbers $\zeta_2(7), \zeta_2(9), \zeta_2(11)$ is irrational.*

Theorem A says that $\dim_{\mathbb{Q}}(\mathbb{Q} + \mathbb{Q}\zeta_2(7) + \mathbb{Q}\zeta_2(9)) \geq 2$; it does not say which of the two numbers is irrational. Theorem A implies Theorem B, which in turn implies the case $s=3$ of Lai’s theorem. The normalization of $\zeta_2(s)$ plays no role: the values of Coleman [11] and Furusho [16] differ from $\zeta_2(s)$ by the non-zero rational factor $(1 - 2^{-s})^{-1}$ (§2.3).

Although Theorem B follows from Theorem A, we prove it separately in Section 3: its proof is shorter, each of its steps extends a published argument, and it was the first of the two results to be formalized. It serves as a warm-up for the proof of Theorem A in Section 4, which uses a different construction.

1.3. **The two proofs.** Both proofs construct linear forms

$$S_n = \rho_0 + \sum_s Z_s \zeta_2(s) \quad (\rho_0, Z_s \in \mathbb{Q})$$

as Volkenborn integrals $\pm \int_{\mathbb{Z}_2} R_n'''(t + \frac{1}{2}) dt$ of the third derivative of explicit rational functions R_n with poles at $0, -1, \dots, -n$, following [23], and conclude with Lai's irrationality criterion (Lemma 2.9). This requires four estimates: the 2-adic size of S_n , a common denominator D_n of ρ_0 and the Z_s , the archimedean size of ρ_0 and the Z_s , and the nonvanishing of S_n for infinitely many n . Table 1 compares the two proofs.

TABLE 1. The two proofs at a glance.

	Theorem B (Section 3)	Theorem A (Section 4)
numbers	$\zeta_2(7), \zeta_2(9), \zeta_2(11)$	$\zeta_2(7), \zeta_2(9)$
rational function $R_n(t)$	$2^{16n}(2t+n) \frac{(t+\frac{1}{2})_n^8}{(t)_{n+1}^8}$	$(2t+n) \frac{\prod_{m=1}^6 (t+\frac{1}{2}-h_m)_{n+2h_m}}{(t)_{n+1}^6}$, $h = \frac{n}{40}(-17, 1, 2, 3, 5, 6)$
indices n	$n = 2^m - 1$	$40 \mid n, n-1$ prime, $n \neq 91080$
2-adic size	$v_2(S_n) = 32n + 14 - 11m$	$v_2(S_n) \geq 12n - 1 - 10 \log_2(1.3n)$
denominators	$d_n^{12}/\Phi_n = e^{11n+o(n)}$; Andrews' transformation	odd part $\leq e^{(10-R^{(0)})n+o(n)}$, $R^{(0)} = 1.0901\dots$; Legendre counts, critical zeros
coefficients	$\leq 3.4 \cdot 10^8(n+1)^2 2^{16n}$; Cauchy's estimate	$\leq 1.56 \cdot 10^{12} n^{-4} e^{-0.7812n}$ for $n \geq 2000$; saddle point
nonvanishing	exact 2-adic valuation	$v_{n-1}(\rho_0) = -9$ (auxiliary prime)
exponent per n	$11 - 16 \log 2 = -0.0904$	$10 - R^{(0)} - 0.7812 - 12 \log 2 = -0.189$
computer assistance	none	value of $R^{(0)}$; three inequalities certified by interval arithmetic
Lean	as in the text (Lemma 3.10 with unspecified constants)	weaker constants: growth rate -0.72 , odd denominators $e^{(9+\epsilon)n}$

Theorem B. Here $R_n(t) = 2^{16n}(2t+n)(t+\frac{1}{2})_n^8/(t)_{n+1}^8$. This is the member $(a, j) = (8, 3)$ (eighth powers, third derivative) of a family whose member $(4, 1)$ is the construction of Lai, Sprang and Zudilin for $\zeta_2(5)$ [23]; the linear forms involve 1, $\zeta_2(7)$, $\zeta_2(9)$ and $\zeta_2(11)$ (Lemma 3.2). The common denominator is $D_n = d_n^{12}/\Phi_n = e^{11n+o(n)}$ (Corollary 3.8). The saving of one power of the prime p for $\max(\sqrt{2n}, 10) < p \leq n$ comes from Andrews' transformation of a terminating very-well-poised hypergeometric series into a multiple sum [2, 18], as in [23, Lemma 5.3], but with depth 8 instead of 4 (Proposition 3.7). The nonvanishing follows Lai's 2-adic dominant-term method [20, §6]: along $n = 2^m - 1$ we compute $v_2(S_n) = 32n + 14 - 11m$ exactly (Proposition 3.13). This needs a new identity (Lemma 3.12), because the linear factor $2t+n$ vanishes at the dominant pole. The final exponent is $11 - 16 \log 2 = -0.0904$ per n : negative, but only barely.

Theorem A. The member $(a, j) = (6, 3)$ of the same family gives linear forms in 1, $\zeta_2(7)$, $\zeta_2(9)$, but its exponent is $9 - 12 \log 2 = +0.68$ (Remark 3.17). We deform it. The six blocks $(t+\frac{1}{2})_n$ of the numerator are replaced by shifted, well-poised blocks $(t+\frac{1}{2}-h_m)_{n+2h_m}$, in the spirit of Rhin and Viola [32], with the fixed shift vector

$$h = \frac{n}{40}(-17, 1, 2, 3, 5, 6) \quad (40 \mid n),$$

found by numerical exploration (*configuration E*, §4.1). The five positive shifts give R_n zeros of order at least 5 at all half-integers in $(-n - \frac{n}{40}, \frac{n}{40})$, which we call *critical zeros*; the negative shift produces prime-selective savings in the denominators. Three of the four estimates need new arguments; the first is a routine extension of the literature:

- (i) $v_2(S_n) \geq 12n - O(\log n)$, by the Δ -calculus of Sprang and Lai applied to binomial coefficients with shifted arguments (Theorem 4.4);
- (ii) the odd part of the common denominator is at most $e^{(10-R^{(0)})n+o(n)}$ with $R^{(0)} = 1.0901\dots$ (Theorem 4.17), by a Legendre-type count of the q -divisible linear factors at each pole; the critical zeros exclude all primes $q > n$;
- (iii) $\max(|\rho_0|, |Z_7|, |Z_9|) \leq 1.56 \cdot 10^{12} n^{-4} e^{-0.7812n}$ for $n \geq 2000$ (Theorem 4.21), although the residues of R_n are exponentially larger. We write the coefficients as contour integrals against periodic kernels with poles only at half-integers, which is possible because of the critical zeros, and bound them at a saddle point;
- (iv) if n is as in Table 1 and $\ell = n - 1$, then $v_\ell(\rho_0) = -9$, while $v_\ell(Z_7) \geq -3$ and $v_\ell(Z_9) \geq -1$ (Theorem 4.32). This gives the nonvanishing, by the auxiliary-prime idea of Lai and Sprang [22, Lemma 2.2]; our $\ell = n - 1$ is a variant of their $\ell(n) = n + 1$ [22, §3, Lemma 5.6].

The final exponent is $10 - 12 \log 2 + g - R^{(0)} = -0.189$ per n , with $g = -0.7812$ (§4.6). Sharper denominator bounds, which are not needed, would improve it to -0.551 (Remark 4.19 and Table 12; proofs not included).

1.4. Computer assistance. The proof of Theorem B uses no computer calculation. The proof of Theorem A, as written in Section 4, is computer-assisted at two places. The first is the numerical value of the savings constant $R^{(0)}$ (Proposition 4.18): an exact rational computation plus a periodic tail; only $R^{(0)} > 0.90103$ is needed, and a crude bound for the tail already gives $R^{(0)} > 0.99$. The second is a set of three inequalities for an explicit harmonic function (Lemma 4.28), which we certified by interval arithmetic with three independent programs. These programs, the computation of $R^{(0)}$ and the data behind the tables are described in §7.4. The numerical data quoted in the remarks of Sections 3 and 4 come mostly from exact rational computations, and from floating-point computations where stated; they illustrate the statements and are not used in the proofs. In the formal proof of Theorem A, all numerical inputs are checked by the Lean kernel (§5.4).

1.5. Formal verification. Both theorems are formalized in Lean 4 [27] with Mathlib [26]. The Lean theorems

`Zeta2.Pair.zeta2_7_9_not_both_rational_unconditional` (Theorem A),
`Zeta2.zeta2_7_9_11_not_all_rational_unconditional` (Theorem B)

have no hypotheses. Their proofs depend only on the three standard axioms of Lean and Mathlib: propositional extensionality, the axiom of choice and the soundness of quotients (`propext`, `Classical.choice`, `Quot.sound`). The formal proofs are complete: they include the prime number theorem (from a Lean proof of the Wiener–Ikehara theorem in open Mathlib pull requests [4]), Andrews’ transformation, and the numerical inequalities used by the formal proof of Theorem A. The one input that is not formalized concerns the reading of the statements. The Lean

developments define $\zeta_2(s)$ through Volkenborn integrals, as $J_{s-1}/((s-1)2^s)$, and the identification of this number with the Kubota–Leopoldt value $L_2(s, \omega^{1-s})$ is Lemma 2.8 of [23] (Lemma 2.3 below), which is cited and not formalized; it only involves a non-zero rational factor. The formal proof of Theorem A follows Section 4 with weaker constants: the growth rate -0.72 in place of -0.7812 , and the rate 9 in place of $10 - R^{(0)} = 8.9099$ for the odd denominators, with the margin $-0.72 + 9 = 8.28 < 12 \log 2 = 8.3178$. Both formalizations are public and are registered in the Palomar registry of Lean-verified results: Theorem A as PALOMAR-2026-09-25-000026 [14, 31] and Theorem B as PALOMAR-2026-09-25-000023 [13, 30]. Section 5 gives the exact statements, the definitions that have to be trusted, and instructions for checking.

1.6. Novelty and related work. To our knowledge, neither Theorem A nor Theorem B has been stated before. This rests on a literature search by AI agents on 24 September 2026 (arXiv, including the papers citing [20]; Zenodo; zbMATH; OpenAlex; Crossref; GitHub; public talks and blogs), repeated briefly on 25 September 2026. Google Scholar, MathSciNet and Lai’s thesis were not searched, so we cannot rule out an earlier statement. We found two unrefereed drafts by C. D. Long, posted on GitHub in August 2026 and based on arithmetic holonomy. The first [24] claims that $\zeta_2(s)$ is irrational for every odd s with $3 \leq s \leq 29$ (together with results for $p = 3, 5, 7$); this claim would imply Theorems A and B. The second [25] claims that for every prime p the values $\zeta_p(s)$ are irrational for a set of odd s of density one. We have not verified these claims, and our proofs do not depend on them.

The methods of this paper are those of [36, 20, 22, 23]: Volkenborn integrals of rational functions, the Δ -calculus, Lai’s criterion and its auxiliary-prime variant, and Andrews’ transformation [2, 18]. The shifted well-poised blocks are inspired by Rhin and Viola [32] (see also the well-poised parameters of [40, §8]), and the saddle-point analysis of the coefficients follows Zudilin’s archimedean analysis of Barnes-type integrals [40, §3, Lemma 5; §8, Lemma 20], [39, §§2–3]. Andrews’ transformation is available at every depth [18, Théorème 8; §9, Corollaires 1–2], and Lai, Sprang and Zudilin used it at depth 4 in the 2-adic setting [23, Lemma 5.3]. To our knowledge the following ingredients are new: for Theorem B, the application of Andrews’ transformation to the 2-adic family $(8, 3)$, with the case analysis of Step 5 of Proposition 3.7, and the identity of Lemma 3.12; for Theorem A, the critical zeros of order ≥ 5 on an extended range, which refine the root trick of [15, Lemma 2] and [20, Lemma 4.6], and their double use for the denominators and for the contour integrals; the contour-integral representation of the coefficients of a 2-adic linear form with half-integer kernels; and the ℓ -adic computation at the auxiliary prime $\ell = n - 1$ (two singular poles, the constants $\beta_1 = -168$ and K_E), a variant of the auxiliary prime $\ell(n) = n + 1$ of [22, §3, Lemma 5.6]. The arithmetic holonomy method of Calegari, Dimitrov and Tang [9] is a different approach, which we do not use.

1.7. Organization. Section 2 collects definitions and published tools. Section 3 proves Theorem B and Section 4 proves Theorem A; the two sections can be read independently, and the symbols R_n , S_n , $r_{i,k}$, c_i , ρ_0 , Z_s , D_n have different meanings in them. Section 5 describes the formalization, Section 6 discusses the limits of the

method, and Section 7 describes how this work was produced with AI agents and what has been verified, and how.

2. PRELIMINARIES

This section collects the definitions and the published tools used in the proofs of Theorems A and B. None of the mathematics in it is new.

2.1. Conventions and notation. Throughout, p and q denote primes. We write v_p for the p -adic valuation on $\mathbb{Q}_p$, with $v_p(0) = +\infty$, and $|x|_p = p^{-v_p(x)}$. For $x \in \mathbb{Q}$ we use $v_q(x)$ for every prime q , and $\mathbb{Z}_{(q)} = \{x \in \mathbb{Q} : v_q(x) \geq 0\}$. The archimedean absolute value is $|\cdot|$, and $\log$ is the natural logarithm. An element of $\mathbb{Q}_2$ is called *rational* if it lies in the image of $\mathbb{Q} \rightarrow \mathbb{Q}_2$.

- $(x)_0 = 1$ and $(x)_k = x(x+1) \cdots (x+k-1)$ for $k \geq 1$ (rising factorial, or Pochhammer symbol).
- For $x \in \mathbb{Z}$ and $N \geq 0$, $\binom{t+x}{N} = (t+x)(t+x-1) \cdots (t+x-N+1)/N!$ is a polynomial in t that takes integer values on $\mathbb{Z}$, hence values in $\mathbb{Z}_p$ on $\mathbb{Z}_p$.
- $s_2(N)$ is the sum of the binary digits of an integer $N \geq 0$. Legendre's formula gives $v_2(N!) = N - s_2(N)$. Kummer's theorem says that $v_p\binom{a+b}{a}$ is the number of carries in the addition of a and b in base p . In particular $p \mid \binom{2a}{a}$ whenever the least non-negative residue of a modulo p is at least $(p+1)/2$, and $\binom{2^m-1}{j}$ is odd for $0 \leq j \leq 2^m - 1$.
- $\mathcal{D}_\lambda = \frac{1}{\lambda!} \frac{d^\lambda}{dt^\lambda}$ for $\lambda \geq 0$, and $[\varepsilon^\mu]F$ is the coefficient of ε^μ in a power series or Laurent series $F(\varepsilon)$.
- $d_n = \text{lcm}(1, 2, \dots, n)$ for $n \geq 1$, and $d_0 = 1$.
- $\theta(x) = \sum_{p \leq x} \log p$ and $\psi(x) = \sum_{p^k \leq x, k \geq 1} \log p$ are the Chebyshev functions. Since $v_p(d_n) = \lfloor \log n / \log p \rfloor$, we have $\log d_n = \psi(n)$.

2.2. The Volkenborn integral. We follow [34, Ch. 5], [20, §2.2] and [23, §2.2]. A function $f: \mathbb{Z}_p \rightarrow \mathbb{Q}_p$ is *Volkenborn integrable* [37] if the Riemann sums

$$(2.1) \quad \frac{1}{p^N} \sum_{k=0}^{p^N-1} f(k)$$

converge p -adically as $N \rightarrow \infty$. The limit is the *Volkenborn integral* $\int_{\mathbb{Z}_p} f(t) dt$. The integral is $\mathbb{Q}_p$ -linear, but it is not translation invariant.

Let K be $\mathbb{Z}_p$ or $\mathbb{Q}_p$. A function $f: \mathbb{Z}_p \rightarrow K$ is *strictly differentiable*, written $f \in S^1(\mathbb{Z}_p, K)$, if $f(x) - f(y) = (x - y)g(x, y)$ for some continuous function g on $\mathbb{Z}_p \times \mathbb{Z}_p$. Every $f \in S^1(\mathbb{Z}_p, \mathbb{Q}_p)$ is Volkenborn integrable [34, p. 264]. Polynomials, power series $\sum_j a_j t^j$ with $a_j \in \mathbb{Z}_p$ and $a_j \rightarrow 0$, and rational functions in $\mathbb{Q}_p(t)$ without poles in $\mathbb{Z}_p$ are strictly differentiable on $\mathbb{Z}_p$ (see [20, §2.2], [23, §2.2]).

Lemma 2.1 (Translation; [20, Lemma 2.2], [23, Lemma 2.4]). *Let $f \in S^1(\mathbb{Z}_p, \mathbb{Q}_p)$ and let k be a positive integer. Then*

$$\int_{\mathbb{Z}_p} f(t+k) dt = \int_{\mathbb{Z}_p} f(t) dt + \sum_{\ell=0}^{k-1} f'(\ell).$$

See [34, Ch. 5, Prop. 2, p. 265]; Lai derives the lemma directly from the definition. For $p = 2$ and an integer $s \geq 1$ we put

$$(2.2) \quad J_s := \int_{\mathbb{Z}_2} \frac{dt}{(t + \frac{1}{2})^s}.$$

The integral exists because the only pole of $(t + \frac{1}{2})^{-s}$ is $-\frac{1}{2} \notin \mathbb{Z}_2$. Applying Lemma 2.1 to $f(t) = (t + \frac{1}{2})^{-s}$, whose derivative is $-s(t + \frac{1}{2})^{-s-1}$, gives for all integers $s \geq 1$ and $k \geq 0$:

$$(2.3) \quad \int_{\mathbb{Z}_2} \frac{dt}{(t + k + \frac{1}{2})^s} = J_s - s A_k^{(s+1)}, \quad A_k^{(s)} := \sum_{\ell=0}^{k-1} \frac{1}{(\ell + \frac{1}{2})^s} = \sum_{\ell=1}^k \frac{1}{(\ell - \frac{1}{2})^s}.$$

2.3. 2-adic zeta values. We recall the definitions of [23, §2.3] (see also [20, §2.3] and [10, Ch. 11]). Put $q_p = p$ for odd p and $q_2 = 4$. Then $\mathbb{Q}_p^\times \cong p^{\mathbb{Z}} \cdot \mu_{\varphi(q_p)}(\mathbb{Z}_p) \times (1 + q_p \mathbb{Z}_p)$, and we write ω and $\langle \cdot \rangle$ for the two projections; ω is the Teichmüller character. For $s \in \mathbb{C}_p \setminus \{1\}$ with $|s|_p < q_p p^{-1/(p-1)}$ and $x \in \mathbb{Q}_p$ with $|x|_p \geq q_p$, the p -adic Hurwitz zeta function is

$$\zeta_p(s, x) := \frac{1}{s-1} \int_{\mathbb{Z}_p} \langle t+x \rangle^{1-s} dt.$$

For a Dirichlet character χ of conductor f and a common multiple M of f and q_p , the Kubota–Leopoldt p -adic L -function is

$$L_p(s, \chi) := \frac{\langle M \rangle^{1-s}}{M} \sum_{\substack{0 \leq j \leq M-1 \\ p \nmid j}} \chi(j) \zeta_p\left(s, \frac{j}{M}\right);$$

it does not depend on the choice of M .

Definition 2.2 ([23, Definition 2.7]). For an integer $s \geq 2$, the p -adic zeta value is $\zeta_p(s) := L_p(s, \omega^{1-s})$.

This is the normalization of [20], [22] and [23]. The values $\zeta_p(s)$ of Coleman [11] and Furusho [16] differ from it by the Euler factor $(1 - p^{-s})^{-1}$ [23, §2.3]. For a fixed integer $s \geq 2$ this factor is a non-zero rational number, so the choice of normalization does not affect any statement about irrationality. By [7, Lemma 2.4] (as quoted in [23, §2.3]), $\zeta_p(s)$ is the p -adic limit of $\zeta(k)$ as $k \rightarrow s$ p -adically through negative integers $k \equiv s \pmod{p-1}$. In particular $\zeta_p(s) = 0$ for every even integer $s \geq 2$.

Lemma 2.3 ([23, Lemma 2.8]). For every integer $s \geq 2$,

$$\frac{1}{s-1} \int_{\mathbb{Z}_2} \frac{dt}{(t + \frac{1}{2})^{s-1}} = 2^s \zeta_2(s).$$

Corollary 2.4. For every integer $s \geq 1$ we have $J_s = s 2^{s+1} \zeta_2(s+1)$. In particular $J_s = 0$ for odd s , and

$$J_6 = 768 \zeta_2(7), \quad J_8 = 4096 \zeta_2(9), \quad J_{10} = 20480 \zeta_2(11).$$

Proof. Apply Lemma 2.3 with $s+1$ in place of s . If s is odd, then $s+1$ is even and $\zeta_2(s+1) = 0$. $\square$

The proofs of Theorems A and B never use the vanishing of J_s for odd s : there the even-index coefficients vanish by a symmetry of the rational functions involved (compare [20, Lemma 3.3]).

Remark 2.5 (The definition used in Lean, and what is cited). Mathlib does not contain the Kubota–Leopoldt L -function. Both Lean developments therefore define the Volkenborn integral through its Riemann sums (2.1) for $p = 2$, define J s as the integral J_s of (2.2), and put `zeta2 s` $= J_{s-1}/((s-1)2^s)$. The six definitions involved are reproduced verbatim in §5.2. By Lemma 2.3, `zeta2 s` $= \zeta_2(s)$ for every $s \geq 2$; for instance `zeta2 7` $= J_6/768$, `zeta2 9` $= J_8/4096$ and `zeta2 11` $= J_{10}/20480$. *This identification is cited from [23]; it is not formalized.* Since it only involves the non-zero rational factor $(s-1)2^s$, the formal statements “`zeta2 7`, `zeta2 9` are not both rational” and “`zeta2 7`, `zeta2 9`, `zeta2 11` are not all rational” are equivalent to the corresponding statements about J_6, J_8, J_{10} . They become Theorems A and B through Lemma 2.3, for $\zeta_2(s)$ as in Definition 2.2 and for every normalization that differs from it by a non-zero rational factor.

2.4. The operators Δ and Δ_m . The following device, introduced by Sprang [36] and developed by Lai [20], bounds the p -adic size of Volkenborn integrals. For an integer $k \geq 1$ let k_- be the integer obtained from k by deleting the leading digit of its base- p expansion; for $p = 2$ this means $k - k_- = 2^{\lfloor \log_2 k \rfloor}$.

Definition 2.6 ([20, Definition 2.3]). Let $f: \mathbb{Z}_p \rightarrow \mathbb{Q}_p$. For every integer $m \geq 0$ put

$$\Delta_m(f) := \inf_{k \geq p^m} v_p \left(\frac{f(k) - f(k_-)}{k - k_-} \right), \quad \Delta(f) := \min\{1 + v_p(f(0)), \Delta_0(f)\}.$$

Clearly $-\infty \leq \Delta(f) \leq \Delta_0(f) \leq \Delta_1(f) \leq \dots \leq +\infty$, and $\Delta(f) > -\infty$ if f is strictly differentiable.

Lemma 2.7 ([20, Lemma 2.4]). *Let $f \in S^1(\mathbb{Z}_p, \mathbb{Q}_p)$. Then*

- (a) $v_p(\int_{\mathbb{Z}_p} f(t) dt) \geq \Delta(f) - 1$;
- (b) *for every integer $m \geq 0$,*

$$\int_{\mathbb{Z}_p} f(t) dt \equiv \frac{1}{p^m} \sum_{k=0}^{p^m-1} f(k) \pmod{p^{\Delta_m(f)-1} \mathbb{Z}_p}.$$

Proof. For $M > m$ one checks

$$\frac{1}{p^M} \sum_{k=0}^{p^M-1} f(k) = \frac{1}{p^m} \sum_{k=0}^{p^m-1} f(k) + \sum_{l=m}^{M-1} \sum_{k=p^l}^{p^{l+1}-1} \frac{f(k) - f(k_-)}{p^{l+1}},$$

and each summand of the double sum has valuation at least $\Delta_m(f) + v_p(k - k_-) - (l+1) = \Delta_m(f) - 1$, since $v_p(k - k_-) = l$. Letting $M \rightarrow \infty$ gives (b). For $m = 0$, (b) says $\int f \equiv f(0) \pmod{p^{\Delta_0(f)-1}}$, and (a) follows from the definition of $\Delta(f)$. $\square$

Lemma 2.8 ([20, Lemma 2.5], [23, Lemma 2.6]). *Let $m \geq 0$.*

- (i) $\Delta(f + g) \geq \min\{\Delta(f), \Delta(g)\}$, and $\Delta(Cf) = \Delta(f) + v_p(C)$ for $C \in \mathbb{Q}_p$.
The same holds with Δ_m in place of Δ .
- (ii) If $f, g \in S^1(\mathbb{Z}_p, \mathbb{Z}_p)$, then $\Delta(fg) \geq \min\{\Delta(f), \Delta(g)\}$, and the same inequality holds with Δ_m in place of Δ .
- (iii) If $f(t) = \sum_{j \geq 0} a_j t^j \in \mathbb{Z}_p[[t]]$ with $|a_j|_p \rightarrow 0$, then $\Delta_m(f) \geq \Delta(f) \geq 0$.

- (iv) If $x \in \mathbb{Z}$, $N \geq 1$ and $f(t) = \binom{t+x}{N}$, then $\Delta_m(f) \geq \Delta(f) \geq -\lfloor \log N / \log p \rfloor$.
If moreover $m > \lfloor \log N / \log p \rfloor$, then $\Delta_m(f^p) \geq 1 - \lfloor \log N / \log p \rfloor$.

Part (i) is immediate from the definitions; parts (ii)–(iv) are [20, Lemma 2.5 (2), (1), (3)]. We shall use the lemma for $p = 2$, where (iv) reads $\Delta(\binom{t+x}{N}) \geq -\lfloor \log_2 N \rfloor$ and $\Delta_m(\binom{t+x}{N}^2) \geq 1 - \lfloor \log_2 N \rfloor$ for $m > \lfloor \log_2 N \rfloor$. The constant function $1 = \binom{t+x}{0}$ has $\Delta(1) = 1$.

2.5. Irrationality criteria.

Lemma 2.9 ([20, Lemma 2.1]). *Let $\alpha_1, \dots, \alpha_s \in \mathbb{Q}_p$, and for $n \geq 1$ let $a_{n,0}, \dots, a_{n,s}$ be integers. Assume that the numbers $L_n := a_{n,0} + \sum_{j=1}^s a_{n,j} \alpha_j$ satisfy*

$$\lim_{n \rightarrow \infty} \max_{0 \leq j \leq s} |a_{n,j}| \cdot |L_n|_p = 0$$

and that $L_n \neq 0$ for infinitely many n . Then at least one of $\alpha_1, \dots, \alpha_s$ is irrational.

Proof. Pass to the subsequence of n with $L_n \neq 0$. If all α_j were rational with common denominator $D \geq 1$, then DL_n would be a non-zero integer, so $|DL_n|_p \geq |DL_n|^{-1} \geq (D(1 + \sum_j |\alpha_j|) \max_j |a_{n,j}|)^{-1}$. Hence $\max_j |a_{n,j}| \cdot |L_n|_p \geq (|D|_p D(1 + \sum_j |\alpha_j|))^{-1}$, a positive constant, contradicting the hypothesis. $\square$

The nonvanishing of L_n is usually the hard part. Lai and Sprang replace it by a condition on auxiliary primes.

Lemma 2.10 ([22, Lemma 2.2]). *Let $\xi_0, \dots, \xi_s \in \mathbb{Q}_p$ with $\xi_0 \neq 0$, and let $L_n = l_{0,n}X_0 + \dots + l_{s,n}X_s \in \mathbb{Z}[X_0, \dots, X_s]$ be non-trivial linear forms. Assume that there are an unbounded set $I \subseteq \mathbb{N}$ and primes $\ell(n)$, $n \in I$, such that*

- (1) $\max_{0 \leq i \leq s} |l_{i,n}| \cdot |L_n(\xi_0, \dots, \xi_s)|_p \rightarrow 0$ as $n \rightarrow \infty$ in I ;
- (2) $v_{\ell(n)}(l_{0,n}) < v_{\ell(n)}(l_{i,n})$ for $i = 1, \dots, s$ and all $n \in I$;
- (3) $\ell(n) \rightarrow \infty$ as $n \rightarrow \infty$ in I .

Then at least one of $\xi_0, \dots, \xi_s$ is irrational.

Proof. Suppose all ξ_i are rational. By (3), for large $n \in I$ we have $v_{\ell(n)}(\xi_0) = 0$ and $v_{\ell(n)}(\xi_i) \geq 0$, so (2) gives $v_{\ell(n)}(L_n(\xi)) = v_{\ell(n)}(l_{0,n}) < \infty$; in particular $L_n(\xi) \neq 0$. The proof of Lemma 2.9, with a common denominator of $\xi_0, \dots, \xi_s$, now contradicts (1). $\square$

Theorem B uses Lemma 2.9; there the nonvanishing is proved directly by an exact 2-adic valuation. The proof of Theorem A uses the variant Lemma 2.10 with $\xi_0 = 1$. The criterion proved in both Lean developments is Lemma 2.9 in the slightly more flexible form “ $|a_{n,j}| \leq B_n$ for all j , $B_n |L_n|_2 \rightarrow 0$ and $L_n \neq 0$ frequently”.

2.6. Least common multiples and the prime number theorem. The only analytic input from prime number theory is the prime number theorem in the form

$$(2.4) \quad \psi(x) = x + o(x), \quad \theta(x) = x + o(x) \quad (x \rightarrow \infty),$$

together with the elementary facts $\log d_n = \psi(n)$ and $0 \leq \psi(x) - \theta(x) = O(\sqrt{x})$. For Theorem B only the upper bound $\psi(n) \leq (1 + o(1))n$ matters (Corollary 3.8). The exponent there is $11 - 16 \log 2 = -0.0904$ per n , far too thin for Chebyshev-type bounds. The $o(n)$ terms coming from (2.4) are ineffective as used here, so neither proof yields an explicit n_0 ; for Theorem B, explicit versions of (2.4) would give one, but we do not pursue this. For Theorem A we also use the explicit bound

$\theta(x) < 1.01624x$ for all $x > 0$ of Rosser and Schoenfeld [35, Theorem 9]. In the Lean developments, (2.4) enters in the form $\psi(x)/x \rightarrow 1$ for Mathlib’s Chebyshev function ψ ; it is proved there from a Lean proof of the Wiener–Ikehara theorem that is taken, with its original authorship, from open Mathlib pull requests [4] (see Section 5).

3. PROOF OF THEOREM B: A WARM-UP

In this section we prove Theorem B.

Theorem 3.1 (Theorem B). *At least one of the three 2-adic numbers $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$ is irrational.*

Theorem A implies Theorem B, so this section is not needed for the main result. We include it because the construction is the simplest one we know that goes beyond Lai’s theorem, the case $s = 3$ of [20, Theorem 1.1], which says that at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$, $\zeta_2(13)$ is irrational. Each ingredient extends a published argument. The linear forms and the denominator bound follow the proof of the irrationality of $\zeta_2(5)$ by Lai, Sprang and Zudilin [23]. The nonvanishing follows Lai’s dominant-term method [20, §6], with one new identity (Lemma 3.12). Theorem B was also the first of the two results to be formalized. The reader may take this section as a warm-up for the proof of Theorem A.

Throughout this section $n \geq 0$ is an integer. The objects R_n , S_n , $r_{i,k}$, c_i , ρ_0 , Z_s , Φ_n , D_n defined below depend on n and belong to this section only; the proof of Theorem A uses a different construction.

3.1. Overview. The construction is the case $(a, j) = (8, 3)$ (eighth powers, third derivative) of the family of rational functions and linear forms

$$R(t) = 2^{2an}(2t+n) \frac{(t+\frac{1}{2})_n^a}{(t)_{n+1}^a}, \quad \pm \int_{\mathbb{Z}_2} R^{(j)}(t+\frac{1}{2}) dt,$$

whose case $(a, j) = (4, 1)$ is the construction of [23, Definitions 3.1 and 3.2] for $\zeta_2(5)$. The proof has four ingredients, listed in Table 2. Table 3 shows how their exponential rates combine; the final exponent $11 - 16 \log 2 = -0.0904$ is negative, but only barely.

TABLE 2. The ingredients of the proof of Theorem B. In the last two rows $n = 2^m - 1$ with $m \geq 2$.

ingredient	statement	method
linear forms (Lemma 3.2)	$S_n = \rho_0 + \sum_s Z_s \zeta_2(s)$, $s \in \{7, 9, 11\}$, with $\rho_0, Z_s \in \mathbb{Q}$	as in [23, Lemma 3.3]
denominators (Cor. 3.8)	$D_n \rho_0, D_n Z_s \in \mathbb{Z}$ for $D_n = d_n^{12} / \Phi_n = e^{11n+o(n)}$	root trick [20, Lemma 4.6]; Andrews’ transformation, as in [23, Lemma 5.3]
size (Lemma 3.10)	$\max(\rho_0 , Z_s) \leq 3.4 \cdot 10^8 (n+1)^2 2^{16n}$	Cauchy’s estimate
nonvanishing (Prop. 3.13)	$v_2(S_n) = 32n + 14 - 11m$ exactly	[20, §6] and Lemma 3.12
conclusion (§3.6)	exponent $11 - 16 \log 2 = -0.0904 < 0$	[20, Lemma 2.1]

TABLE 3. Exponential rates along $n = 2^m - 1$, per unit of n (up to $o(1)$). Here $a_{n,j}$ are the integer coefficients of $D_n S_n$.

quantity	rate
$\frac{1}{n} \log \max(\rho_0 , Z_7 , Z_9 , Z_{11})$	$\leq 16 \log 2 = 11.0904$
$\frac{1}{n} \log D_n$	$= 11$
$\frac{1}{n} \log \max_j a_{n,j} $	$\leq 11 + 16 \log 2 = 22.0904$
$-\frac{1}{n} \log D_n S_n _2$	$= 32 \log 2 = 22.1807$
$\frac{1}{n} \log (\max_j a_{n,j} \cdot D_n S_n _2)$	$\leq 11 - 16 \log 2 = -0.0904$

No step of the proof relies on a computer calculation. The numerical data quoted below were computed in exact rational arithmetic (unless stated otherwise) and serve only as corroboration. The whole proof, including Andrews' transformation and the prime number theorem, has been formalized in Lean and checked by the Lean kernel; the only cited, unformalized input is the identification of the Lean definition of ζ_2 with the Kubota–Leopoldt value (Remark 2.5).

3.2. The construction and the linear forms. For $n \geq 0$ put

$$(3.1) \quad R_n(t) := 2^{16n} (2t + n) \frac{(t + \frac{1}{2})_n^8}{(t)_{n+1}^8} \in \mathbb{Q}(t), \quad S_n := - \int_{\mathbb{Z}_2} R_n'''(t + \frac{1}{2}) dt.$$

The poles of $R_n(t + \frac{1}{2})$ are the points $-\frac{1}{2} - k$ with $0 \leq k \leq n$. They have 2-adic absolute value 2, so none lies in $\mathbb{Z}_2$, and $S_n \in \mathbb{Q}_2$ is well defined (§2.2). The sign follows [23, Definition 3.2]. Since $\deg R_n = 1 + 8n - 8(n+1) = -7$,

$$(3.2) \quad R_n(t) = \sum_{i=1}^8 \sum_{k=0}^n \frac{r_{i,k}}{(t+k)^i}, \quad r_{i,k} \in \mathbb{Q}, \quad c_i := \sum_{k=0}^n r_{i,k}.$$

For $0 \leq k \leq n$ we have $(-k + \varepsilon)_{n+1} = (-1)^k \varepsilon (1 - \varepsilon)_k (1 + \varepsilon)_{n-k}$ and $(-k + \varepsilon + \frac{1}{2})_n = (-1)^k (\frac{1}{2} - \varepsilon)_k (\frac{1}{2} + \varepsilon)_{n-k}$. Hence the partial-fraction coefficients are Taylor coefficients:

$$(3.3) \quad G_k(\varepsilon) := \varepsilon^8 R_n(-k + \varepsilon) = 2^{16n} (n - 2k + 2\varepsilon) \Psi_k(\varepsilon)^8,$$

where

$$\Psi_k(\varepsilon) := \frac{(\frac{1}{2} - \varepsilon)_k (\frac{1}{2} + \varepsilon)_{n-k}}{(1 - \varepsilon)_k (1 + \varepsilon)_{n-k}},$$

and $r_{8-\mu,k} = [\varepsilon^\mu] G_k(\varepsilon)$ for $0 \leq \mu \leq 7$. With $(\frac{1}{2})_j / j! = 4^{-j} \binom{2j}{j}$ this gives, for instance, $r_{8,k} = G_k(0) = (n - 2k) \binom{2k}{k} \binom{2n-2k}{n-k}^8 \in \mathbb{Z}$.

Replacing t by $-t - n$ gives $R_n(-t - n) = -R_n(t)$. Comparing partial fractions,

$$(3.4) \quad r_{i,n-k} = (-1)^{i+1} r_{i,k} \quad (1 \leq i \leq 8, 0 \leq k \leq n),$$

so $c_i = 0$ for even i . Moreover $c_1 = \lim_{t \rightarrow \infty} t R_n(t) = 0$ because $\deg R_n \leq -2$.

Lemma 3.2 (Linear forms). *For every $n \geq 0$,*

$$(3.5) \quad S_n = \rho_0 + Z_7 \zeta_2(7) + Z_9 \zeta_2(9) + Z_{11} \zeta_2(11),$$

where $Z_{i+4} := (i)_4 2^{i+4} c_i$ for $i \in \{3, 5, 7\}$, that is,

$$Z_7 = 46080 c_3, \quad Z_9 = 860160 c_5, \quad Z_{11} = 10321920 c_7,$$

and

$$(3.6) \quad \rho_0 := - \sum_{i=1}^8 \sum_{k=1}^n (i)_4 r_{i,k} A_k^{(i+4)}, \quad A_k^{(s)} = \sum_{\ell=0}^{k-1} \frac{1}{(\ell + \frac{1}{2})^s}.$$

Equivalently, $S_n = \rho_0 + 60 c_3 J_6 + 210 c_5 J_8 + 504 c_7 J_{10}$ with J_s as in (2.2).

Proof. By (3.2), $R_n'''(t) = - \sum_{i,k} (i)_3 r_{i,k} (t+k)^{-i-3}$. By linearity and (2.3) with $s = i+3$,

$$S_n = \sum_{i=1}^8 \sum_{k=0}^n (i)_3 r_{i,k} \int_{\mathbb{Z}_2} \frac{dt}{(t+k+\frac{1}{2})^{i+3}} = \sum_{i=1}^8 (i)_3 c_i J_{i+3} - \sum_{i=1}^8 \sum_{k=1}^n (i)_4 r_{i,k} A_k^{(i+4)}.$$

The last sum is $-\rho_0$. The terms with $i=1$ and with even i in the first sum vanish because $c_1 = 0$ and $c_i = 0$ for even i . Since $(3)_3 = 60$, $(5)_3 = 210$ and $(7)_3 = 504$, this is the second form of the lemma. By Corollary 2.4, $(i)_3 c_i J_{i+3} = (i)_4 2^{i+4} c_i \zeta_2(i+4)$. $\square$

The proof does not use $\zeta_2(\text{even}) = 0$, and Lemma 2.3 is used only to rewrite $J_6 = 768 \zeta_2(7)$, $J_8 = 4096 \zeta_2(9)$ and $J_{10} = 20480 \zeta_2(11)$. For example, $R_0(t) = 2t^{-7}$ and $S_0 = 20643840 \zeta_2(11)$. For $n=1$,

$$\rho_0 = 14155776, \quad (Z_7, Z_9, Z_{11}) = (-849346560, 21139292160, 52848230400).$$

In exact arithmetic we confirmed $c_1 = 0$ and $c_i = 0$ for even i for all $n \leq 300$ and $n \in \{383, 447, 511\}$. The same computation for $(a, j) = (4, 1)$ reproduces the values $\rho_{1,0} = -1024$, $\rho_{1,3} = 73728$ of [23] and their recurrence [23, (1.1)] for $n \leq 11$.

3.3. Denominators.

Lemma 3.3 (Building blocks; [19, Lemma 4.2], [40, Lemma 16]). *Let $n \geq 0$, $F(t) := 4^n(t + \frac{1}{2})_n/n!$ and $G(t) := n!/(t)_{n+1}$. For every integer $\lambda \geq 0$,*

$$d_n^\lambda \mathcal{D}_\lambda F(t)|_{t=-k} \in \mathbb{Z} \quad (k \in \mathbb{Z}), \quad d_n^\lambda \mathcal{D}_\lambda((t+k)G(t))|_{t=-k} \in \mathbb{Z} \quad (0 \leq k \leq n).$$

The first statement is [19, Lemma 4.2] with $(a, b) = (1, 2)$, in the form stated in [21, Lemma 5.2]; the second is [40, Lemma 16] ([21, Lemma 5.1]). The same two building blocks are used in [23, §5].

Lemma 3.4. $d_n^{8-i} r_{i,k} \in \mathbb{Z}$ for $1 \leq i \leq 8$ and $0 \leq k \leq n$. Hence $d_n^{8-i} c_i \in \mathbb{Z}$, and $d_n^5 Z_7, d_n^3 Z_9, d_n Z_{11} \in \mathbb{Z}$.

Proof. We have $(t+k)^8 R_n(t) = (2t+n) F(t)^8 ((t+k)G(t))^8$ and $r_{i,k} = \mathcal{D}_{8-i}((t+k)^8 R_n(t))|_{t=-k}$. By the Leibniz rule, $r_{i,k}$ is a sum of products of $\mathcal{D}_{\lambda_0}(2t+n)$, eight factors $\mathcal{D}_\lambda F$ and eight factors $\mathcal{D}_\lambda((t+k)G)$ at $t=-k$, with total order $8-i$. The first factor is $n-2k$, 2 or 0. Multiply by d_n^{8-i} and apply Lemma 3.3. The statement on $Z_{i+4} = (i)_4 2^{i+4} c_i$ follows. $\square$

Numerically the Z_s are one power of d_n better: $d_n^4 Z_7, d_n^2 Z_9$ and Z_{11} are integers for all $n \leq 40$ and $n \in \{60, 100, 150, 200\}$. We do not need this, because ρ_0 dominates the denominators.

Lemma 3.5 (Root trick; compare [15, Lemma 2], [20, Lemma 4.6], [23, Lemma 5.1]). $d_n^{12} \rho_0 \in \mathbb{Z}$.

Proof. By (3.6), $\rho_0 = -\sum_{k=1}^n \sum_{\ell_0=0}^{k-1} X(k, \ell_0)$ with

$$X(k, \ell_0) := \sum_{i=1}^8 \frac{(i)_4 r_{i,k}}{(\ell_0 + \frac{1}{2})^{i+4}},$$

and it suffices to show that $d_n^{12} X(k_0, \ell_0) \in \mathbb{Z}$ for $1 \leq k_0 \leq n$ and $0 \leq \ell_0 < k_0$. At the prime 2 this follows from Lemma 3.4, since $v_2((\ell_0 + \frac{1}{2})^{-i-4}) = i+4 \geq 0$ and $12 \geq 8-i$. Let q be an odd prime, and suppose that $v_q(d_n^{12} X(k_0, \ell_0)) < 0$. Some summand then has negative q -adic valuation after multiplication by d_n^{12} . By Lemma 3.4 that valuation is at least $(i+4)(v_q(d_n) - v_q(2\ell_0 + 1))$, so $v_q(2\ell_0 + 1) > v_q(d_n)$.

Put $t_0 := -k_0 + \ell_0 + \frac{1}{2}$. Since $t_0 + \frac{1}{2} + (k_0 - \ell_0 - 1) = 0$ and $0 \leq k_0 - \ell_0 - 1 \leq n-1$, the point t_0 is a zero of $(t + \frac{1}{2})_n$, hence a zero of R_n of order at least 8. In particular

$$0 = R_n^{(4)}(t_0) = \sum_{i=1}^8 \sum_{k=0}^n \frac{(i)_4 r_{i,k}}{(t_0 + k)^{i+4}} = X(k_0, \ell_0) + \sum_{k \neq k_0} \sum_{i=1}^8 \frac{(i)_4 r_{i,k}}{(\ell_0 - k_0 + k + \frac{1}{2})^{i+4}}.$$

So some $k_1 \neq k_0$ gives a summand of negative q -adic valuation after multiplication by d_n^{12} , and as before $v_q(2(\ell_0 - k_0 + k_1) + 1) > v_q(d_n)$. Subtracting, $q^{v_q(d_n)+1}$ divides $2(k_1 - k_0)$, hence $k_1 - k_0$. This is impossible, because $0 < |k_1 - k_0| \leq n < q^{v_q(d_n)+1}$. $\square$

In particular, no prime $q \in (n, 2n)$ divides the denominator of ρ_0 . For primes $p \leq n$ with $p^2 > 2n$ we have $v_p(d_n) = 1$, and Lemma 3.5 gives only $v_p(\rho_0) \geq -12$. The saving of one power at these primes rests on Andrews' transformation, which we state in the form proved in Lean.

Theorem 3.6 (Andrews [2, Theorem 4], $q \rightarrow 1$; [18, Théorème 8]). *Let $m, N \geq 0$ be integers, K a field of characteristic 0, and $a, b_1, c_1, \dots, b_{m+1}, c_{m+1} \in K$ with $a \neq 0$. Assume that $(1+a-b_k)_N$ and $(1+a-c_k)_N$ for $1 \leq k \leq m+1$, $(1+a+N)_N$ and $(b_{m+1}+c_{m+1}-a-N)_N$ are non-zero. Then*

$$\begin{aligned} & \sum_{\kappa=0}^N \frac{a+2\kappa}{a} \frac{(a)_\kappa}{\kappa!} \frac{(-N)_\kappa}{(1+a+N)_\kappa} \prod_{k=1}^{m+1} \frac{(b_k)_\kappa (c_k)_\kappa}{(1+a-b_k)_\kappa (1+a-c_k)_\kappa} \\ &= \frac{(1+a)_N (1+a-b_{m+1}-c_{m+1})_N}{(1+a-b_{m+1})_N (1+a-c_{m+1})_N} \sum_{0 \leq i_1 \leq \dots \leq i_m \leq N} \frac{(-N)_{i_m}}{(b_{m+1}+c_{m+1}-a-N)_{i_m}} \\ & \quad \times \prod_{k=1}^m \frac{(1+a-b_k-c_k)_{i_k-i_{k-1}} (b_{k+1})_{i_k} (c_{k+1})_{i_k}}{(i_k-i_{k-1})! (1+a-b_k)_{i_k} (1+a-c_k)_{i_k}}, \end{aligned}$$

where $i_0 = 0$; for $m = 0$ the (empty) multiple sum is interpreted as 1.

The left-hand side is the terminating very-well-poised series

$${}_{2m+5}F_{2m+4} \left[\begin{matrix} a, \frac{a}{2} + 1, b_1, c_1, \dots, b_{m+1}, c_{m+1}, -N \\ \frac{a}{2}, 1+a-b_1, 1+a-c_1, \dots, 1+a-c_{m+1}, 1+a+N \end{matrix}; 1 \right]$$

of [18, (6.1)], written with $(\frac{a}{2}+1)_\kappa / (\frac{a}{2})_\kappa = (a+2\kappa)/a$. For fixed m and N , (6.1) is an identity of rational functions of the parameters. Since $(x)_\kappa$ divides $(x)_N$ for $\kappa \leq N$, the hypotheses make every denominator on both sides non-zero, so the identity holds at every such point. The Lean development contains a complete proof of this statement, via the $q = 1$ Bailey chain and a polynomial Pfaff–Saalschütz identity.

Proposition 3.7 (One-power saving). *Let $p \geq 11$ be a prime with $p^2 > 2n$. Then $v_p(\rho_0) \geq -11$.*

This is the analogue for $(a, j) = (8, 3)$ of [23, Lemma 5.3], and we follow its proof, with Andrews' transformation of depth $m = 8$ in place of depth 4.

Proof. Step 1 (a residue formula). For $x \neq 0$ we have, in $\mathbb{Q}[[\varepsilon]]$,

$$24(x - \varepsilon)^{-5} = \sum_{i \geq 1} (i)_4 x^{-(i+4)} \varepsilon^{i-1}.$$

The Laurent expansion of $R_n(-k + \varepsilon)$ is $\sum_{i=1}^8 r_{i,k} \varepsilon^{-i}$ plus a power series in ε . Comparing the coefficients of ε^{-1} gives

$$\sum_{i=1}^8 \frac{(i)_4 r_{i,k}}{x^{i+4}} = 24[\varepsilon^7] \left(G_k(\varepsilon) (x - \varepsilon)^{-5} \right).$$

Take $x = \ell - \frac{1}{2}$ with $1 \leq \ell \leq k$. Since $(\frac{1}{2} - \varepsilon)_k = (\frac{1}{2} - \varepsilon)_{\ell-1} (\ell - \frac{1}{2} - \varepsilon) (\ell + \frac{1}{2} - \varepsilon)_{k-\ell}$, the factor $(\ell - \frac{1}{2} - \varepsilon)^{-5}$ cancels five of the eight factors $\ell - \frac{1}{2} - \varepsilon$ of $\Psi_k(\varepsilon)^8$. Summing over $1 \leq \ell \leq k \leq n$ and using $A_k^{(s)} = \sum_{\ell=1}^k (\ell - \frac{1}{2})^{-s}$, we get

$$(3.7) \quad \rho_0 = -24 \sum_{\ell=1}^n [\varepsilon^7] T_{n,\ell}(\varepsilon),$$

$$T_{n,\ell}(\varepsilon) := 2^{16n} (\ell - \frac{1}{2} - \varepsilon)^3 (\frac{1}{2} - \varepsilon)_{\ell-1}^8 \sum_{k=\ell}^n (n - 2k + 2\varepsilon) \frac{(\ell + \frac{1}{2} - \varepsilon)_{k-\ell}^8 (\frac{1}{2} + \varepsilon)_{n-k}^8}{(1 - \varepsilon)_k^8 (1 + \varepsilon)_{n-k}^8}.$$

Step 2 (very-well-poised form). Fix $1 \leq \ell \leq n$, put $N := n - \ell$, and work in the field $K = \mathbb{Q}((\varepsilon))$. Let

$$a := 2\ell - n - 2\varepsilon, \quad b := -N - \varepsilon, \quad c := \ell + \frac{1}{2} - \varepsilon,$$

so that $1 + a - b = \ell + 1 - \varepsilon$, $1 + a - c = \frac{1}{2} - N - \varepsilon$, $1 + a - b - c = \frac{1}{2}$, $a + N = \ell - 2\varepsilon$, and $n - 2(\ell + \kappa) + 2\varepsilon = -(a + 2\kappa)$. Write $k = \ell + \kappa$ with $0 \leq \kappa \leq N$. Using $(x)_{N-\kappa} = (-1)^\kappa (x)_N / (1 - x - N)_\kappa$ for $x = \frac{1}{2} + \varepsilon$ and $x = 1 + \varepsilon$, and $(1 - \varepsilon)_{\ell+\kappa} = (1 - \varepsilon)_\ell (1 + a - b)_\kappa$, one finds

$$(3.8) \quad T_{n,\ell}(\varepsilon) = -a Q_\ell(\varepsilon) \sum_{\kappa=0}^N \frac{a + 2\kappa}{a} \left(\frac{(b)_\kappa (c)_\kappa}{(1 + a - b)_\kappa (1 + a - c)_\kappa} \right)^8,$$

where

$$Q_\ell(\varepsilon) := 2^{16n} \frac{(\ell - \frac{1}{2} - \varepsilon)^3 (\frac{1}{2} - \varepsilon)_{\ell-1}^8 (\frac{1}{2} + \varepsilon)_N^8}{(1 - \varepsilon)_\ell^8 (1 + \varepsilon)_N^8}.$$

Step 3 (Andrews' transformation). Apply Theorem 3.6 in K with $m = 8$, the above N and a , $(b_k, c_k) = (b, c)$ for $1 \leq k \leq 8$, and $(b_9, c_9) = (1, 1 + a + N) = (1, \ell + 1 - 2\varepsilon)$. The hypotheses hold: $a \neq 0$, and $(\ell + 1 - \varepsilon)_N$, $(\frac{1}{2} - N - \varepsilon)_N$, $(a)_N$, $(-N)_N = (-1)^N N!$, $(\ell + 1 - 2\varepsilon)_N$ and $(b_9 + c_9 - a - N)_N = (2)_N$ are non-zero in K . The factor of index $k = 9$ on the left-hand side is $(1)_\kappa (1 + a + N)_\kappa / ((a)_\kappa (-N)_\kappa)$; it cancels $(a)_\kappa / \kappa!$ and $(-N)_\kappa / (1 + a + N)_\kappa$, so the left-hand side is exactly the sum over κ in (3.8). On the right-hand side the prefactor is $(1 + a)_N (-N - 1)_N / ((a)_N (-N)_N) = (N + 1)(a + N)/a$, and $(-N)_{i_8} (1)_{i_8} / (2)_{i_8} = (-N)_{i_8} / (i_8 + 1)$. With $d_k := i_k - i_{k-1}$ we obtain

$$(3.9) \quad T_{n,\ell}(\varepsilon) = \sum_{0 \leq i_1 \leq \dots \leq i_8 \leq N} F_i(\varepsilon),$$

$$F_{\mathbf{i}}(\varepsilon) := -(N+1)(\ell-2\varepsilon) Q_{\ell}(\varepsilon) \prod_{k=1}^8 \frac{(\frac{1}{2})_{d_k}}{d_k!} \prod_{k=1}^7 \frac{(-N-\varepsilon)_{i_k} (\ell + \frac{1}{2} - \varepsilon)_{i_k}}{(\ell+1-\varepsilon)_{i_k} (\frac{1}{2} - N - \varepsilon)_{i_k}} \\ \times \frac{(\ell+1-2\varepsilon)_{i_8} (-N)_{i_8}}{(i_8+1)(\ell+1-\varepsilon)_{i_8} (\frac{1}{2} - N - \varepsilon)_{i_8}}.$$

Every $F_{\mathbf{i}}$ lies in $\mathbb{Q}[[\varepsilon]]$. (In [23, proof of Lemma 5.3] the corresponding last parameter, there $c_5(\delta) = \ell+1-2\varepsilon-\delta$, is sent to its final value by $\delta \rightarrow 0$; here this is not necessary, since all denominators in Theorem 3.6 are non-zero at the chosen point.)

Step 4 (the value at $\varepsilon = 0$). We use $(\frac{1}{2})_j/j! = 4^{-j} \binom{2j}{j}$, $(\frac{1}{2})_{\ell-1}/\ell! = 4^{1-\ell} \binom{2\ell-2}{\ell-1}/\ell$, $\binom{2\ell}{\ell} = 2(2\ell-1) \binom{2\ell-2}{\ell-1}/\ell$, and, for $0 \leq J \leq N$,

$$\frac{(-N)_J (\ell + \frac{1}{2})_J}{(\ell+1)_J (\frac{1}{2} - N)_J} = \frac{\binom{2(\ell+J)}{\ell+J} \binom{2(N-J)}{N-J}}{\binom{2\ell}{\ell} \binom{2N}{N}}, \quad \frac{(-N)_J}{(\frac{1}{2} - N)_J} = \frac{4^J \binom{2(N-J)}{N-J}}{\binom{2N}{N}}.$$

All powers of 2 and all factors $\binom{2N}{N}$ cancel, and
(3.10)

$$F_{\mathbf{i}}(0) = -\frac{64 \binom{2\ell-2}{\ell-1}}{(2\ell-1)^4} \cdot \frac{N+1}{i_8+1} \prod_{k=1}^8 \binom{2d_k}{d_k} \prod_{k=1}^7 \binom{2(\ell+i_k)}{\ell+i_k} \prod_{k=1}^8 \binom{2(N-i_k)}{N-i_k} \neq 0.$$

Step 5 (valuation at $\varepsilon = 0$). Every non-zero integer of absolute value at most $2n$ has $v_p \leq 1$, because $p^2 > 2n$. So (3.10) gives $v_p(F_{\mathbf{i}}(0)) \geq -4v_p(2\ell-1) - v_p(i_8+1) \geq -5$. We claim that $v_p(F_{\mathbf{i}}(0)) \geq -4$. This is clear unless $p \mid 2\ell-1$ and $p \mid i_8+1$, that is, $\ell \equiv \frac{p+1}{2}$ and $i_8 \equiv -1 \pmod{p}$. Assume this, let $x \in [0, p-1]$ be the residue of i_1 modulo p , and recall that $p \mid \binom{2A}{A}$ whenever the residue of A modulo p is at least $\frac{p+1}{2}$ (§2.1).

- If $x \geq \frac{p+1}{2}$, then $p \mid \binom{2i_1}{i_1} = \binom{2d_1}{d_1}$.
- If $x \leq \frac{p-3}{2}$, then $\ell+i_1$ has residue $\frac{p+1}{2} + x \in [\frac{p+1}{2}, p-1]$, so $p \mid \binom{2(\ell+i_1)}{\ell+i_1}$.
- If $x = \frac{p-1}{2}$, then $N-i_1 = n-\ell-i_1 \equiv n \pmod{p}$. If the residue of n is at least $\frac{p+1}{2}$, then $p \mid \binom{2(N-i_1)}{N-i_1}$. If it equals $\frac{p-1}{2}$, then $p \mid N+1 = n-\ell+1$. If it is at most $\frac{p-3}{2}$, then $N-i_8 \equiv n + \frac{p+1}{2} \pmod{p}$ has residue in $[\frac{p+1}{2}, p-1]$, so $p \mid \binom{2(N-i_8)}{N-i_8}$.

In each case a factor of the numerator of (3.10) is divisible by p , which proves the claim.

Step 6 (derivatives). Call $H \in \mathbb{Q}[[\varepsilon]]$ p -tame if $v_p([\varepsilon^j]H) \geq -j$ for all $j \geq 0$. Products of p -tame series are p -tame, and $(1+u\varepsilon)^e$ is p -tame for all $u \in \mathbb{Q}$ with $v_p(u) \geq -1$ and all $e \in \mathbb{Z}$ (for $e < 0$ expand the geometric series). Every factor of $F_{\mathbf{i}}(\varepsilon)$ in Step 3 that depends on ε is a power of a linear polynomial $c' + d'\varepsilon$ with $d' \in \{\pm 1, \pm 2\}$, $2c' \in \mathbb{Z} \setminus \{0\}$ and $|2c'| \leq 2n$. Indeed, these factors are $\ell - \frac{1}{2} - \varepsilon$, $\ell - 2\varepsilon$, factors $r + \frac{1}{2} \pm \varepsilon$, $r+1 \pm \varepsilon$ and $r+1-2\varepsilon$ with $0 \leq r \leq n-1$, and factors $r - N - \varepsilon$ and $r + \frac{1}{2} - N - \varepsilon$ with $0 \leq r \leq N-1$. Writing $c' + d'\varepsilon = c'(1 + (d'/c')\varepsilon)$ with $v_p(d'/c') = -v_p(2c') \geq -1$, we see that $F_{\mathbf{i}}(\varepsilon)/F_{\mathbf{i}}(0)$ is p -tame. Hence $v_p([\varepsilon^7]F_{\mathbf{i}}) \geq v_p(F_{\mathbf{i}}(0)) - 7 \geq -11$ for every chain $\mathbf{i}$. By (3.9), $v_p([\varepsilon^7]T_{n,\ell}) \geq -11$ for every ℓ , and by (3.7), $v_p(\rho_0) \geq -11$ because $v_p(24) = 0$. $\square$

Step 6 replaces the induction argument of [40, Lemma 17] that is used in [23]. The steps were also checked in exact arithmetic: (3.7) for $n \in \{1, \dots, 7, 9\}$; (3.9) at random rational ε for $3 \leq n \leq 8$ and all ℓ ; (3.10) for all chains and all ℓ ,

$n \in \{3, 5, 6, 7\}$; the bound $v_p(F_i(0)) \geq -4$ on 200000 random chains biased towards the critical case, $n \leq 400$ (the value -4 occurs); and $v_p([\varepsilon^7]T_{n,\ell}) \geq -11$ for $n \in \{18, 22, 26, 30\}$, all ℓ and all admissible p (the value -11 occurs).

Corollary 3.8 (Common denominator). *Put*

$$\Phi_n := \prod_{\substack{p \text{ prime} \\ \max(\sqrt{2n}, 10) < p \leq n}} p, \quad D_n := \frac{d_n^{12}}{\Phi_n}.$$

Then D_n is an integer, and $D_n \rho_0, D_n Z_7, D_n Z_9, D_n Z_{11} \in \mathbb{Z}$. Moreover $D_n = d_n^{11} E_n$ for an integer E_n with $1 \leq E_n \leq n^{\sqrt{2n}+10}$ ($n \geq 1$), so that

$$\log D_n = 11 \psi(n) + O(\sqrt{n} \log n) = 11n + o(n).$$

If $n = 2^m - 1$, then $v_2(D_n) = 12(m - 1)$.

Proof. Φ_n is a product of distinct primes $p \leq n$, so $\Phi_n \mid d_n$. Let $p \mid \Phi_n$. Then $p \leq n < p^2$, so $v_p(d_n) = 1$ and $v_p(D_n) = 11$; hence $v_p(D_n \rho_0) \geq 0$ by Proposition 3.7, and $v_p(D_n Z_{i+4}) \geq 11 - (8 - i) > 0$ by Lemma 3.4. For every other prime p , $v_p(D_n) = 12 v_p(d_n)$, and Lemmas 3.4 and 3.5 apply. Next, $D_n = d_n^{11} \prod p^{v_p(d_n)}$, the product over the primes $p \leq n$ with $p \nmid \Phi_n$. These satisfy $p \leq \max(\sqrt{2n}, 10)$, so there are at most $\sqrt{2n} + 10$ of them, and $p^{v_p(d_n)} \leq n$. The asymptotic formula follows from $\log d_n = \psi(n)$ and (2.4); only $\psi(n) \leq (1 + o(1))n$ is needed. Finally $2 \nmid \Phi_n$, and $v_2(d_n) = m - 1$ for $n = 2^m - 1$. $\square$

Without Proposition 3.7 one would only have the denominator $d_n^{12} = e^{12n+o(n)}$, and the final exponent would be $12 - 16 \log 2 = +0.91 > 0$.

Remark 3.9 (Denominators in exact data). For all $n \leq 300$ and $n \in \{383, 447, 511\}$ the odd part of the exact common denominator of ρ_0, Z_7, Z_9, Z_{11} divides d_n^{11} , and ρ_0 really needs d_n^{11} : the least e such that d_n^e is divisible by the odd part of the denominator of ρ_0 is $e = 11$ for 298 of these 303 values of n , the exceptions being $n \in \{1, 2, 3, 4, 6\}$. For primes $p \in (\sqrt{2n}, n]$ the exact exponent of p in the denominator of ρ_0 is 11 for all but 0 to 2 sporadic primes per n (for example $p = 41$ and $p = 233$ for $n = 511$, with exponent 10). The resulting extra saving is about $0.02n$ for the tested $n \leq 511$ and shows no pattern. Unlike Lai's construction [20, Lemma 4.3], the present family seems to admit no prime-selective saving depending on the fractional parts $\{n/p\}$, so the exponent $11 - 16 \log 2$ cannot be improved through the denominators alone. These observations are numerical and are not used.

3.4. The size of the coefficients.

Lemma 3.10. *For all $n \geq 0$, $1 \leq i \leq 8$ and $0 \leq k \leq n$,*

$$|r_{i,k}| \leq 4^{8-i} (n + \tfrac{1}{2}) 2^{16n},$$

and

$$\max(|\rho_0|, |Z_7|, |Z_9|, |Z_{11}|) \leq 3.4 \cdot 10^8 (n + 1)^2 2^{16n}.$$

Proof. The poles of $R_n(-k + \varepsilon)$ other than $\varepsilon = 0$ lie in $\mathbb{Z} \setminus \{0\}$, so G_k is holomorphic for $|\varepsilon| < 1$. On $|\varepsilon| = \frac{1}{4}$, each factor of $|(\frac{1}{2} - \varepsilon)_k / (1 - \varepsilon)_k| = \prod_{r < k} |r + \frac{1}{2} - \varepsilon| / |r + 1 - \varepsilon|$ is at most $(r + \frac{3}{4}) / (r + \frac{3}{4}) = 1$; the same holds for $(\frac{1}{2} + \varepsilon)_{n-k} / (1 + \varepsilon)_{n-k}$; and $|n - 2k + 2\varepsilon| \leq n + \frac{1}{2}$. Hence $|G_k| \leq (n + \frac{1}{2}) 2^{16n}$ on $|\varepsilon| = \frac{1}{4}$, and Cauchy's estimate gives $|r_{8-\mu,k}| = |[\varepsilon^\mu] G_k| \leq 4^\mu (n + \frac{1}{2}) 2^{16n}$.

Next, $|c_i| \leq (n+1) \max_k |r_{i,k}|$, and for $s \geq 5$, $0 < A_k^{(s)} < \sum_{\ell \geq 0} (\ell + \frac{1}{2})^{-s} = (2^s - 1)\zeta(s) \leq 1.037 \cdot 2^s$. Therefore

$$|Z_{i+4}| \leq (i)_4 2^{20-i} (n+1)(n+\frac{1}{2}) 2^{16n}, \quad |\rho_0| \leq 1.037 \sum_{i=1}^8 (i)_4 2^{20-i} \cdot n(n+\frac{1}{2}) 2^{16n}.$$

Finally $\sum_{i=1}^8 (i)_4 2^{20-i} = 324599808$, and $1.037 \cdot 324599808 < 3.4 \cdot 10^8$. $\square$

Numerically the bound for $r_{i,k}$ is far from sharp: $\max_{i,k} |r_{i,k}| / (4^{8-i}(n+1)2^{16n}) \leq 0.004883$ for $n \leq 40$ and $n \in \{60, 100, 150, 200\}$. The growth rate appears sharp numerically: the exact values of $\frac{1}{n} \log \max(|\rho_0|, |Z_7|, |Z_9|, |Z_{11}|)$ are about 11.082 for $250 \leq n \leq 511$, against $16 \log 2 = 11.0904$ (Table 5).

3.5. The exact 2-adic valuation along $n = 2^m - 1$. We show that $S_n \neq 0$ along $n = 2^m - 1$ by computing $v_2(S_n)$ exactly. As in [20, §6], one expands the derivative by the Leibniz rule and finds a single term whose 2-adic valuation is strictly smaller than that of all the others. Lai's analysis rests on the following lemma.

Lemma 3.11 ([20, Lemma 6.1]). *Let $m \geq 2$, $n = 2^m - 1$ and $k_0 = 2^{m-1}$. For $1 \leq l \leq n$ put $W_l := v_2((l-1)!(n-l)!)$. Then $W_{k_0} = n+1-2m$ and $W_l \geq n+2-2m$ for $l \neq k_0$. Moreover $v_2(n!) = n - m$.*

Proof. By Legendre's formula, $W_l = n - 1 - s_2(l-1) - s_2(n-l)$ and $v_2(n!) = n - s_2(n) = n - m$. Both $l-1$ and $n-l$ lie in $[0, 2^m - 2]$, so each has at most $m-1$ binary digits equal to 1. If both have exactly $m-1$ such digits, then $\min(l-1, n-l) \leq 2^{m-1} - 1$ forces $\min(l-1, n-l) = 2^{m-1} - 1$, i.e. $l = k_0$. Hence $s_2(l-1) + s_2(n-l) \leq 2m - 2$, with equality exactly for $l = k_0$, which gives both claims. $\square$

In our construction a new difficulty appears. At the half-integer shift, the linear factor $2t + n$ of R_n becomes $2t + 1 + n = 2(t + 2^{m-1})$, which vanishes exactly at the point $t = -k_0$ where the dominant factor $(t + k_0)^8$ sits. Terms in which this linear factor is not differentiated are not small enough for the argument of [20]. They are handled by the following identity.

Lemma 3.12 (Key identity). *Let $m \geq 1$, $n = 2^m - 1$ and $k_0 = 2^{m-1}$. Then*

$$(t + k_0) \binom{t+n}{n} = 2^{m-1} \Lambda(t), \quad \Lambda(t) := \binom{t+n}{n} + 2 \binom{t+n}{n+1},$$

and $\Delta_\mu(\Lambda) \geq \Delta(\Lambda) \geq 1 - m$ for every $\mu \geq 0$.

Proof. Since $t \binom{t+n}{n} = (n+1) \binom{t+n}{n+1}$ and $n+1 = 2^m$, we get $(t + 2^{m-1}) \binom{t+n}{n} = 2^m \binom{t+n}{n+1} + 2^{m-1} \binom{t+n}{n}$. By Lemma 2.8 (iv), $\Delta(\binom{t+n}{n}) \geq -\lfloor \log_2 n \rfloor = 1 - m$ and $\Delta(\binom{t+n}{n+1}) \geq -\lfloor \log_2 2^m \rfloor = -m$. Conclude with Lemma 2.8 (i). $\square$

Proposition 3.13 (Exact valuation). *Let $m \geq 2$ and $n = 2^m - 1$. Then*

$$v_2(S_n) = 32n + 14 - 11m.$$

In particular $S_n \neq 0$.

Proof. Put $k_0 := 2^{m-1}$, $u(t) := t + k_0$, $P(t) := (t+1)_n = \prod_{l=1}^n (t+l)$, and $h(t) := \prod_{k=0}^n (2t+2k+1)^{-8}$. Since $(t + \frac{1}{2})_{n+1} = 2^{-n-1} \prod_{k=0}^n (2t+2k+1)$ and $2t+1+n = 2u(t)$, we have

$$R_n(t + \frac{1}{2}) = 2^{24n+8} f(t), \quad f := 2u P^8 h,$$

hence $S_n = -2^{24n+8} \int_{\mathbb{Z}_2} f'''(t) dt$, and we have to show that

$$v_2\left(\int_{\mathbb{Z}_2} f'''(t) dt\right) = 8n + 6 - 11m.$$

The function h . We have $h(t) = H(2t)$ with $H(x) = \prod_{k=0}^n (x+2k+1)^{-8} \in \mathbb{Z}_2[[x]]$. Hence $\mathcal{D}_\beta h = 2^\beta h_\beta$, where $h_\beta(t) := (\mathcal{D}_\beta H)(2t)$ is a power series in t with coefficients in $\mathbb{Z}_2$ tending to 0. So $h_\beta \in S^1(\mathbb{Z}_2, \mathbb{Z}_2)$ (§2.2), and $\Delta_\mu(h_\beta) \geq \Delta(h_\beta) \geq 0$ for all μ by Lemma 2.8 (iii); moreover $h_0(0) = h(0)$ is a 2-adic unit.

Leibniz expansion. Since $\mathcal{D}_\gamma(2u) = 2u^{1-\gamma}$ for $\gamma \in \{0, 1\}$ (and 0 for $\gamma \geq 2$), and $\mathcal{D}_\mu(t+l)^8 = \binom{8}{\mu}(t+l)^{8-\mu}$, we have

$$f''' = 3! \mathcal{D}_3 f = \sum_{\tau} 2 \cdot 3! u^{1-\gamma} 2^\beta h_\beta \prod_{l=1}^n \binom{8}{M(l)} (t+l)^{8-M(l)},$$

where $\tau = (\gamma, \beta, M)$ runs over $\gamma \in \{0, 1\}$, $\beta \geq 0$ and functions $M: \{1, \dots, n\} \rightarrow \mathbb{Z}_{\geq 0}$ with $\gamma + \beta + |M| = 3$, $|M| := \sum_l M(l)$. We view M as a multiset of size $|M|$. Write $P = n! \binom{t+n}{n}$ and $P/(t+l) = (l-1)!(n-l)! Y_l$ with $Y_l(t) := \binom{t+l-1}{l-1} \binom{t+n}{n-l}$. Then $\prod_l (t+l)^{8-M(l)} = P^{8-|M|} \prod_l (P/(t+l))^{M(l)}$, and the term indexed by τ equals $K_\tau g_\tau(t)$ with

$$K_\tau := 2 \cdot 3! \prod_l \binom{8}{M(l)} \cdot n!^{8-|M|} \prod_l ((l-1)!(n-l)!)^{M(l)},$$

$$g_\tau(t) := u(t)^{1-\gamma} 2^\beta h_\beta(t) \binom{t+n}{n}^{8-|M|} \prod_l Y_l(t)^{M(l)}.$$

Each g_τ is a product of functions in $S^1(\mathbb{Z}_2, \mathbb{Z}_2)$, and $\int f''' = \sum_{\tau} K_\tau \int g_\tau$ is a finite sum.

Bounds for the integrals. Every binomial polynomial occurring in g_τ has lower index at most n , so by Lemma 2.8 (iv) its Δ is at least $-\lfloor \log_2 n \rfloor = -(m-1)$ (constants 1 have $\Delta = 1$).

- If $\gamma = 1$, then this bound, $\Delta(h_\beta) \geq 0$ and Lemma 2.8 (i), (ii) give $\Delta(g_\tau) \geq \beta - (m-1)$, and by Lemma 2.7 (a), $v_2(\int g_\tau) \geq \beta - m$.
- If $\gamma = 0$, then $8 - |M| \geq 5$, and we pair u with one factor $\binom{t+n}{n}$: by Lemma 3.12, $g_\tau = 2^{m-1+\beta} h_\beta \Lambda \binom{t+n}{n}^{7-|M|} \prod_l Y_l^{M(l)}$ with $\Delta(\Lambda) \geq 1 - m$. Hence $\Delta(g_\tau) \geq (m-1+\beta) + (1-m) = \beta$ and $v_2(\int g_\tau) \geq \beta - 1$.

Bounds for the constants. We have $v_2(2 \cdot 3!) = 2$, $v_2\binom{8}{1} = 3$, $v_2\binom{8}{2} = 2$, $v_2\binom{8}{3} = 3$, and by Lemma 3.11, $v_2(n!) = n - m$ and $v_2((l-1)!(n-l)!) \geq n+1-2m + [l \neq k_0]$, where $[l \neq k_0]$ is 1 if $l \neq k_0$ and 0 otherwise. Let $\sigma(M) := \sum_l v_2\binom{8}{M(l)}$ and let $\nu(M) := \sum_{l \neq k_0} M(l)$ be the number of elements of the multiset M different from k_0 . Then

$$\begin{aligned} v_2(K_\tau) &\geq 2 + \sigma(M) + (8 - |M|)(n - m) + |M|(n + 1 - 2m) + \nu(M) \\ &= 8n - 8m + 2 + |M|(1 - m) + \sigma(M) + \nu(M). \end{aligned}$$

Combining this with the bounds for the integrals and $|M| = 3 - \gamma - \beta$, both cases $\gamma = 1$ and $\gamma = 0$ give

$$(3.11) \quad v_2\left(K_\tau \int_{\mathbb{Z}_2} g_\tau\right) \geq 8n + 4 - 11m + E(\tau), \quad E(\tau) := \sigma(M) + \nu(M) + \beta m.$$

All terms but one are small. Table 4 lists $E(\tau)$ for every shape of τ ; there l, l', l'' denote pairwise distinct elements of $\{1, \dots, n\}$, required to differ from k_0 in the rows with $(\gamma, \beta) = (1, 0)$. Since $m \geq 2$, we get $E(\tau) \geq 3$, and hence $v_2(K_\tau \int g_\tau) \geq 8n + 7 - 11m$, for every τ except $\tau_0 := (1, 0, \{k_0, k_0\})$. The minimum $E = 3$ is attained at $(0, 0, \{k_0, k_0, k_0\})$.

TABLE 4. The quantity $E(\tau) = \sigma(M) + \nu(M) + \beta m$ of (3.11) for the terms of the Leibniz expansion ($m \geq 2$).

(γ, β, M)	M	$\sigma(M)$	$\nu(M)$	βm	$E(\tau)$
$(1, 0, 2)$	$\{k_0, k_0\}$	2	0	0	2 (dominant term τ_0)
$(1, 0, 2)$	$\{k_0, l\}$	6	1	0	7
$(1, 0, 2)$	$\{l, l\}$	2	2	0	4
$(1, 0, 2)$	$\{l, l'\}$	6	2	0	8
$(1, 1, 1)$	$\{l\}$	3	≥ 0	m	≥ 5
$(1, 2, 0)$	$\emptyset$	0	0	$2m$	≥ 4
$(0, 0, 3)$	$\{l, l, l\}$	3	≥ 0	0	≥ 3
$(0, 0, 3)$	$\{l, l, l'\}$	5	≥ 0	0	≥ 5
$(0, 0, 3)$	$\{l, l', l''\}$	9	≥ 0	0	≥ 9
$(0, 1, 2)$	$\{l, l\}$ or $\{l, l'\}$	≥ 2	≥ 0	m	≥ 4
$(0, 2, 1)$	$\{l\}$	3	≥ 0	$2m$	≥ 7
$(0, 3, 0)$	$\emptyset$	0	0	$3m$	≥ 6

The dominant term. For τ_0 we have $K_{\tau_0} = 2 \cdot 3! \binom{8}{2} n!^6 ((k_0 - 1)!(n - k_0)!)^2 = 336 n!^6 ((k_0 - 1)!(n - k_0)!)^2$, so by Lemma 3.11

$$v_2(K_{\tau_0}) = 4 + 6(n - m) + 2(n + 1 - 2m) = 8n + 6 - 10m.$$

Since $n - k_0 = k_0 - 1$, we have $g_{\tau_0} = h \binom{t+n}{n}^6 Y_{k_0}^2$ with $Y_{k_0} = \binom{t+k_0-1}{k_0-1} \binom{t+n}{k_0-1}$. We claim $v_2(\int g_{\tau_0}) = -m$. By Lemma 2.8 (iv) with $\lfloor \log_2 n \rfloor = m - 1 < m$, we have $\Delta_m(\binom{t+n}{n}^2) \geq 1 - (m - 1) = 2 - m$. The two binomials in Y_{k_0} have lower index $k_0 - 1 = 2^{m-1} - 1$, so their Δ_m is at least $-(m - 2) = 2 - m$. Also $\Delta_m(h) \geq 0$. By Lemma 2.8 (ii), $\Delta_m(g_{\tau_0}) \geq 2 - m$, and Lemma 2.7 (b) gives

$$\int_{\mathbb{Z}_2} g_{\tau_0}(t) dt \equiv \frac{1}{2^m} \sum_{x=0}^{2^m-1} g_{\tau_0}(x) \pmod{2^{1-m}\mathbb{Z}_2}.$$

For $1 \leq x \leq 2^m - 1$, adding x and $n = 2^m - 1$ in base 2 produces a carry, so $\binom{x+n}{n}$ is even by Kummer's theorem, and $g_{\tau_0}(x) \equiv 0 \pmod{2^6}$. On the other hand, $g_{\tau_0}(0) = h(0) \binom{2^m-1}{2^m-1-1}^2$ is odd, because $h(0)$ is a unit and $\binom{2^m-1}{j}$ is odd for all $0 \leq j \leq 2^m - 1$. Hence the right-hand side is 2^{-m} times a 2-adic unit, so $v_2(\int g_{\tau_0}) = -m$, and

$$v_2\left(K_{\tau_0} \int_{\mathbb{Z}_2} g_{\tau_0}\right) = 8n + 6 - 11m.$$

Conclusion. All other terms have valuation at least $8n + 7 - 11m$, so $v_2(\int f''') = 8n + 6 - 11m$ and $v_2(S_n) = 24n + 8 + 8n + 6 - 11m = 32n + 14 - 11m$. $\square$

Remark 3.14 (Numerical confirmation). The formula $v_2(S_n) = 32n + 14 - 11m$ agrees with exact computations for $m = 2, \dots, 9$ (Table 5), for instance $v_2(S_{127}) = 4001$ and $v_2(S_{511}) = 16267$. These values were computed from Lemma 3.2 with the $\zeta_2(s)$ evaluated to 17000 bits. An independent evaluation of $\int f'''$ from Mahler expansions, which uses no ζ_2 -values, gives $v_2(\int f''') = 8, 29, 82, 199 = 8n + 6 - 11m$ for $m = 2, 3, 4, 5$; there the normalized dominant integral $\int g_{\tau_0}$ has valuation exactly $-m$, and the sum of all other terms has valuation exactly $8n + 7 - 11m$. An enumeration of all shapes of τ with the bounds (3.11) confirms $E(\tau) \geq 3$ for $\tau \neq \tau_0$ for $m = 2, \dots, 40$. A cruder version of the same argument, which we have checked but not written out, gives $v_2(S_n) \geq 32n + 5 - 12 \log_2(n+1)$ for all $n \geq 1$ (compare [23, Lemma 6.2]); this bound holds on all our data, but it is not needed, not proved here, and not formalized.

3.6. Proof of Theorem B. For $m \geq 2$ let $n = 2^m - 1$, and define integers $a_{m,j}$ by

$$(a_{m,0}, a_{m,1}, a_{m,2}, a_{m,3}) := D_n \cdot (\rho_0, Z_7, Z_9, Z_{11}).$$

By Lemma 3.2,

$$L_m := D_n S_n = a_{m,0} + a_{m,1} \zeta_2(7) + a_{m,2} \zeta_2(9) + a_{m,3} \zeta_2(11),$$

and the $a_{m,j}$ are indeed integers by Corollary 3.8. By Proposition 3.13, $L_m \neq 0$ and $|L_m|_2 \leq |S_n|_2 = 2^{-(32n+14-11m)}$. By Lemma 3.10 and Corollary 3.8,

$$\max_{0 \leq j \leq 3} |a_{m,j}| \leq D_n \cdot 3.4 \cdot 10^8 (n+1)^2 2^{16n} = \exp((11 + 16 \log 2) n + o(n)).$$

Since $11m = O(\log n)$, it follows that

$$\max_{0 \leq j \leq 3} |a_{m,j}| \cdot |L_m|_2 \leq \exp((11 - 16 \log 2) n + o(n)),$$

which tends to 0 as $m \rightarrow \infty$ because $11 - 16 \log 2 = -0.09035\dots < 0$. By Lemma 2.9, applied to $(\alpha_1, \alpha_2, \alpha_3) = (\zeta_2(7), \zeta_2(9), \zeta_2(11))$ and the sequence indexed by m , at least one of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$ is irrational. $\square$

3.7. Remarks.

Remark 3.15 (Exact data along $n = 2^m - 1$). Table 5 lists, for $m = 2, \dots, 9$, the exact value of $v_2(S_n)$, the exponential size of the coefficients, the size of the exact common denominator D_n^{ex} of ρ_0, Z_7, Z_9, Z_{11} (which divides D_n and is odd for these n), and the resulting exact exponent $\frac{1}{n} \log(D_n^{\text{ex}} \max(|\rho_0|, |Z_s|) \cdot |D_n^{\text{ex}} S_n|_2)$. All entries come from exact rational computations; the logarithms are rounded to five decimals. At these small n the last column still fluctuates considerably; compare the asymptotic exponent $11 - 16 \log 2 = -0.0904$ of the proved bound.

Remark 3.16 (The proved bound for finite n). Along $n = 2^m - 1$ the proof bounds $\log(\max_j |a_{m,j}| \cdot |L_m|_2)$ by

$$M(n) := \log(3.4 \cdot 10^8 (n+1)^2) + \log D_n + 16n \log 2 - (32n + 14 - 11m + 12(m-1)) \log 2,$$

where $12(m-1) = v_2(D_n)$. Table 6 gives $M(n)$, computed in double precision from the exact values of θ and ψ (sieve up to 2^{27}). $M(n)$ is negative for $m = 8$ and for all $11 \leq m \leq 27$, and positive for $m \leq 7$ and for $m = 9, 10$: for small n the lower-order

TABLE 5. Exact data for $n = 2^m - 1$. Here $\text{coef} = \max(|\rho_0|, |Z_7|, |Z_9|, |Z_{11}|)$.

m	n	$v_2(S_n)$	$\frac{1}{n} \log \text{coef}$	$\frac{1}{n} \log D_n^{\text{ex}}$	exact exponent
2	3	88	14.75457	2.92963	-2.64811
3	7	205	12.32848	6.53964	-1.43120
4	15	450	11.52161	7.46725	-1.80555
5	31	951	11.23008	10.12910	0.09521
6	63	1964	11.12564	10.03409	-0.44885
7	127	4001	11.09137	10.48211	-0.26338
8	255	8086	11.08268	10.58088	-0.31600
9	511	16267	11.08245	10.89994	-0.08302

terms and the excess of $\frac{1}{n} \log D_n$ over 11 outweigh the margin $0.0904n$. This is an illustration only: the proof uses (2.4), whose error term is not made explicit.

TABLE 6. The proved bound $M(n)$ of Remark 3.16 along $n = 2^m - 1$ (selected m). The limit of $M(n)/n$ is $11 - 16 \log 2 = -0.09035$.

m	n	$\frac{1}{n} \log D_n$	$M(n)$	$M(n)/n$
7	127	11.12799	27.89	0.21961
8	255	10.95517	-10.67	-0.04184
9	511	11.17080	65.60	0.12838
10	1023	11.09057	25.41	0.02484
11	2047	11.01566	-127.03	-0.06206
12	4095	11.06789	-65.40	-0.01597
13	8191	11.00758	-650.74	-0.07945
16	65535	10.99793	-6027.76	-0.09198
20	1048575	11.00088	-93785.85	-0.08944
24	16777215	11.00017	-1513054.08	-0.09019
27	134217727	11.00008	-12116603.04	-0.09028

Remark 3.17 (Other parameters). For even a and odd $j \leq a - 3$, the analogous forms $\pm \int_{\mathbb{Z}_2} R^{(j)}(t + \frac{1}{2}) dt$ with $R(t) = 2^{2an}(2t + n)(t + \frac{1}{2})_n^a / (t)_{n+1}^a$ are linear forms in 1 and $\zeta_2(j + 4), \zeta_2(j + 6), \dots, \zeta_2(j + a)$. The arguments of Lemmas 3.2, 3.4, 3.5, Proposition 3.7 (giving the denominator d_n^{a+j} after the saving) and Lemma 3.10 extend (we have not written out the details), and the exponent becomes $(a + j) - 2a \log 2$, provided the analogue of Proposition 3.13 holds. Numerically, for $m \leq 7$, $v_2(S_{2^m-1}) = 4an + c - (a + j)m$ with $(a, j, c) = (4, 1, 5), (8, 3, 14)$ and $(10, 3, 14)$; the case analysis of Proposition 3.13 has been carried out only for $(8, 3)$. Among these families only $(8, 3)$ gives a new result. For example, $(8, 1)$ gives $\{5, 7, 9\}$, which contains $\zeta_2(5)$, already known to be irrational; $(10, 3)$ gives Lai's set $\{7, 9, 11, 13\}$ again, with exponent $13 - 20 \log 2 = -0.863$; and $(6, 3)$ would give the pair $\{\zeta_2(7), \zeta_2(9)\}$ but has exponent $9 - 12 \log 2 = +0.68 > 0$. Theorem A reaches that pair with a different, shifted family.

Remark 3.18 (Limits of the construction). The conclusion of Theorem B is that the $\mathbb{Q}$ -vector space spanned by $1, \zeta_2(7), \zeta_2(9), \zeta_2(11)$ has dimension at least 2; the argument gives nothing more. "At least two of the three are irrational" would follow from dimension at least 3. For forms with $|L|_2 \approx e^{-\alpha n}$ and coefficients

of size $e^{\beta n}$, the p -adic Nesterenko-type criterion [36, Theorem 1.4] could give at most the lower bound α/β for the dimension, even granting its hypotheses (forms for all large n , and two-sided bounds for $|L_n|_p$, whereas here the exact valuation is known only along $n = 2^m - 1$). Here $\alpha = 32 \log 2$ and $\beta = 11 + 16 \log 2$, so $\alpha/\beta = 22.18/22.09 = 1.004$, whereas dimension 3 would need a ratio above 2. Nor is anything lost 2-adically: along $n = 2^m - 1$ the loss is $11m - 14 = O(\log n)$, and a common power of 2 in the coefficients would not help, since $\max_j |a_{m,j}| \cdot |L_m|_2$ does not change when the form is multiplied by a power of 2.

Remark 3.19 (Formal counterpart). The Lean development for Theorem B (repository `gmDevi/zeta2-7-9-11-lean` [13]; see also Section 5) mirrors this section: each result is a proposition `Stmt_X` in `Zeta2Lean/Statements.lean`, proved in its own file. The correspondence is as follows (all names in the namespace `Zeta2`).

this paper	Lean
Lemma 2.1, (2.2)	<code>Stmt_Translation</code> , <code>Stmt_JConv</code>
Lemma 3.2	<code>Stmt_PF</code> , <code>Stmt_CoeffVanish</code> , <code>Stmt_L1</code>
Lemmas 3.3, 3.4, 3.5	<code>Stmt_BlockF</code> , <code>Stmt_L2a</code> , <code>Stmt_L2b</code>
Theorem 3.6	<code>Andrews.Stmt</code> , proved as <code>Andrews.proof</code>
Proposition 3.7, Steps 1–6	<code>Stmt_ResidueForm</code> , <code>Stmt_AndrewsApplied</code> , <code>Stmt_FJClosed</code> , <code>Stmt_FJKummer</code> , <code>Stmt_L2c</code>
Corollary 3.8	<code>Stmt_L2cor</code>
Lemma 3.10	<code>Stmt_L4</code> , with unspecified constants in $C(n+1)^{A2^{16n}}$
Lemmas 2.7, 2.8	<code>Stmt_Delta</code> , <code>Stmt_DeltaFun</code>
Lemma 3.11	<code>Stmt_Digit</code>
Proposition 3.13	<code>Stmt_Leibniz</code> , <code>Stmt_LeibTermBound</code> , <code>Stmt_L5Dom</code> , <code>Stmt_L5</code>
§3.6	<code>Stmt_Asymptotic</code> , <code>PNT.Stmt</code> (proved as <code>PNT.proof</code>), <code>Stmt_Criterion</code>
Theorem 3.1	<code>zeta2_7_9_11_not_all_rational_unconditional</code>

Steps 3 and 6 of Proposition 3.7 are written above as they are formalized. The formal proof differs from the text in two inessential ways: the Δ -calculus and Proposition 3.13 are proved at the level of Riemann sums, so that no term of the Leibniz expansion needs its own convergence proof, and Lemma 3.10 is formalized with unspecified constants. The lower bound for $v_2(S_n)$ for all n in Remark 3.14 is not formalized.

4. PROOF OF THEOREM A

In this section we prove Theorem A: at least one of the two 2-adic numbers $\zeta_2(7)$, $\zeta_2(9)$ is irrational. As for Theorem B, we construct linear forms

$$S_n = \rho_0 + Z_7 \zeta_2(7) + Z_9 \zeta_2(9) \quad (\rho_0, Z_7, Z_9 \in \mathbb{Q})$$

as Volkenborn integrals of an explicit rational function R_n , and we conclude with an irrationality criterion. The function R_n is a deformation of the member $(a, j) = (6, 3)$ of the family of §1.3, whose member $(4, 1)$ is the construction of Lai, Sprang and Zudilin [23]: the six Pochhammer symbols $(t + \frac{1}{2})_n$ of the numerator are replaced by shifted, well-poised blocks $(t + \frac{1}{2} - h_m)_{n+2h_m}$, in the spirit of Rhin and Viola [32] (compare also the well-poised parameters of [40, §8]). For the undeformed member ($h = 0$) the count that proves Theorem B gives the exponent $9 - 12 \log 2 = +0.682 > 0$, so this approach does not give irrationality there. The

shifts lower the archimedean size of the coefficients and, prime by prime, their common denominator, while the 2-adic rate $12 \log 2$ is unchanged.

The price is that three of the four quantitative inputs need new arguments; the first is a routine extension of the literature.

- (i) *2-adic size* (§4.2). The Δ -calculus of Sprang and Lai is applied to binomial coefficients with shifted arguments (Lemma 2.8(iv) already allows arbitrary integer offsets); the balance $\sum_m h_m = 0$ keeps the rate $12 \log 2$.
- (ii) *Denominators* (§4.3). The building-block lemmas behind Theorem B do not apply to blocks with $h_m < 0$; indeed, in the computed range primes $q > \sqrt{n}$ occur in the common denominator with exponent $a+j+2 = 11$. We bound the q -adic valuation at each pole by counting the q -divisible linear factors of the local expansion. The zeros of order ≥ 5 of R_n at half-integers (*critical zeros*) keep every prime $q > n$ out of the denominator.
- (iii) *Archimedean size* (§4.4). The residues of R_n are exponentially larger than the coefficients: each of ρ_0, Z_7, Z_9 hides about 1.60 nats per n of cancellation between different poles. We write the coefficients as contour integrals against periodic kernels with poles at the half-integers (admissible because of the critical zeros) and bound them by a saddle point. Three inequalities for an explicit harmonic function are certified by interval arithmetic.
- (iv) *Nonvanishing* (§4.5). Lai's 2-adic dominant-term method, used for Theorem B, has no unique dominant term here. Instead, the auxiliary prime $\ell = n - 1$ separates ρ_0 from Z_7, Z_9 in the sense of Lemma 2.10. This is a variant of the auxiliary prime $\ell(n) = n + 1$ of Lai and Sprang [22, §3, Lemma 5.6]; what is new is the ℓ -adic computation for this family (two singular poles, the constant $\beta_1 = -168$, and the constant K_E).

Everything is proved for one fixed shift vector (configuration E, (4.1)) and combined in §4.6 along one fixed infinite set $\mathcal{N}$ of indices n . Table 7 summarizes the ingredients. Along $\mathcal{N}$, the quantity controlled by Lemma 2.9 decays like $e^{-0.189n+o(n)}$ (Table 12).

Status of the arguments. Computer assistance enters at two places: the numerical value of the savings constant $R^{(0)}$ of Theorem 4.17 (an exact rational computation plus a periodic tail, of which only a crude bound is needed), and the three landscape inequalities of Lemma 4.28, which are certified by interval arithmetic in three independent ways. In the Lean formalization both are replaced by weaker versions whose certificates are checked by the Lean kernel: the formal proof uses the rates $\delta = 9$ for the denominators and $g = -0.72$ for the growth, in place of $10 - R^{(0)} = 8.9099$ and -0.7812 . The informal arguments of this section were produced and checked by AI agents (adversarial verification, and exact computations at many values of n); they have not been refereed by a human expert.

4.1. The construction and the linear forms. We use the notation of Section 2. Throughout this section n is a positive multiple of 40, and

$$(4.1) \quad \begin{aligned} h &= (h_1, \dots, h_6) := \frac{n}{40} (-17, 1, 2, 3, 5, 6) \in \mathbb{Z}^6, \\ \eta &:= \frac{h}{n} = (-0.425, 0.025, 0.05, 0.075, 0.125, 0.15). \end{aligned}$$

TABLE 7. The ingredients of the proof of Theorem A. The entries marked * are computer-assisted (Proposition 4.18 and Lemma 4.28); the others are proved by hand. The last column gives the name X of the corresponding Lean statement `Stmt_X` (namespace `Zeta2.Pair`; the nonvanishing statement is proved through `Stmt_LaiSprangCond`), with the weaker constant where the formal version uses one.

Ingredient	Result (configuration E, $40 \mid n$)	Lean
Linear form (Lem. 4.2)	$S_n = \rho_0 + Z_7\zeta_2(7) + Z_9\zeta_2(9)$	L1
2-adic size (Cor. 4.5)	$v_2(S_n) \geq 12n - 1 - 10 \log_2(1.3n)$	Valuation
Denominators (Thm. 4.17)*	$\log D_n \leq (10 - R^{(0)})n + o(n)$, with $10 - R^{(0)} = 8.9098758$	Denominators , $\delta = 9$
Growth (Thm. 4.21)*	$\max(\rho_0 , Z_7 , Z_9) \leq$ $1.56 \cdot 10^{12} n^{-4} e^{-0.7812n}$ for $n \geq 2000$	Growth , $g = -0.72$
Nonvanishing (Thm. 4.32)	if $\ell = n - 1$ is prime and $\ell \neq 91079$ (i.e. $n \in \mathcal{N}$): $v_\ell(\rho_0) = -9$, $v_\ell(Z_7) \geq -3$, $v_\ell(Z_9) \geq -1$	Nonvanishing

We call this *configuration E*. Then $\sum_m h_m = 0$; the numbers $N_m := n + 2h_m$ are

$$(N_1, \dots, N_6) = n \cdot \left(\frac{3}{20}, \frac{21}{20}, \frac{11}{10}, \frac{23}{20}, \frac{5}{4}, \frac{13}{10} \right), \quad \sum_{m=1}^6 N_m = 6n;$$

the largest shift is $h_6 = 3n/20$, and $\max_m N_m = 13n/10$. Put $C_m := [-h_m, n + h_m - 1] \cap \mathbb{Z}$. Then $|C_m| = N_m$, $(t + \frac{1}{2} - h_m)_{N_m} = \prod_{c \in C_m} (t + \frac{1}{2} + c)$, and

$$(4.2) \quad C_1 = \left[\frac{17n}{40}, \frac{23n}{40} - 1 \right], \quad C_m \supseteq \left[-\frac{n}{40}, n - 1 + \frac{n}{40} \right] \quad (2 \leq m \leq 6).$$

We define

$$(4.3) \quad \begin{aligned} R_n(t) &:= (2t + n) \frac{\prod_{m=1}^6 (t + \frac{1}{2} - h_m)_{n+2h_m}}{(t)_{n+1}^6} \\ &= \frac{(2t + n) \prod_{m=1}^6 \prod_{c \in C_m} (t + \frac{1}{2} + c)}{\prod_{k=0}^n (t + k)^6}, \\ S_n &:= \int_{\mathbb{Z}_2} R_n'''(t + \frac{1}{2}) dt. \end{aligned}$$

Unlike the family used for Theorem B, R_n carries no power of 2 and S_n is defined without a minus sign. Neither matters: powers of 2 cancel exactly in §4.6, and only absolute values and valuations of S_n are used.

By Lemma 4.1(a) below, R_n has a partial-fraction expansion

$$(4.4) \quad R_n(t) = \sum_{k=0}^n R_{n,k}(t), \quad R_{n,k}(t) := \sum_{i=1}^6 \frac{r_{i,k}}{(t+k)^i} \quad (r_{i,k} \in \mathbb{Q}),$$

and we put $c_i := \sum_{k=0}^n r_{i,k}$. For $0 \leq k \leq n$ we also put $G_k(\varepsilon) := \varepsilon^6 R_n(-k + \varepsilon)$.

Lemma 4.1 (Basic properties). *The function R_n has the following properties.*

- (a) $\deg R_n = -5$. The poles of R_n are the integers $-k$, $0 \leq k \leq n$; they have order 6, except $-n/2$, which has order 5. Hence (4.4) holds without polynomial part, and $r_{6-\lambda,k} = [\varepsilon^\lambda]G_k(\varepsilon)$ for $0 \leq \lambda \leq 5$, where

$$(4.5) \quad G_k(\varepsilon) = (n - 2k + 2\varepsilon) \prod_{m=1}^6 \prod_{c \in C_m} (\varepsilon + c - k + \tfrac{1}{2}) \prod_{\substack{0 \leq j \leq n \\ j \neq k}} (\varepsilon + j - k)^{-6}.$$

For $k = n/2$ we write $G_{n/2} = 2\varepsilon H$; then $r_{6-\lambda,n/2} = 2[\varepsilon^{\lambda-1}]H$.

- (b) (Critical zeros.) For $c \in \mathbb{Z}$, R_n is holomorphic at $-\frac{1}{2} - c$ and vanishes there to the order $\mu(c) := \#\{m : c \in C_m\}$. We have $\mu(c) \geq 5$ for $-\frac{n}{40} \leq c \leq n - 1 + \frac{n}{40}$, that is,

$$\text{ord}_x R_n \geq 5 \quad \text{for every } x \in \mathbb{Z} + \tfrac{1}{2} \text{ with } -n - \tfrac{n}{40} < x < \tfrac{n}{40},$$

and $\mu(\frac{n}{2} - 1) = \mu(\frac{n}{2}) = 6$.

- (c) (Symmetry.) We have $R_n(-n - t) = -R_n(t)$. Consequently, for $0 \leq k \leq n$ and $1 \leq i \leq 6$,

$$(4.6) \quad \begin{aligned} R_{n,n-k}(t) &= -R_{n,k}(-n - t), \\ r_{i,n-k} &= (-1)^{i+1} r_{i,k}, \quad G_{n-k}(\varepsilon) = -G_k(-\varepsilon), \end{aligned}$$

and $c_i = 0$ for even i .

- (d) $c_1 = 0$.

Proof. (a) The degree is $1 + \sum_m N_m - 6(n+1) = -5$. The zeros of the numerator are half-integers, except the zero $-n/2$ of $2t + n$, which lowers the order of the pole at $-n/2$ to 5. As $\deg R_n < 0$, there is no polynomial part. Formula (4.5) is (4.3) at $t = -k + \varepsilon$, multiplied by ε^6 , and $G_k(\varepsilon) = \sum_i r_{i,k} \varepsilon^{6-i} + O(\varepsilon^6)$.

(b) A factor $t + \frac{1}{2} + c'$ vanishes only at $-\frac{1}{2} - c'$, and neither $2t + n$ (n is even) nor the denominator vanishes at a half-integer. By (4.2), every c in the stated range lies in $C_2, \dots, C_6$; and $\frac{n}{2} - 1, \frac{n}{2} \in C_1$ because $\frac{3n}{40} \geq 1$.

(c) The map $c \mapsto n - 1 - c$ maps each C_m onto itself, and $-n - t + \frac{1}{2} + c = -(t + \frac{1}{2} + (n - 1 - c))$; so the numerator blocks change by the sign $(-1)^{\sum_m N_m} = (-1)^{6n} = 1$. The denominator changes by $((-1)^{n+1})^6 = 1$, and $2(-n - t) + n = -(2t + n)$. Comparing principal parts at $-(n - k)$ gives (4.6), hence $r_{i,n-k} = (-1)^{i+1} r_{i,k}$, and summing over k gives $c_i = (-1)^{i+1} c_i$. Finally $G_{n-k}(\varepsilon) = \varepsilon^6 R_n(-n - (-k - \varepsilon)) = -G_k(-\varepsilon)$.

- (d) $c_1 = \lim_{t \rightarrow \infty} t R_n(t) = 0$, because $\deg R_n \leq -2$. $\square$

Lemma 4.2 (Linear form). *The rational function $R_n(t + \frac{1}{2})$ has no pole in $\mathbb{Z}_2$, so S_n is defined, and*

$$(4.7) \quad \begin{aligned} S_n &= \rho_0 + Z_7 \zeta_2(7) + Z_9 \zeta_2(9), \\ Z_7 &:= -(3)_4 2^7 c_3 = -46080 c_3, \quad Z_9 := -(5)_4 2^9 c_5 = -860160 c_5, \end{aligned}$$

where, with $A_k^{(s)}$ as in (2.3),

$$(4.8) \quad \begin{aligned} \rho_0 &:= \sum_{i=1}^6 \sum_{k=1}^n (i)_4 r_{i,k} A_k^{(i+4)} = \sum_{k=1}^n \sum_{l=0}^{k-1} X(k, l), \\ X(k, l) &:= \sum_{i=1}^6 \frac{(i)_4 r_{i,k}}{(l + \frac{1}{2})^{i+4}}. \end{aligned}$$

Moreover,

$$(4.9) \quad X(k, l) = R_{n,k}^{(4)}\left(l + \frac{1}{2} - k\right) = 24[\varepsilon^5] \frac{G_k(\varepsilon)}{(l + \frac{1}{2} - \varepsilon)^5}.$$

Proof. The poles $-\frac{1}{2} - k$ of $R_n(t + \frac{1}{2})$ satisfy $|\frac{1}{2} - k|_2 = 2$. By (4.4), $R_n'''(t) = -\sum_{i,k} (i)_3 r_{i,k}(t+k)^{-i-3}$, and every $(t+k+\frac{1}{2})^{-s}$ is integrable. By (2.3) and Corollary 2.4,

$$S_n = -\sum_{i=1}^6 (i)_3 c_i J_{i+3} + \sum_{i,k} (i)_4 r_{i,k} A_k^{(i+4)} = \rho_0 - \sum_{i=1}^6 (i)_4 2^{i+4} c_i \zeta_2(i+4),$$

where we used $(i)_3(i+3) = (i)_4$ and $A_0^{(s)} = 0$. The term $i = 1$ vanishes by Lemma 4.1(d), and the terms with even i by Lemma 4.1(c). This gives (4.7), with $(3)_4 2^7 = 360 \cdot 128$ and $(5)_4 2^9 = 1680 \cdot 512$. The first identity in (4.9) is $R_{n,k}^{(4)}(t) = \sum_i (i)_4 r_{i,k}(t+k)^{-i-4}$. The second follows from $r_{i,k} = [\varepsilon^{6-i}]G_k$ and $24(\xi - \varepsilon)^{-5} = \sum_{i \geq 1} (i)_4 \xi^{-i-4} \varepsilon^{i-1}$. $\square$

Remark 4.3. (1) The constants $46080 = 2^{10} \cdot 3^2 \cdot 5$, $860160 = 2^{13} \cdot 3 \cdot 5 \cdot 7$ and $(i)_4 \in \{24, 120, 360, 840, 1680, 3024\}$ have no prime factor larger than 7. (2) Identity (4.7) was also checked on exact data. Seven independent implementations of the partial-fraction expansion agree on ρ_0, c_3, c_5 ; and two evaluations of S_n directly from the definition of the Volkenborn integral (by Mahler series, and by Bernoulli numbers), which use no partial fractions, agree with the right-hand side of (4.7), for instance modulo $2^{v_2(S_n)+60}$ at $n = 40$ and $n = 80$. In every computed case $c_1 = c_2 = c_4 = c_6 = 0$. (3) The Lean files use $S_n := -\int_{\mathbb{Z}_2} R_n'''(t + \frac{1}{2}) dt$. Their ρ_0, Z_7, Z_9 are therefore the negatives of ours, and (4.7) appears there as $S_n = \rho_0 + 60 c_3 J_6 + 210 c_5 J_8$ (statement `Stmt_L1`; recall $J_6 = 768 \zeta_2(7)$ and $J_8 = 4096 \zeta_2(9)$).

4.2. The 2-adic size of S_n . We prove a bound that holds for shift vectors of either sign, and then specialize it to configuration E. The tools are the operators of §2.4 for $p = 2$.

Theorem 4.4. *Let $n \geq 0$, $a \geq 1$, $j \geq 0$ and $\iota \in \{0, 1\}$ be integers, and let $h \in \mathbb{Z}^a$ satisfy $\sum_m h_m = 0$ and $N_m := n + 2h_m \geq 0$ for all m (there is no sign condition on the h_m). Put*

$$R(t) := (2t + n)^\iota \frac{\prod_{m=1}^a (t + \frac{1}{2} - h_m)_{N_m}}{(t)_{n+1}^a}, \quad S := \int_{\mathbb{Z}_2} R^{(j)}(t + \frac{1}{2}) dt,$$

$\sigma := \sum_m s_2(N_m)$, $N^* := \max(1, N_1, \dots, N_a)$ and $\lambda := \lfloor \log_2 N^* \rfloor + 1$. Then

$$v_2(S) \geq 2an + a + j - \sigma - (j+1)\lambda.$$

Corollary 4.5. *Let $40 \mid n$ and let h be configuration E. Then*

$$(4.10) \quad v_2(S_n) \geq 12n + 9 - \sigma_n - 4\lambda_n \geq 12n + 9 - 10\lambda_n \geq 12n - 1 - 10\log_2(1.3n),$$

where $\sigma_n := \sum_m s_2(N_m)$ and $\lambda_n := \lfloor \log_2(13n/10) \rfloor + 1$.

Proof. Take $(a, j, \iota) = (6, 3, 1)$, so that $2an + a + j = 12n + 9$ and $N^* = 13n/10$. Then use $s_2(N_m) \leq \lambda_n$ and $\lambda_n \leq 1 + \log_2(1.3n)$. $\square$

The proof of Theorem 4.4 uses four lemmas.

Lemma 4.6. *We have $R(t + \frac{1}{2}) = 2^{a(n+1)} E(t) \prod_{m=1}^a N_m! \binom{t+n+h_m}{N_m}$, where $E(t) := (2t+1+n)^t \prod_{k=0}^n (2t+2k+1)^{-a}$.*

Proof. For $N = n+2h \geq 0$ and either sign of h we have $(t+1-h)_N = \prod_{l=1-h}^{n+h} (t+l) = N! \binom{t+n+h}{N}$; for $h \geq 1$ the block contains the harmless zeros $t = 0, \dots, h-1$. Moreover $(t + \frac{1}{2})_{n+1} = 2^{-n-1} \prod_{k=0}^n (2t+2k+1)$ and $2(t + \frac{1}{2}) + n = 2t+1+n$. $\square$

In particular $R^{(j)}(t + \frac{1}{2})$ has no pole in $\mathbb{Z}_2$, and S is defined.

Lemma 4.7. *For $\beta \geq 0$, the function $E_\beta := E^{(\beta)}/(\beta! 2^\beta)$ lies in $S^1(\mathbb{Z}_2, \mathbb{Z}_2)$ and satisfies $\Delta(E_\beta) \geq 0$.*

Proof. Apply the Leibniz rule for $\mathcal{D}_\beta$ to the product of the $\iota + a(n+1)$ factors $(2t+1+n)^\iota$ and $(2t+2k+1)^{-1}$ (each of the latter taken a times). Since $\mathcal{D}_b(2t+c)^{-1} = (-2)^b(2t+c)^{-1-b}$, $\mathcal{D}_1(2t+1+n) = 2$ and $\mathcal{D}_2(2t+1+n) = 0$, the function E_β is a $\mathbb{Z}$ -linear combination of products of functions $(2t+c)^{-e}$ (c odd, $e \geq 1$) and $(2t+1+n)^0$ or 1 . For $t \in \mathbb{Z}_2$ we have $(2t+c)^{-e} = \sum_{r \geq 0} \binom{-e}{r} 2^r c^{-e-r} t^r$, whose coefficients lie in $\mathbb{Z}_2$ and tend to 0. Conclude with Lemma 2.8(i)–(iii). $\square$

Lemma 4.8. *Let $x \in \mathbb{Z}$ and $N, \mu \geq 0$. Then*

$$\frac{d^\mu}{dt^\mu} \binom{t+x}{N} = \sum_{r=\mu}^N c_{\mu,r} \binom{t+x}{N-r}, \quad c_{\mu,r} := [z^r](\log(1+z))^\mu,$$

and $v_2(c_{\mu,r}) \geq -\mu \lfloor \log_2 r \rfloor$ for $\mu, r \geq 1$.

Proof. In $\mathbb{Q}[t][[z]]$ we have $\sum_{N \geq 0} \binom{t+x}{N} z^N = \exp((t+x) \log(1+z))$: every z -coefficient of either side is a polynomial in t , and both sides equal $(1+z)^{t+x}$ for all integers $t \geq -x$. Differentiating μ times in t multiplies the right-hand side by $(\log(1+z))^\mu$. Finally, $c_{\mu,r}$ is a sum of products $\prod_{k=1}^\mu (-1)^{i_k-1}/i_k$ with $1 \leq i_k \leq r$. $\square$

Lemma 4.9. *We have $R^{(j)}(t + \frac{1}{2}) = \sum \mathcal{K}(\beta, \mu, r) \Phi_{\beta,r}(t)$, summed over $\beta \geq 0$ and $\mu \in \mathbb{Z}_{\geq 0}^a$ with $\beta + \sum_m \mu_m = j$, and over $r \in \mathbb{Z}^a$ with $r_m = 0$ if $\mu_m = 0$ and $\mu_m \leq r_m \leq N_m$ otherwise; here*

$$\mathcal{K}(\beta, \mu, r) := 2^{a(n+1)} \frac{j!}{\prod_m \mu_m!} 2^\beta \prod_m N_m! \prod_m c_{\mu_m, r_m},$$

$$\Phi_{\beta,r}(t) := E_\beta(t) \prod_m \binom{t+n+h_m}{N_m - r_m}.$$

Moreover $v_2(\mathcal{K}(\beta, \mu, r)) \geq 2an + a + \beta - \sigma - (j - \beta)(\lambda - 1)$, and $\Phi_{\beta,r} \in S^1(\mathbb{Z}_2, \mathbb{Z}_2)$ with $\Delta(\Phi_{\beta,r}) \geq -(\lambda - 1)$.

Proof. The expansion is the Leibniz rule applied to Lemma 4.6, with $E^{(\beta)} = \beta! 2^\beta E_\beta$ and Lemma 4.8. In $\mathcal{K}$, the powers of 2 contribute $a(n+1) + \beta$; the number $j!/\prod_m \mu_m!$ is $\beta!$ times a multinomial coefficient, hence an integer; by Legendre's formula and $\sum_m h_m = 0$,

$$v_2\left(\prod_m N_m!\right) = \sum_m (N_m - s_2(N_m)) = an + 2 \sum_m h_m - \sigma = an - \sigma$$

(this is the only place where $\sum_m h_m = 0$ is used); and, by Lemma 4.8, for every m with $\mu_m \geq 1$ (so that $1 \leq r_m \leq N_m$),

$$v_2(c_{\mu_m, r_m}) \geq -\mu_m \lfloor \log_2 N_m \rfloor \geq -\mu_m(\lambda - 1),$$

while $c_{0,0} = 1$; here $\sum_m \mu_m = j - \beta$. In $\Phi_{\beta, r}$, the factors with $r_m = N_m$ equal 1. The other factors satisfy $\Delta \geq -\lfloor \log_2 N^* \rfloor = -(\lambda - 1)$ by Lemma 2.8(iv), which allows offsets of either sign. Conclude with Lemma 4.7 and Lemma 2.8(ii). $\square$

Proof of Theorem 4.4. The sum in Lemma 4.9 is finite, and each of its terms is a strictly differentiable function on $\mathbb{Z}_2$. By linearity,

$$S = \sum \mathcal{K}(\beta, \mu, r) \int_{\mathbb{Z}_2} \Phi_{\beta, r}(t) dt.$$

By Lemma 2.7(a), each integral has $v_2 \geq -\lambda$. Hence

$$\begin{aligned} v_2\left(\mathcal{K}(\beta, \mu, r) \int_{\mathbb{Z}_2} \Phi_{\beta, r}\right) &\geq 2an + a + \beta - \sigma - (j - \beta)(\lambda - 1) - \lambda \\ &= 2an + a + j - \sigma - (j + 1)\lambda + \beta\lambda, \end{aligned}$$

and the ultrametric inequality gives the claim. $\square$

Remark 4.10. (1) Only the odd part of a common denominator of ρ_0, Z_7, Z_9 matters later (§4.6), and Theorem A only needs $v_2(S_n) \geq 12n - O(\log n)$. (2) Table 8 compares (4.10) with the exact values. Two further exact values are $v_2(S_{520}) = 6199$ and $v_2(S_{600}) = 7159$, against the bounds 6184 and 7139; so $12n - v_2(S_n)$ lies between 23 and 41 for all sixteen multiples of 40 up to 640. Numerically $v_2(S_n) = 12n - \sigma_n + O(\log n)$, so the rate $12 \log 2$ cannot be improved for this family. (3) In Lean, the bound is proved in the uniform form $v_2(S) \geq 12n - 10 \lfloor \log_2 6n \rfloor$ for every *admissible* pair (n, h) , that is, $h \in \mathbb{Z}^6$ with $\sum_m h_m = 0$, $N_m = n + 2h_m \geq 0$ for all m , and $h_m \geq 0$ for at least five m (the Lean predicate `Admissible`); hence $\|S\|_2 2^{12n} \leq 6^{10}(n+1)^{10}$. The Lean statement `Stmt.Valuation` is the existential form “ $\|S\|_2 2^{12n} \leq c(n+1)^A$ for some c and A ”, and it is proved with $c = 6^{10}$ and $A = 10$. The formal proof uses the same Δ -calculus, organized through the Taylor coefficients of the blocks (Vandermonde’s identity) instead of Lemma 4.8, with the uniform loss $\lfloor \log_2 6n \rfloor \geq \lfloor \log_2 N_m \rfloor$ per derivative.

4.3. The odd common denominator. Let D_n be the least positive odd integer such that $D_n \rho_0, D_n Z_7, D_n Z_9 \in \mathbb{Z}[\frac{1}{2}]$. The power of 2 in a common denominator is irrelevant, since it cancels exactly in §4.6. We show $\log D_n \leq (10 - R^{(0)})n + o(n)$ with $R^{(0)} = 1.0901 \dots$ (Theorem 4.17). In general D_n does not divide d_n^{10} (in the computed range, primes $q > \sqrt{n}$ occur in D_n with exponent 11), so we work directly with the linear factors of the local expansions (4.5). Throughout, q is an odd prime.

Local data at a pole. Write $G_k(\varepsilon) = G_k(0) \Pi_k(\varepsilon)$ with $\Pi_k(\varepsilon) = \prod_{\beta \in B_k} (1 + \varepsilon/\beta)^{e_\beta}$, where B_k is the multiset consisting of

- the numerator values $\beta = c - k + \frac{1}{2}$ ($c \in C_m$, $1 \leq m \leq 6$), with $e_\beta = 1$;
- the value $\beta = (n - 2k)/2$, with $e_\beta = 1$ (only for $k \neq n/2$);
- the pole values $\beta = j - k$ ($0 \leq j \leq n$, $j \neq k$), with $e_\beta = -6$.

For $k = n/2$ we use $H = G_{n/2}/(2\varepsilon)$ instead of $G_{n/2}$ and omit the value $(n - 2k)/2$. All β lie in $\frac{1}{2}\mathbb{Z} \setminus \{0\}$, so $v_q(\beta) \geq 0$. Put $s_k := v_q(G_k(0))$ for $k \neq n/2$ and $s_{n/2} := v_q(H(0))$.

TABLE 8. The bound (4.10) and the exact values of $v_2(S_n)$ for configuration E (exact rational arithmetic; the rows $n \leq 160$ were recomputed independently from the definition of the Volkenborn integral, by Mahler series).

n	σ_n	λ_n	bound	$12n + 9 - \sigma_n - 4\lambda_n$	$v_2(S_n)$	$12n - v_2(S_n)$
40	18	6		447	457	23
80	18	7		923	935	25
120	21	8		1396	1415	25
160	18	8		1879	1893	27
200	26	9		2347	2365	35
240	21	9		2832	2853	27
280	25	9		3308	3323	37
320	18	9		3795	3811	29
360	29	9		4264	4283	37
400	26	10		4743	4763	37
440	29	10		5220	5246	34
480	21	10		5708	5731	29
560	25	10		6664	6681	39
640	18	10		7631	7649	31

Lemma 4.11 (Legendre form).

$$s_k = \sum_{m=1}^6 v_q \left(\prod_{c \in C_m} (2(c-k)+1) \right) + [k \neq \frac{n}{2}] v_q(n-2k) - 6(v_q(k!) + v_q((n-k)!)),$$

and $s_k \geq w_q(n)$, where

$$w_q(n) := v_q \left(\frac{\prod_m N_m!}{(n!)^{6^e}} \right) = \sum_{e \geq 1} \left(\sum_m \left\lfloor \frac{N_m}{q^e} \right\rfloor - 6 \left\lfloor \frac{n}{q^e} \right\rfloor \right).$$

Each bracket in the last sum is ≥ -5 , and it vanishes when $q^e > 13n/10$.

Proof. The identity follows from $\prod_{j \neq k} |j-k| = k!(n-k)!$, since powers of 2 are q -units. The N_m numbers $2(c-k)+1$, $c \in C_m$, are consecutive odd integers, so their product has $v_q \geq \sum_e \lfloor N_m/q^e \rfloor = v_q(N_m!)$; moreover $v_q(k!) + v_q((n-k)!) \leq v_q(n!)$, and the middle term is ≥ 0 . Finally $\sum_m \lfloor N_m/q^e \rfloor > \sum_m N_m/q^e - 6 = 6n/q^e - 6 \geq 6 \lfloor n/q^e \rfloor - 6$, and the bracket is an integer. $\square$

Lemma 4.12 (First-order Taylor bound). *Let $\Pi(\varepsilon) = \prod_\beta (1 + \varepsilon/\beta)^{e_\beta}$ (a finite product) with $v_q(\beta) \geq 0$ and $e_\beta \in \mathbb{Z}$. Then $v_q([\varepsilon^\lambda] \Pi) \geq -\text{Top}_\lambda$, where*

$$\text{Top}_\lambda := \max_\beta \sum j_\beta v_q(\beta),$$

the maximum being taken over all (j_β) with $j_\beta \geq 0$, $\sum_\beta j_\beta = \lambda$, and $j_\beta \leq e_\beta$ whenever $e_\beta \geq 0$. In particular $\text{Top}_\lambda \leq \lambda \max_\beta v_q(\beta)$. If every β with $v_q(\beta) > 0$ has $v_q(\beta) = 1$ and $e_\beta = 1$, and there are s such factors (with multiplicity), then $\text{Top}_\lambda \leq \min(\lambda, s)$.

Proof. $[\varepsilon^\lambda] \Pi = \sum_{\sum j_\beta = \lambda} \prod_\beta \binom{e_\beta}{j_\beta} \beta^{-j_\beta}$, where $\binom{e}{j} \in \mathbb{Z}$, and $\binom{e}{j} = 0$ if $0 \leq e < j$. $\square$

Theorem 4.13 (Denominators for every n). *Let $40 \mid n$ and let q be an odd prime.*

(a) *If $q > n$, then $v_q(D_n) = 0$.*

(b) *If $q^2 \leq 2.3n - 1$, then*

$$v_q(D_n) \leq 5\lfloor \log_q 1.3n \rfloor + 5\lfloor \log_q 2n \rfloor + 5\lfloor \log_q 2.3n \rfloor \leq 15 \log_q(3n).$$

(c) *If $\sqrt{2.3n-1} < q \leq n$, then*

$$v_q(D_n) \leq E_q^{\text{res}}(n) := \max\left(0, 10 - \min_{k \neq n/2} s_k, 9 - s_{n/2}\right) \leq 15.$$

Proof. Call q large if $q^2 > 2.3n - 1$. For large q , every $\beta \in B_k$ and every $l + \frac{1}{2}$ with $0 \leq l \leq n-1$ has $v_q \in \{0, 1\}$: numerator values satisfy $|2\beta| \leq 2(n+h_6)-1 = 2.3n-1$, pole values $|\beta| \leq n$, the value $(n-2k)/2$ satisfies $|n-2k| \leq n$, and $|2l+1| \leq 2n-1$.

(c) By Lemma 4.12 with $\text{Top}_\lambda \leq \lambda$, $v_q(r_{6-\lambda,k}) \geq s_k - \lambda$ for $k \neq n/2$, and $v_q(r_{6-\lambda,n/2}) \geq s_{n/2} - (\lambda - 1)$. Hence every term of $X(k, l) = \sum_i (i)_4 r_{i,k} (l + \frac{1}{2})^{-i-4}$ has $v_q \geq s_k - (6-i) - (i+4) = s_k - 10$, or $\geq s_{n/2} - 9$ if $k = n/2$. The same bounds hold, with room to spare, for c_3 and c_5 , and the constants $(i)_4$, 46080, 860160 are integers. So $v_q(\rho_0)$, $v_q(Z_7)$, $v_q(Z_9) \geq -E_q^{\text{res}}(n)$. Since $q^2 > 13n/10$, Lemma 4.11 gives $s_k \geq w_q(n) \geq -5$, so $E_q^{\text{res}}(n) \leq 15$.

(b) Put $M := \lfloor \log_q 2.3n \rfloor$; then $v_q(\beta) \leq M$ for every β , and $\nu := v_q(2l+1) \leq \lfloor \log_q 2n \rfloor \leq M$. Lemma 4.12 gives $v_q(r_{6-\lambda,k}) \geq s_k - \lambda M$, hence $v_q(X(k, l)) \geq s_k - \max_i ((6-i)M + (i+4)\nu) = s_k - 5M - 5\nu$; the case $k = n/2$ is similar, and the bounds for c_3, c_5 are weaker requirements. Finally $s_k \geq w_q(n) \geq -5\lfloor \log_q 1.3n \rfloor$ by Lemma 4.11.

(a) This is where the critical zeros enter. Let $q > n$; then q is large, and no pole value ($|j-k| \leq n$) and no nonzero value $n-2k$ ($|n-2k| \leq n$) is divisible by q . So the q -divisible factors of G_k (or of H) are exactly the numerator factors $\varepsilon + c - k + \frac{1}{2}$ with $q \mid 2(c-k) + 1$. Each has $e_\beta = 1$ and $v_q(\beta) = 1$, and there are s_k of them. By the last clause of Lemma 4.12, $v_q(r_{6-\lambda,k}) \geq s_k - \min(\lambda, s_k) \geq 0$; so c_3, c_5 are q -integral, and so is $X(k, l)$ when $q \nmid 2l+1$. Now let $q \mid 2l+1$, put $\xi := l + \frac{1}{2}$ (so $v_q(\xi) = 1$) and $c := k-1-l \in [0, n-1]$. Then $\varepsilon - \xi = \varepsilon + c - k + \frac{1}{2}$ is a numerator factor of G_k , which by Lemma 4.1(b) occurs $\mu(c) \geq 5$ times. The function $\tilde{G}_k := G_k/(\varepsilon - \xi)^5$ satisfies $v_q(\tilde{G}_k(0)) = s_k - 5$ and has $s_k - 5$ remaining q -divisible factors, all with exponent 1. By (4.9), $X(k, l) = 24[\varepsilon^5]G_k(\varepsilon)(\xi - \varepsilon)^{-5} = -24[\varepsilon^5]\tilde{G}_k(\varepsilon)$, and Lemma 4.12 gives $v_q(X(k, l)) \geq (s_k - 5) - \min(5, s_k - 5) \geq 0$. For $k = n/2$ the same argument applied to H gives $X(k, l) = -48[\varepsilon^4]\tilde{H}$. Hence every $X(k, l)$, and therefore ρ_0 , is q -integral. $\square$

Remark 4.14. The exact odd denominators were computed for the 26 values $n = 40, 80, \dots, 800$ and $n = 1200, 1600, 2000, 2400, 2800, 3200$. At every odd prime the true exponent is at most the bound of Theorem 4.13(b) or (c), and no prime $q > n$ divides D_n .

The continuum variables. For large $q \leq n$ put $x := n/q \geq 1$ and $y := k/q \in [0, x]$. Define, for real $x \geq 1$ and y ,

$$(4.11) \quad v(x, y) := \sum_{m=1}^6 \# \{ \sigma \in 2\mathbb{Z} + 1 : -2(\eta_m x + y) < \sigma < 2((1 + \eta_m)x - y) \} \\ - 6(\lfloor y \rfloor + \lfloor x - y \rfloor).$$

Lemma 4.15 (Dictionary). *For large $q \leq n$ and $0 \leq k \leq n$, $s_k = v(x, y) + [q \mid n - 2k]$, $k \neq \frac{n}{2}$; in particular $s_{n/2} = v(x, x/2)$.*

Proof. For large q every q -divisible factor has v_q exactly 1, so s_k is a count. Numerator: $2(c-k)+1 = \sigma q$ with $c \in C_m$ holds iff $-2(h_m+k)+1 \leq \sigma q \leq 2(n+h_m-k)-1$; as σq is odd and the two bounds are even integers ± 1 , this is equivalent to $-2(h_m+k) < \sigma q < 2(n+h_m-k)$, and we divide by q . Poles: $j-k = \tau q$ with $0 \leq j \leq n$ and $\tau \neq 0$ iff $-[y] \leq \tau \leq [x-y]$ and $\tau \neq 0$, and each pole factor has exponent -6 . The value $n-2k$ contributes 0 or 1. $\square$

Let $\mathcal{S}_E := \{0, 1\} \cup \{-\eta_m, 1+\eta_m : 1 \leq m \leq 6\}$, $\mathcal{A} := \{|\gamma-\gamma'| : \gamma, \gamma' \in \mathcal{S}_E, \gamma \neq \gamma'\}$ and

$$\mathcal{C} := \bigcup_{\alpha \in \mathcal{A}} \frac{1}{2\alpha} \mathbb{Z},$$

a discrete subset of $\mathbb{R}$. For real $x \geq 1$ put $\varpi(x) := \min_{0 \leq y < 1} v(x, y)$ and $f^{(0)}(x) := \max(0, 10 - \varpi(x))$.

Lemma 4.16 (Structure of ϖ). *The functions v and ϖ have the following properties.*

- (i) (Periodicity.) *For all x and y we have $v(x, y+1) = v(x, y)$ and $v(x+40, y) = v(x, y)$. Hence $\varpi(x+40) = \varpi(x)$; moreover $\varpi(x)$ is the minimum of $v(x, \cdot)$ on all of $\mathbb{R}$.*
- (ii) *Let $x \geq 1$, $x \notin \mathcal{C}$. The 14 points $0, \{x\}, \{\frac{1}{2} - \eta_m x\}, \{(1 + \eta_m)x - \frac{1}{2}\}$ ($1 \leq m \leq 6$) of $[0, 1)$ are pairwise distinct; $y \mapsto v(x, y)$ is constant on each open interval between consecutive points, and at each of the 14 points it takes the value of an adjacent interval. In particular $\varpi(x)$ is the least of these constant values.*
- (iii) *ϖ is constant on each connected component of $[1, \infty) \setminus \mathcal{C}$. On every interval $[1, X]$ it is a step function with finitely many steps.*
- (iv) *If $q \leq n$ is large and $x = n/q$, then $\min_k s_k \geq \varpi(x)$ (including $k = n/2$).*

Proof. (i) Replacing y by $y+1$ moves each interval in (4.11) by -2 and does not change $[y] + [x-y]$. Replacing x by $x+40$ moves the lower end of interval m by $-80\eta_m \in 2\mathbb{Z}$ and the upper end by $80(1+\eta_m) \in 2\mathbb{Z}$, so interval m gains $40+80\eta_m$ odd integers; these gains add up to $240 = 6 \cdot 40$ because $\sum_m \eta_m = 0$, which is exactly the growth of $6([y] + [x-y])$.

(ii) The count for interval m changes only where one of its ends crosses an odd integer, i.e. at $y \equiv \frac{1}{2} - \eta_m x$ or $y \equiv (1 + \eta_m)x - \frac{1}{2} \pmod{1}$; and $[y] + [x-y]$ equals $[x]$ for $y \in [0, \{x\}]$ and $[x] - 1$ for $y \in (\{x\}, 1)$. Each of the 14 points has the form $\{\gamma x + d\}$ with $\gamma \in \mathcal{S}_E$ and $d \in \{0, \frac{1}{2}\}$, and two of them coincide only if $(\gamma - \gamma')x \in \frac{1}{2}\mathbb{Z}$, i.e. only if $x \in \mathcal{C}$. Since the intervals in (4.11) are open, at a crossing point of a single end the count agrees with the count on one side; the same holds at 0 and $\{x\}$ by the formula for the pole count.

(iii) The combinatorial type in (ii) (the cyclic order of the 14 points and the values on the intervals) can only change where two points coincide, i.e. on $\mathcal{C}$; and $\mathcal{C} \cap [1, X]$ is finite.

(iv) By Lemma 4.15 and (i), $s_k \geq v(x, k/q) \geq \varpi(x)$. $\square$

Theorem 4.17 (Denominators, asymptotically). *As $n \rightarrow \infty$ through multiples of 40,*

$$\log D_n \leq (10 - R^{(0)})n + o(n), \quad \text{where}$$

$$R^{(0)} := \int_1^\infty \min(10, \varpi(x)) \frac{dx}{x^2} = 1.090124192761300787 \dots$$

Proof. Fix $X \geq 1$ and split the odd primes q into five classes.

- (1) $q > n$. No contribution, by Theorem 4.13(a).
- (2) $q^2 \leq 2.3n - 1$. By Theorem 4.13(b), these primes contribute at most $15\pi(\sqrt{2.3n})\log(3n)$ in total, which is $O(\sqrt{n})$.
- (3) q large and $n/q \in \mathcal{C} \cap [1, X]$. There are at most $\#(\mathcal{C} \cap [1, X])$ such primes, and each contributes at most $15\log n$ by Theorem 4.13(c).
- (4) q large and $n/q \in [1, X] \setminus \mathcal{C}$. Here $v_q(D_n) \leq f^{(0)}(n/q)$, by Theorem 4.13(c) and Lemma 4.16(iv). By Lemma 4.16(iii), the function $f^{(0)}$ takes a constant value $f_i \geq 0$ on each of the finitely many components (a_i, b_i) of $[1, X] \setminus \mathcal{C}$. So, by (2.4),

$$\begin{aligned} \sum_q f^{(0)}(n/q) \log q &\leq \sum_i f_i (\theta(n/a_i) - \theta(n/b_i)) \\ &= n \int_1^X f^{(0)}(x) \frac{dx}{x^2} + o(n). \end{aligned}$$

- (5) q large and $n/q > X$. By Theorem 4.13(c) and [35, Theorem 9], these primes contribute at most $15\theta(n/X) < 15.3n/X$.

Hence $\limsup_n \frac{1}{n} \log D_n \leq \int_1^X f^{(0)} \frac{dx}{x^2} + 15.3/X$. Letting $X \rightarrow \infty$ gives the bound $\int_1^\infty f^{(0)} \frac{dx}{x^2} = 10 - R^{(0)}$. $\square$

Proposition 4.18 (The value of $R^{(0)}$; computer-assisted). *On $[1, 41] \setminus \mathcal{C}$ the function ϖ takes values in $\{-2, -1, \dots, 5\}$, so $\min(10, \varpi) = \varpi$ there, and*

$$\begin{aligned} \int_1^{41} \varpi(x) \frac{dx}{x^2} &= \frac{2275273804717574138460377}{2189991449667171005856000} = 1.03894186667412 \dots, \\ \int_{41}^\infty \varpi(x) \frac{dx}{x^2} &= 0.051182326087180157451 \dots \end{aligned}$$

Hence $R^{(0)} = 1.090124192761300787 \dots$; in particular $R^{(0)} \geq 1.0389 - 2/41 > 0.99$.

Justification. By Lemma 4.16, $\mathcal{C} \cap [1, 41]$ is finite and ϖ is constant on the intervals (a_i, b_i) between consecutive points; the head is the finite sum $\sum_i \varpi_i(1/a_i - 1/b_i)$, computed in exact rational arithmetic after evaluating $v(x, \cdot)$ at one point of each interval. By 40-periodicity, the tail equals $\frac{1}{40} \sum_i \varpi_i(F(\frac{b_i}{40}) - F(\frac{a_i}{40}))$, summed over the intervals of one period $[41, 81]$, since $\sum_{j \geq 0} (\frac{1}{w+j} - \frac{1}{z+j}) = F(z) - F(w)$, where F is the digamma function, the logarithmic derivative of Euler's gamma function. Three independent implementations (two written for the denominator analysis, one written from Lemmas 4.15 and 4.16 alone, which also re-evaluates ϖ at a second point of every interval) give identical heads and totals; splitting at 81 instead of 41 gives a different head and tail but the same total to 22 digits. The tail is evaluated in floating point with 30 to 40 digits. This does not matter for Theorem A, which only needs $R^{(0)} > 10 - 12\log 2 - 0.7812 = 0.90103 \dots$ (§4.6): the minimum of ϖ

on $[1, 41] \setminus \mathcal{C}$, hence by periodicity on $[41, \infty) \setminus \mathcal{C}$, is -2 (exact computation), so the tail is $\geq -2/41$, and the exact head gives $R^{(0)} \geq 1.0389 - 0.0488 > 0.99$. $\square$

TABLE 9. Top: the first steps of the residue function ϖ (on $(\frac{20}{13}, \frac{5}{3})$ it takes the values 1, 2, 3 on three consecutive steps). Bottom: the dx/x^2 -measure of $\{x \in [1, 41] : \varpi(x) = j\}$. Exponents above 10 ($f^{(0)} = 11, 12$) and the savings both come from the negative shift h_1 .

x	$(1, \frac{20}{17})$	$(\frac{20}{17}, \frac{40}{29})$	$(\frac{40}{29}, \frac{20}{13})$	$(\frac{20}{13}, \frac{5}{3})$	$(\frac{5}{3}, 2)$	$(2, \frac{30}{13})$	$(\frac{30}{13}, \frac{40}{17})$
$\varpi(x)$	0	-1	0	1, 2, 3	4	-2	-1
$f^{(0)}(x)$	10	11	10	9, 8, 7	6	12	11
j	-2	-1	0	1	2	3	4
weight	0.080	0.143	0.309	0.049	0.099	0.104	0.180
							5
							0.012

Remark 4.19 (Sharper denominators; not needed for Theorem A). Theorem 4.17 uses only the leading q -adic valuation at each pole. Three further mechanisms improve the constant. We state the results of this remark as claims: their proofs are not given in this paper, and they have been checked only by AI agents (§7.2). The proofs and exact checks are in the project's working notes, which are available from the author; a summary is §4.6 of the informal proof `docs/proof.md` that accompanies the Lean development of Theorem A (Section 5).

- (1) *Second-order Taylor lemma.* Let q be large. Then

$$\Pi(\varepsilon) = P(\varepsilon/q)U(\varepsilon), \quad P(u) := \prod_{v_q(\beta)=1} \left(1 + \frac{u}{\beta'}\right)^{e_\beta}, \quad \beta' := \beta/q,$$

where U has q -integral coefficients, and

$$v_q([\varepsilon^\lambda]\Pi) \geq -\max_{j \leq \lambda} (j - v_q([u^j]P)).$$

The identity

$$[u^{r-1}](1-u)^{-r}(1-2u) = \binom{2r-2}{r-1} - 2\binom{2r-3}{r-1} = 0$$

shows, for instance, that $P = (1-u)^{-6}(1-2u)$ has $[u^5]P = 0$ and $[u^4]P = 14$; so a coefficient that Lemma 4.12 puts at q^{-5} is in fact $O(q^{-4})$. This gains exactly one power of q for $x \in (1, \frac{20}{17}) \cup (\frac{10}{7}, \frac{20}{13})$, and adds exactly $(1 - \frac{17}{20}) + (\frac{7}{10} - \frac{13}{20}) = \frac{1}{5}$ to R :

$$R^{(2)} = R^{(0)} + \frac{1}{5} = 1.290124192761300787 \dots$$

- (2) *Anti-diagonal root identity.* For $0 \leq c \leq n-1$, $\sum_{k=0}^n X(k, k-1-c) = 0$, because $R_n^{(4)}(-\frac{1}{2}-c) = 0$ by Lemma 4.1(b) (Lai's root trick [20, Lemma 4.6], applied to each anti-diagonal). So on each anti-diagonal the better of two one-sided bounds may be used.
- (3) *Row cancellation.* A Wilson-type translation lemma relates the unit parts of $G_k(0)$ and $G_{k+q}(0)$ modulo q , and the symmetry $k \mapsto n-k$ (Lemma 4.1(c)) relates those of $G_k(0)$ and $G_{n-k}(0)$. Together they give, under an explicit combinatorial criterion, one extra power of q for the row sums $\sum_{k \geq l} X(k, l)$

with $q \mid 2l + 1$. On the undeformed $(6, 3)$ family the resulting exponent bound equals $a + j = 9$ for $x = n/q \in [1, 12]$ (a continuum computation), so for primes $q \geq n/12$ this recovers the one-power saving that [23, Lemma 5.3] and Proposition 3.7 obtain from Andrews' transformation for all primes $p > \sqrt{2n}$.

With these, the claim is that $\log D_n \leq (10 - R_E)n + o(n)$, where

$$R_E = \frac{9876623266560421249}{7052098086803952000} + \int_{41}^{\infty} \varpi(x) \frac{dx}{x^2} = 1.451704999621994665 \dots$$

The rational number is the integral over $[1, 41]$ of 10 minus the third-order exponent bound, which is constant between the 1217 points of $\mathcal{C} \cap [1, 41]$; beyond 41 the residue-level tail of Proposition 4.18 is used. (Using the third-order bound only on $[1, 11]$ gives $1.4322937593058673 \dots$) With R_E the exponent of the proof would improve from -0.189 to -0.551 per n (Table 12). These arguments are considerably longer and more intricate than the ones above, and Theorem A does not depend on them. Two checks: the continuum bound was compared with the true exponents at 2081 large primes ($200 \leq n \leq 3200$) without a violation; and at $\ell = n - 1$ with $n \in \mathcal{N}$ (§4.5) we have $x = n/\ell \in (1, \frac{20}{17})$, where the residue level gives $v_\ell(D_n) \leq 10$ and the second-order lemma gives ≤ 9 , while Theorem 4.32 gives $v_\ell(\rho_0) = -9$; so $v_\ell(D_n) = 9$, and the second-order bound is sharp there.

Remark 4.20 (Formalization). The Lean statement `Stmt.Denominators` is proved for configuration E at the residue level with $\delta = 9$. The odd part of the common denominator is $\prod_q q^{E_q}$ over the odd primes $q \leq n$, where E_q is the first-order bound with valuations capped at $\lfloor \log_q 2n \rfloor$ (the factor $n - 2k$ is not used); for $q^2 > 2n$ this is the residue level $10 - \min_k v(n/q, k/q)$. Lower bounds for ϖ on $[1, 41]$ are given by 766 interval certificates checked by the kernel (numerically they coincide with ϖ , whose values there lie in $[-2, 5]$); primes $q \leq n/81$ are bounded using $\varpi \geq -2$; and the prime number theorem enters through $\theta(x)/x \rightarrow 1$. The resulting constant is $T = 8.9607 \dots \leq 8.97 < 9$. The residue-level constant with the full tail would be $10 - R^{(0)} = 8.9099$, so the truncation at $x = 81$ costs 0.05.

4.4. The archimedean size of the coefficients.

Theorem 4.21 (Growth). *The following bounds hold.*

(a) *For every $n \equiv 0 \pmod{40}$ with $n \geq 2000$,*

$$\max(|\rho_0|, |Z_7|, |Z_9|) \leq 1.56 \cdot 10^{12} n^{-4} e^{-0.7812n}.$$

(b) *For every $n \equiv 0 \pmod{40}$ with $n \geq 10^4$,*

$$\max(|\rho_0|, |Z_7|, |Z_9|) \leq 3.31 \cdot 10^{11} n^{-4} e^{-0.78127n}.$$

In particular $\limsup_n \frac{1}{n} \log \max(|\rho_0|, |Z_7|, |Z_9|) \leq -0.78127$.

The proof occupies the rest of this subsection; it is computer-assisted through Lemma 4.28.

Why the naive bound fails. By Stirling's formula,

$$\log |r_{6,k}| = nP(-k/n) + O(\log n),$$

where P is the real profile of (4.15) below. The maximum of P is $P(-0.3485) = 0.8210$, while the coefficients are of size $e^{-0.781n}$. So each of ρ_0, Z_7, Z_9 involves a cancellation of about 1.60 nats per n between the contributions of different poles,

and every bound of the form $\sum_k |\cdots|$ (for instance, Cauchy estimates for the residues) gives a positive exponential rate.

Idea. We use kernels that are 1-periodic and holomorphic at the integers, so that $\sum_k \text{Res}_{t=-k}$ is a fixed linear combination of the c_i ; that have poles only at the half-integers, which is harmless because of the critical zeros; and that decay like $e^{-2\pi|\text{Im } t|}$. The residue theorem then expresses the coefficients as contour integrals, which we bound by a saddle point. The archimedean template is Zudilin's treatment of Barnes-type integrals for well-poised forms: the contour-integral representation [40, §3, Lemma 5] and the saddle-point asymptotics [40, §8, Lemma 20], [39, §§2–3]; here it is applied to the coefficients of a 2-adic linear form.

Lemma 4.22 (Kernels). *Put $w := e^{2\pi it}$ and*

$$K_2(t) := \left(\frac{\pi}{\cos \pi t} \right)^2 = \frac{4\pi^2 w}{(1+w)^2}, \quad K_4(t) := \left(\frac{\pi}{\cos \pi t} \right)^4 = \frac{16\pi^4 w^2}{(1+w)^4},$$

and $W(t) := \frac{d^4}{dt^4} \pi \tan \pi t$.

- (a) *They have period 1 and are holomorphic on $\mathbb{C} \setminus (\mathbb{Z} + \frac{1}{2})$, in particular at the integers.*
- (b) *For $k \in \mathbb{Z}$: $K_2(-k+u) = \pi^2 + \pi^4 u^2 + \frac{2}{3}\pi^6 u^4 + O(u^6)$ and $K_4(-k+u) = \pi^4 + 2\pi^6 u^2 + \frac{7}{3}\pi^8 u^4 + O(u^6)$.*
- (c) *At $x \in \mathbb{Z} + \frac{1}{2}$, K_2 and K_4 have poles of order 2 and 4, and $W(x+u) = -24u^{-5} + O(1)$.*
- (d) *For $|\text{Im } t| \geq 1$ we have $|K(t)| \leq \kappa_K e^{-2\pi|\text{Im } t|}$, where*

$$\kappa_{K_2} = \frac{4\pi^2}{(1 - e^{-2\pi})^2} = 39.626\dots, \quad \kappa_{K_4} = \frac{16\pi^4 e^{-2\pi}}{(1 - e^{-2\pi})^4} = 2.9323\dots,$$

and $\kappa_W = 2\pi(2\pi)^4 \sum_{k \geq 1} k^4 e^{-2\pi(k-1)} \leq 10088.01$.

- (e) *If $\text{Re } t \in \mathbb{Z}$, then $|K_2(t)| \leq \pi^2$, $|K_4(t)| \leq \pi^4$ and $|W(t)| \leq 16\pi^5$.*
- (f) *$K(\bar{t}) = \overline{K(t)}$; for even n , K_2 and K_4 are invariant under $t \mapsto -n-t$, and $W(-n-t) = -W(t)$.*

Proof. With $\vartheta = \tan \pi t$ one has $W = 8\pi^5 \vartheta(1 + \vartheta^2)(2 + 3\vartheta^2)$, since $\tan''' = 8 \tan(1 + \tan^2)(2 + 3 \tan^2)$. (a), (c) and (f) are clear, and (b) follows from $\sec^2 v = 1 + v^2 + \frac{2}{3}v^4 + O(v^6)$ and $\sec^4 v = 1 + 2v^2 + \frac{7}{3}v^4 + O(v^6)$. For (d) use the displayed w -expansions and $\pi \tan \pi t = i\pi + 2\pi i \sum_{k \geq 1} (-1)^k w^k$, which gives $W = 2\pi i (2\pi)^4 \sum_{k \geq 1} (-1)^k k^4 w^k$, together with $|w| = e^{-2\pi \text{Im } t} \leq e^{-2\pi}$ (and conjugation for $\text{Im } t < 0$). For (e), $|\cos \pi t| = \cosh(\pi \text{Im } t) \geq 1$ and $\vartheta = i \tanh(\pi \text{Im } t)$ when $\text{Re } t \in \mathbb{Z}$. $\square$

Proposition 4.23 (Contour representations). *Let $x_0 := n/40 \in \mathbb{Z}$, and let Γ be a piecewise smooth path from $-i\infty$ to $+i\infty$ that is symmetric under complex conjugation, meets $\mathbb{R}$ only at x_0 , and lies in a fixed vertical strip for large $|\text{Im } t|$. Put*

$$V_2 := \pi^4 c_3 + \frac{2}{3}\pi^6 c_5, \quad V_4 := 2\pi^6 c_3 + \frac{7}{3}\pi^8 c_5, \quad I_n := \sum_{\nu \geq 0} R_n^{(4)}(\nu + \frac{1}{2}).$$

Then

$$V_2 = \frac{1}{\pi i} \int_{\Gamma} R_n K_2 dt, \quad V_4 = \frac{1}{\pi i} \int_{\Gamma} R_n K_4 dt, \quad I_n = \frac{1}{2\pi i} \int_{\Gamma} R_n W dt,$$

and

$$(4.12) \quad I_n = 45720 \zeta(7) c_3 + 858480 \zeta(9) c_5 - \rho_0,$$

where $45720 = (3)_4(2^7 - 1)$ and $858480 = (5)_4(2^9 - 1)$. Consequently

$$(4.13) \quad \begin{aligned} c_5 &= \frac{V_4 - 2\pi^2 V_2}{\pi^8}, & c_3 &= \frac{\frac{7}{3}V_2 - \frac{2}{3}\pi^{-2}V_4}{\pi^4}, \\ \rho_0 &= 45720 \zeta(7) c_3 + 858480 \zeta(9) c_5 - I_n. \end{aligned}$$

Proof. Independence of the path. The poles of R_n and of the kernels are real, and R_n and the kernels are holomorphic at $x_0 > 0$. By Lemma 4.1(a) and Lemma 4.22(d), $R_n K$ is $O(|t|^{-5} e^{-2\pi |\operatorname{Im} t|})$ for $|\operatorname{Im} t| \geq 1$. By Cauchy's theorem, applied between two admissible paths in each half-plane and truncated at $|\operatorname{Im} t| = Y \rightarrow \infty$, all admissible paths give the same integrals. We take $\Gamma = \{\operatorname{Re} t = x_0\}$.

V_2 and V_4 . Let $K \in \{K_2, K_4\}$. In the strip $-n - x_0 < \operatorname{Re} t < x_0$ the half-integers are removable singularities of $R_n K$, because $\operatorname{ord} R_n \geq 5$ there (Lemma 4.1(b)) while the poles of K have order ≤ 4 . At $-k$, Lemma 4.22(b) gives

$$\operatorname{Res}_{t=-k} R_n K_2 = \pi^2 r_{1,k} + \pi^4 r_{3,k} + \frac{2}{3} \pi^6 r_{5,k},$$

and similarly for K_4 . Summing over k and using $c_1 = 0$ gives V_2 , respectively V_4 . The residue theorem on the rectangle $[-n - x_0, x_0] \times [-Y, Y]$, whose horizontal sides contribute $O(ne^{-2\pi Y})$, gives

$$V = \frac{1}{2\pi i} \left(\int_{\operatorname{Re} t = x_0} - \int_{\operatorname{Re} t = -n - x_0} \right) R_n K dt,$$

with both lines oriented upwards. The substitution $t = -n - s$ turns the second integral into $-\int_{\operatorname{Re} s = x_0} R_n K ds$, because $R_n K$ is odd under $t \mapsto -n - t$ (Lemma 4.1(c), Lemma 4.22(f)).

I_n . To the right of x_0 the only singularities of $R_n W$ are the half-integers $x > x_0$, with residue $-24[u^4]R_n(x+u) = -R_n^{(4)}(x)$. Apply the residue theorem to $\{x_0 < \operatorname{Re} t < N, |\operatorname{Im} t| < Y\}$ with $N \in \mathbb{Z}$: on $\operatorname{Re} t = N$ we have $|R_n| = O(N^{-5})$, and $|W| \leq 16\pi^5$ and $|W| \leq \kappa_W e^{-2\pi |\operatorname{Im} t|}$ by Lemma 4.22(d),(e). Letting $Y, N \rightarrow \infty$ gives

$$\frac{1}{2\pi i} \int_{\operatorname{Re} t = x_0} R_n W dt = \sum_{x \in \mathbb{Z} + \frac{1}{2}, x > x_0} R_n^{(4)}(x).$$

At the half-integers $0 < x < x_0$ we have $R_n^{(4)}(x) = 0$, since $\operatorname{ord}_x R_n \geq 5$: *this is where the critical zeros are used*. So the right-hand side is I_n . Finally, $R_n^{(4)}(t) = \sum_{i,k} (i)_4 r_{i,k} (t+k)^{-i-4}$ and $\sum_{\nu \geq 0} (\nu + k + \frac{1}{2})^{-s} = (2^s - 1)\zeta(s) - A_k^{(s)}$ give $I_n = \sum_i (i)_4 (2^{i+4} - 1)\zeta(i+4) c_i - \rho_0$, which is (4.12) because $c_1 = c_2 = c_4 = c_6 = 0$. Solving the linear system for V_2, V_4 (its determinant is π^{12}) gives (4.13). $\square$

Remark 4.24. By (4.7) and (4.12), $I_n = -(\rho_0 + (1 - 2^{-7})Z_7 \zeta(7) + (1 - 2^{-9})Z_9 \zeta(9))$: a real linear form in $1, \zeta(7), \zeta(9)$ with the coefficients of S_n , up to sign and the Euler factors at 2 (compare [23, Remark 3.4]). The identities of Proposition 4.23 were checked numerically against the exact c_3, c_5, ρ_0 : they hold to relative error about 10^{-40} at $n = 40$ and $n = 80$, and were confirmed independently on three different paths; a path crossing $\mathbb{R}$ at $x_0 + 1$ breaks only the identity for I_n , as predicted; and along the path Γ_n used below, the right-hand sides of (4.13) reproduce $\frac{1}{n} \log |\rho_0|$, $\frac{1}{n} \log |Z_7|$, $\frac{1}{n} \log |Z_9|$ to six digits for $n = 120, 160, \dots, 640$.

Pointwise bounds. Let $F(z) := z \log z - z$ (principal branch), which is continuous on the closed upper half-plane $\overline{\mathbb{H}}$ with $F(0) = 0$. For $\text{Im } t > 0$ and $\tau \in \overline{\mathbb{H}}$ put

$$\begin{aligned} \mathcal{L}(t) &:= \log |2t + n| + \sum_{m=1}^6 \text{Re}[F(t + n + h_m) - F(t - h_m)] \\ &\quad - 6 \text{Re}[F(t + n + \tfrac{1}{2}) - F(t - \tfrac{1}{2})], \\ \varphi(\tau) &:= \sum_{m=1}^6 [F(\tau + 1 + \eta_m) - F(\tau - \eta_m)] - 6[F(\tau + 1) - F(\tau)]. \end{aligned}$$

Thus $\varphi(\tau) = \sum_j \varsigma_j F(\tau + a_j)$, where the pairs (ς_j, a_j) are

$$(1, 1 + \eta_m), \quad (-1, -\eta_m) \quad (1 \leq m \leq 6), \quad (-6, 1), \quad (6, 0).$$

They satisfy $\sum_j \varsigma_j = \sum_j \varsigma_j a_j = \sum_j \varsigma_j a_j^2 = 0$, the last two identities because $\sum_m \eta_m = 0$. In particular, F may be replaced by $z \log z$ in the definition of φ .

Lemma 4.25 (Midpoint rule). *For $\text{Im } t \geq 1$, $|\log |R_n(t)| - \mathcal{L}(t)| \leq (\pi + 3)/2$.*

Proof. Consider a block $\prod_{l=0}^{L-1} (t + b + l)$ with $b \in \mathbb{R}$, and $f(s) := \log |t + b + s| = \frac{1}{2} \log(u^2 + y^2)$ with $u = \text{Re } t + b + s$, $y = \text{Im } t \geq 1$. For each cell, $|\int_{l-1/2}^{l+1/2} f - f(l)| \leq \frac{1}{24} \max_{\text{cell}} |f''|$, and $|f''| = |y^2 - u^2|/(u^2 + y^2)^2 \leq (u^2 + y^2)^{-1}$. This majorant is unimodal in u with maximum y^{-2} ; bounding its maximum on each cell by its integral over the neighboring cell on the side of the peak, except for at most three cells near the peak, gives $\sum_l \max_{\text{cell}} |f''| \leq \pi/y + 3/y^2 \leq \pi + 3$. Moreover $\int_{-1/2}^{L-1/2} f = \text{Re}[F(t + b + L - \frac{1}{2}) - F(t + b - \frac{1}{2})]$, because $F' = \log$. The six numerator blocks ($b = \frac{1}{2} - h_m$, $L = N_m$, with end points $t - h_m$ and $t + n + h_m$) and the six copies of $(t)_{n+1}$ ($b = 0$, $L = n + 1$, end points $t - \frac{1}{2}$ and $t + n + \frac{1}{2}$) each contribute an error $\leq (\pi + 3)/24$. $\square$

Lemma 4.26 (Scaling). *For $\text{Im } \tau > 0$, $\text{Re } \tau \geq 0$ and $|\tau| > 1/(2n)$,*

$$\mathcal{L}(n\tau) \leq n \text{Re } \varphi(\tau) - 5 \log n + \log |2\tau + 1| - 3 \log(|\tau| - \tfrac{1}{2n}).$$

Proof. We have $F(nz) = nF(z) + nz \log n$, and the linear terms contribute

$$n \log n \left[\sum_m (1 + 2\eta_m) - 6 \left(1 + \frac{1}{n} \right) \right] = -6 \log n.$$

Hence, with $\epsilon_n := 1/(2n)$,

$$\begin{aligned} \mathcal{L}(n\tau) - n \text{Re } \varphi(\tau) &= -5 \log n + \log |2\tau + 1| \\ &\quad - 6n \text{Re}[F(\tau + 1 + \epsilon_n) - F(\tau + 1)] \\ &\quad + 6n \text{Re}[F(\tau - \epsilon_n) - F(\tau)]. \end{aligned}$$

Since $\text{Re } F'(z) = \log |z|$, the first bracket term equals $-6n \int_0^{\epsilon_n} \log |\tau + 1 + s| ds \leq 0$, and the second equals $-6n \int_{-\epsilon_n}^0 \log |\tau + s| ds \leq -3 \log(|\tau| - \epsilon_n)$. $\square$

Lemma 4.27 (Crossing segment). *For an integer $x_0 \geq 1$ and $|y| \leq 1$, $|R_n(x_0 + iy)| \leq e^{\pi^2/2} |R_n(x_0 + i)|$.*

Proof. Every numerator factor, and $|2x_0 + n + 2iy|$, is largest at $|y| = 1$. For the denominator, $|x_0 + l + iy| \geq x_0 + l$ and $|x_0 + l + i| \leq (x_0 + l)e^{1/(2(x_0+l)^2)}$, so the ratio of the denominators is at most $\exp(3 \sum_{l \geq 0} (x_0 + l)^{-2}) \leq e^{\pi^2/2}$. $\square$

Combining Lemmas 4.25 and 4.26: for $t = n\tau$ with $\operatorname{Im} t \geq 1$ and $\operatorname{Re} \tau \geq 0$,

$$(4.14) \quad \begin{aligned} \log |R_n(n\tau)| &\leq n \operatorname{Re} \varphi(\tau) + E_n(\tau), \quad \text{where} \\ E_n(\tau) &:= -5 \log n + \log |2\tau + 1| - 3 \log(|\tau| - \frac{1}{2n}) + \frac{\pi + 3}{2}. \end{aligned}$$

As $r \mapsto \log(1 + 2r) - 3 \log(r - \frac{1}{2n})$ is decreasing, $E_n(\tau) \leq E_{\max} - 5 \log n$ whenever $|\tau| \geq 1/40$ and $n \geq 2000$, where $E_{\max} := \log(1.051) - 3 \log(0.02475) + \frac{\pi+3}{2} \leq 14.218$.

The landscape. Put $\Psi(\tau) := \operatorname{Re} \varphi(\tau) - 2\pi \operatorname{Im} \tau$ on $\overline{\mathbb{H}}$, and let

$$(4.15) \quad P(x) := \operatorname{Re} \varphi(x) \quad (x \in \mathbb{R})$$

be the *real profile* (boundary values from $\overline{\mathbb{H}}$); one has $P(-1-x) = P(x)$. By (4.14) and Lemma 4.22(d), $|R_n(n\tau)K(n\tau)| \leq \kappa_K \exp(n\Psi(\tau) + E_n(\tau))$ for $\operatorname{Im}(n\tau) \geq 1$ and $\operatorname{Re} \tau \geq 0$.

The saddle point (numerical, explanatory only). The critical points of $\varphi(\tau) + 2\pi i\tau$ satisfy $\varphi'(\tau) = -2\pi i$. The relevant one is

$$\tau^* = 0.0695900497 + 0.0312802763 i, \quad \Psi(\tau^*) = -0.7812761397.$$

On the real axis, at the crossing point $x_0/n = 0.025$, we have $P(0.025) = -0.8005 < \Psi(\tau^*)$; so it is the saddle, not the crossing point, that sets the rate.

In the next lemma, $[x', x''] \times [y', y'']$ denotes the box of the points $x + iy$ with $x' \leq x \leq x''$ and $y' \leq y \leq y''$.

Lemma 4.28 (Landscape inequalities; computer-assisted). *The following inequalities hold.*

- (i) $P(x) \leq 0.85$ for $x \in [-0.5, 0.15]$;
- (ii) $\Psi \leq -0.7812$ on $[0.025, 0.070] \times [0, 0.001]$;
- (iii) $\Psi \leq -0.7812$ on $[0.069, 0.070] \times [0, 0.26]$.

The following sharp versions also hold.

- (i') $P(x) \leq 0.8215$ for $x \in [-0.5, 0.15]$;
- (ii') $\Psi \leq -0.78127$ on $[0.025, 0.0697] \times [0, 10^{-4}]$;
- (iii') $\Psi \leq -0.78127$ on $[0.0695, 0.0697] \times [0, 0.26]$.

For comparison, the true maximum in (i) is 0.820998, at $x = -0.34848$; the true maxima in (iii) and (iii') are -0.781259 and -0.7812757 .

Certification. Each of the six inequalities was certified by three programs with different bounding methods. All three bisect boxes and use the formula

$$\operatorname{Re}(z \log z) = x \log |z| - y \arg z \quad (z = x + iy).$$

- (1) Interval arithmetic in `mpmath.iv` [17] (96 bits, outward rounding). On a box, $\log |z|$ is enclosed from the exact extrema of $x^2 + y^2$ and $\arg z$ from the corners (it is monotone in x and in y); boxes containing a branch point get a separate enclosure; the bound is the minimum of the plain enclosure and a centered (mean-value) form.
- (2) Exact rational boxes, with a second-order Taylor bound for the harmonic function Ψ ($|\varphi''| \leq \sum_j |\varsigma_j|/d_j$, where d_j is the distance from $-a_j$ to the box) and a monotone-corner bound near the branch points; point values of $\log$ and $\arctan$ from `mpmath.iv` at 120 bits. (The first program uses 53-bit box end points, so the points $x = 1/40$ and $x = 0.15$ are covered only by the rational boxes of the second and third programs.)

- (3) A program without floating point and without `mpmath` (about 440 lines), using exact rationals and integer intervals at scale 2^{-128} . The logarithm is computed as $k \log 2 + 2 \operatorname{artanh} \frac{m-1}{m+1}$, the arctangent by alternating series, π by Machin's formula and $1/e$ by its series, all with explicit remainders. The bound on a box is the minimum of two bounds: exact per-term extrema of $\operatorname{Re}(z \log z)$, obtained from $\partial_y \operatorname{Re}(z \log z) = -\arg z \leq 0$ and from the sign of $\partial_x \operatorname{Re}(z \log z) = \log |z| + 1$; and, near the saddle, a centered mean-value bound.

In all three programs, control runs with targets slightly below the true maxima fail and targets slightly above pass; for program (3) the failing targets were 0.8209 for (i), -0.7930 for (ii), and -0.7813 and -0.78126 for (iii), and they fail at the expected places. In addition, 12 000 random point tests and 2427 random box tests found no violation of the enclosures used. Table 10 lists the numbers of boxes used by program (3).

TABLE 10. The inequalities of Lemma 4.28 as certified by the floating-point-free program (3), with the number of boxes in parentheses.

version	(i): $P \leq$	(ii): $\Psi \leq$	(iii): $\Psi \leq$
standard	0.85 (33)	-0.7812 (23)	-0.7812 (283)
sharp	0.8215 (53)	-0.78127 (19)	-0.78127 (157)

Lemma 4.29 (The upper half-plane). $\operatorname{Re} \varphi \leq 0.85$ on $\overline{\mathbb{H}}$ (≤ 0.8215 with (i')). Consequently $\Psi(\tau) \leq -0.7836$ for $\operatorname{Im} \tau \geq 0.26$ (≤ -0.8121 with (i')).

Proof. For $x \geq 0.15$ all the arguments $x+1+\eta_m$, $x-\eta_m$, $x+1$, x are real and ≥ 0 . The map $\eta \mapsto F(x+1+\eta) - F(x-\eta)$ has second derivative $\frac{1}{x+1+\eta} - \frac{1}{x-\eta} < 0$ on $(-\frac{1}{2}, x)$, so it is concave on $[-\frac{1}{2}, x]$, which contains all η_m and their mean 0. By Jensen's inequality, $P(x) \leq 6[F(x+1) - F(x)] - 6[F(x+1) - F(x)] = 0$. From $\operatorname{Re} F(-\bar{w}) = -\operatorname{Re} F(w) - \pi \operatorname{Im} w$ and $\sum_j \varsigma_j = 0$ one gets $\operatorname{Re} \varphi(-1-\bar{\tau}) = \operatorname{Re} \varphi(\tau)$; hence also $P(x) \leq 0$ for $x \leq -1.15$, and $\sup_{\mathbb{R}} P \leq 0.85$ by Lemma 4.28(i). The function $u := \operatorname{Re} \varphi$ is harmonic on $\mathbb{H}$ and continuous on $\overline{\mathbb{H}}$, and $\varphi(\tau) = O(|\tau|^{-2})$ as $\tau \rightarrow \infty$, because $\sum_j \varsigma_j = \sum_j \varsigma_j a_j = \sum_j \varsigma_j a_j^2 = 0$. For $\epsilon > 0$, the harmonic function $u - \epsilon \log |\tau + i|$ is ≤ 0.85 on $\mathbb{R}$ and on large half-circles, hence on $\mathbb{H}$ by the maximum principle [1, Ch. 4, §6]; let $\epsilon \rightarrow 0$. Finally $0.85 - 2\pi \cdot 0.26 = -0.7836 \dots$ and $0.8215 - 2\pi \cdot 0.26 = -0.8121 \dots$ $\square$

Proof of Theorem 4.21. Fix $n \equiv 0 \pmod{40}$ with $n \geq 2000$, put $x_0 = n/40$ and $x_1 := \lfloor 0.0696n \rfloor$, so that $x_1/n \in [0.0691, 0.0696]$. Let Γ_n^+ be the path

$$A: x_0 \rightarrow x_0 + i, \quad B: x_0 + i \rightarrow x_1 + i, \quad C: x_1 + i \rightarrow x_1 + i\infty,$$

and $\Gamma_n := \overline{\Gamma_n^+} \cup \Gamma_n^+$, oriented upwards. It is admissible in Proposition 4.23, and by conjugation symmetry $\int_{\Gamma_n} R_n K dt = 2i \operatorname{Im} \int_{\Gamma_n^+} R_n K dt$. Hence, with $\mathcal{J}_K := \int_{\Gamma_n^+} |R_n(t)K(t)| |dt|$,

$$|V_2| \leq \frac{2}{\pi} \mathcal{J}_{K_2}, \quad |V_4| \leq \frac{2}{\pi} \mathcal{J}_{K_4}, \quad |I_n| \leq \frac{1}{\pi} \mathcal{J}_W.$$

Along the whole path $|\tau| = |t/n| \geq 1/40$, so $E_n \leq E_{\max} - 5 \log n$.

A: By Lemma 4.27 and Lemma 4.22(e), $|R_n K| \leq \kappa_K^A e^{\pi^2/2} |R_n(x_0 + i)|$ on A , with $\kappa_{K_2}^A = \pi^2$, $\kappa_{K_4}^A = \pi^4$, $\kappa_W^A = 16\pi^5$. The point $\tau_0 = \frac{1}{40} + \frac{i}{n}$ lies in the box of Lemma 4.28(ii), so by (4.14)

$$\begin{aligned} |R_n(x_0 + i)| &\leq \exp(n\Psi(\tau_0) + 2\pi + E_n(\tau_0)) \\ &\leq \exp(-0.7812n + 2\pi + E_{\max} - 5\log n). \end{aligned}$$

The length of A is 1.

B: Here $t = x + i$ with $x_0 \leq x \leq x_1$, so $\tau = t/n$ lies in $[0.025, 0.0696] \times \{\frac{1}{n}\}$, and Lemma 4.28(ii) gives $|R_n K| \leq \kappa_K e^{-0.7812n + E_{\max} n^{-5}}$. The length of B is at most $0.0446n$.

C: Here $t = x_1 + iy$ with $y \geq 1$. For $y \leq 0.26n$, Lemma 4.28(iii) gives $|R_n K| \leq \kappa_K e^{-0.7812n + E_{\max} n^{-5}}$. For $y \geq 0.26n$, Lemma 4.29 gives $|R_n K| \leq \kappa_K e^{0.85n - 2\pi y + E_{\max} n^{-5}}$, and $\int_{0.26n}^{\infty} e^{0.85n - 2\pi y} dy \leq e^{-0.7836n} / (2\pi)$.

Altogether, for $n \geq 2000$,

$$\mathcal{J}_K \leq n^{-5} e^{E_{\max}} e^{-0.7812n} \left[\kappa_K^A e^{\pi^2/2 + 2\pi} + \kappa_K \left(0.3046n + \frac{1}{2\pi} \right) \right].$$

Inserting this into (4.13), with $\zeta(7) \leq 1.0084$, $\zeta(9) \leq 1.0021$, $Z_7 = -46080c_3$ and $Z_9 = -860160c_5$, gives the constants of Table 11 and hence (a). Part (b) is the same argument with the sharp versions (i')–(iii') and $n \geq 10^4$ (so that $x_1/n \in [0.0695, 0.0696]$ and $1/n \leq 10^{-4}$), using $0.8215 - 2\pi \cdot 0.26 = -0.8121 < -0.78127$; its constant is $3.31 \cdot 10^{11}$. $\square$

TABLE 11. Constants C in the bounds $|\cdot| \leq C n^{-4} e^{-0.7812n}$ ($n \geq 2000$, $40 \mid n$) of the proof of Theorem 4.21(a). An independent interval recomputation gives $1.5548 \cdot 10^{12}$ for $|\rho_0|$; the constant of part (b) was recomputed as $3.3070 \cdot 10^{11}$.

$ V_2 $	$ V_4 $	$ I_n $	$ Z_7 $	$ Z_9 $	$ \rho_0 $
$3.61 \cdot 10^8$	$3.45 \cdot 10^9$	$8.82 \cdot 10^{10}$	$5.09 \cdot 10^{11}$	$9.59 \cdot 10^{11}$	$1.56 \cdot 10^{12}$

Remark 4.30 (Sharpness and robustness). (1) The exponent -0.78127 is an upper bound, and it is not proved that it is the true rate. Numerically, the contour representation (which reproduces the exact coefficients for $n \leq 640$) gives $\frac{1}{n} \log \max(|\rho_0|, |Z_7|, |Z_9|) = -0.7954, -0.7900, -0.7864, -0.7829, -0.7818$ at $n = 10^3, 2 \cdot 10^3, 4 \cdot 10^3, 1.6 \cdot 10^4, 6.4 \cdot 10^4$, converging to $\Psi(\tau^*) = -0.7812761397$. Part (a) of Theorem 4.21 gives $-0.78237, -0.78218, -0.78145$ at $n = 2000, 8000, 64000$, against the values $-0.78996, -0.78427, -0.78179$ computed from the contour representation; the integral $\mathcal{J}_K$ itself tracks the true size, so the path Γ_n is essentially a path of steepest descent. (2) The proof of Theorem A needs only $g < -0.5921$ (§4.6). Even the weakest set of inequalities certified without the centered form and without `mpmath` gives $g = -0.75$: namely $P \leq 0.9$ on $[-0.5, 0.15]$, $\Psi \leq -0.75$ on $[0.025, 0.070] \times [0, 0.001]$ and $\Psi \leq -0.75$ on $[0.069, 0.070] \times [0, 0.27]$; with the height 0.27 one has $0.9 - 2\pi \cdot 0.27 = -0.7965 < -0.75$ above the box. (3) Configurations with a zero shift do not work in this way: without a half-integer zero of order 5 to the right of 0, the kernel W cannot cross the real axis at a point $x_0 > \frac{1}{2}$, and the rate of ρ_0 is then pinned at $\text{Re } \varphi(0)$ (for instance -0.3664 and -0.4132 for the two

configurations $(-0.4, 0, 0.025, 0.075, 0.125, 0.175)$ and $(-0.4, 0, 0.05, 0.1, 0.1, 0.15)$. This is why all five non-negative shifts of configuration E are strictly positive.

Remark 4.31 (Formalization). In Lean, `Stmt_Growth configE gE` is proved with $g_E = -18/25 = -0.72$ by a variant of the argument above: the kernels are $K_s(t) = \sum_{\nu \in \mathbb{Z}} (t - \nu - \frac{1}{2})^{-s}$ for $s = 2, 4, 5$; the line $\text{Re } t = n/40$ is used up to height 1, and above height 1 the contour is shifted to $\text{Re } t = 0.07n$; the pointwise bound is $|R_n(t)| \leq |2t + n|(|t| + 2n)^6 \exp(n \text{Re } \varphi(t/n))$ for $\text{Re } t \geq 0$, $\text{Im } t \geq 1$. The two landscape inequalities needed are $P(\xi) \leq -18/25$ for $\xi \in [1/40, 7/100]$ (true maximum -0.80049) and $\Psi(\frac{7}{100} + i\eta) \leq -18/25$ for $\eta \in (0, 1]$ (true maximum -0.78127). They are proved in Lean by explicit subdivisions with rational enclosures of log and arctan, checked by the kernel. So the growth statement used by the formal proof is machine-checked, including its numerical part.

4.5. Nonvanishing: the auxiliary prime $\ell = n - 1$. Let

$$\begin{aligned} \mathcal{N} &:= \{n \in 40\mathbb{Z}_{>0} : n - 1 \text{ is prime, } n \neq 91080\} \\ &= \{p + 1 : p \text{ prime, } p \equiv 39 \pmod{40}, p \neq 91079\}. \end{aligned}$$

By Dirichlet's theorem $\mathcal{N}$ is infinite; its smallest element is 80.

Theorem 4.32 (ℓ -adic separation). *Let $n \in \mathcal{N}$ and $\ell := n - 1$. Then*

$$v_\ell(\rho_0) = -9, \quad v_\ell(Z_7) \geq -3, \quad v_\ell(Z_9) \geq -1.$$

More precisely, $\ell^9 \rho_0 \equiv 168 U_0 K_E \pmod{\ell}$, where

$$\begin{aligned} K_E &:= -8 + 4 \sum_{m=1}^6 \frac{1}{1 + 2\eta_m} = \frac{12386744}{345345} = \frac{2^3 \cdot 17 \cdot 91079}{3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 23}, \\ U_0 &:= \frac{n \prod_{m=1}^6 \prod_{c \in C_m, c \neq n/2-1} (c + \frac{1}{2})}{\prod_{1 \leq j \leq n, j \neq n-1} j^6}, \end{aligned}$$

and U_0 is an ℓ -adic unit.

Here $4 \sum_m (1 + 2\eta_m)^{-1} = 80 \Sigma_E$ with $\Sigma_E = \frac{1}{3} + \frac{1}{21} + \frac{1}{22} + \frac{1}{23} + \frac{1}{25} + \frac{1}{26} = \frac{946844}{1726725}$, since $20(1 + 2\eta_m) \in \{3, 21, 22, 23, 25, 26\}$. The number 91079 is prime and $\equiv 39 \pmod{40}$; this is why $n = 91080$ is excluded.

Throughout the proof, $n \in \mathcal{N}$, $\ell = n - 1 \geq 79$, and $l^* := \frac{n}{2} - 1$, so that $l^* + \frac{1}{2} = \frac{\ell}{2}$. All integers below lie in $(-3\ell, 3\ell)$; hence each has $v_\ell \in \{0, 1\}$, and an odd one is divisible by ℓ only if it equals $\pm\ell$.

- (S1) Pole values: $0 < |j - k| \leq n = \ell + 1$, so $\ell \mid j - k$ only if $\{j, k\} = \{0, \ell\}$ or $\{1, n\}$.
- (S2) Numerator values $2(c - k) + 1$ ($c \in C_m$) are odd with $|2(c - k) + 1| \leq 2.3n - 1 < 3\ell$; they equal ℓ iff $c = k + \frac{n}{2} - 1$, and $-\ell$ iff $c = k - \frac{n}{2}$. These values of c lie in C_m iff $k \leq \frac{n}{2} + h_m$, respectively $k \geq \frac{n}{2} - h_m$ (the other end-point conditions hold automatically, since $\frac{n}{2} - 1 \geq \frac{17n}{40}$ and $\frac{3n}{2} - \frac{17n}{40} - 1 \geq n$).
- (S3) $n - 2k$ is even and $|n - 2k| \leq n < 2\ell$, so it is an ℓ -unit unless $k = \frac{n}{2}$.
- (S4) $2l + 1 \in [1, 2n - 1] = [1, 2\ell + 1]$, so $\ell \mid 2l + 1$ only for $l = l^*$.
- (S5) $(i)_4$ ($1 \leq i \leq 6$), 46080, 860160, 168 and 40 are ℓ -units.

Lemma 4.33 (Local shape at ℓ). *For $0 \leq k \leq n$ put $e_k := \sum_{m=1}^6 ([k \leq \frac{n}{2} + h_m] + [k \geq \frac{n}{2} - h_m])$.*

- (a) If $2 \leq k \leq n-2$, then $v_\ell(r_{i,k}) \geq i-1$ for $1 \leq i \leq 6$.
 (b) If $k \in \{0, 1\}$, then

$$(4.16) \quad G_k(\varepsilon) = Q(\varepsilon) U_k(\varepsilon), \quad Q(\varepsilon) := \left(\frac{\varepsilon + \ell/2}{\varepsilon + \ell} \right)^6 = 2^{-6} \left(\frac{1 + 2\varepsilon/\ell}{1 + \varepsilon/\ell} \right)^6,$$

with $U_k \in \mathbb{Z}_{(\ell)}[[\varepsilon]]$ and $U_k(0) \in \mathbb{Z}_{(\ell)}^\times$, and $U_0(0) = U_0$ is the unit of Theorem 4.32. If $k \in \{n-1, n\}$, then $G_k(\varepsilon) = Q(-\varepsilon)U_k(\varepsilon)$ with U_k of the same kind. In these four cases $v_\ell(r_{i,k}) \geq i-6$.

- (c) $v_\ell(c_i) \geq i-6$ for all i ; hence $v_\ell(Z_7) \geq -3$ and $v_\ell(Z_9) \geq -1$.

Proof. (a) By (S1) no pole value is divisible by ℓ . If $k \leq \frac{n}{2}$, every m with $h_m \geq 0$ satisfies $k \leq \frac{n}{2} + h_m$, and if $k \geq \frac{n}{2}$, every such m satisfies $k \geq \frac{n}{2} - h_m$; so $e_k \geq 5$, and by (S2) G_k contains e_k numerator factors $\varepsilon \pm \frac{\ell}{2}$. Write $G_k = \prod_{s=1}^{e_k} (\varepsilon \pm \frac{\ell}{2}) \cdot V_k$: all remaining factors ($n-2k+2\varepsilon$, or 2ε if $k = \frac{n}{2}$; $\varepsilon + \beta$ with $\beta \in \mathbb{Z}_{(\ell)}$; and $(\varepsilon + j - k)^{-6} = (j - k)^{-6} (1 + \frac{\varepsilon}{j-k})^{-6}$ with $j - k$ an ℓ -unit) lie in $\mathbb{Z}_{(\ell)}[[\varepsilon]]$. The coefficient of ε^ν in the first product is an elementary symmetric function of degree $e_k - \nu$ in numbers of valuation 1. Hence $v_\ell([\varepsilon^\mu]G_k) \geq e_k - \mu \geq 5 - \mu$, i.e. $v_\ell(r_{i,k}) \geq i-1$.

(b) Let $k \in \{0, 1\}$. Since $\frac{n}{2} + h_1 = \frac{3n}{40} \geq 6$ and $\frac{n}{2} - h_6 = \frac{7n}{20} > 1$, each block contributes exactly one factor $\varepsilon + \frac{\ell}{2}$ (from $c = k + \frac{n}{2} - 1$) and none equal to $\varepsilon - \frac{\ell}{2}$; the only pole value divisible by ℓ is $j - k = \ell$ (from $j = k + \ell \leq n$), with exponent -6 . This gives (4.16) with $U_k \in \mathbb{Z}_{(\ell)}[[\varepsilon]]$. Its constant term is an ℓ -unit: $n - 2k \in \{n, n-2\}$ is $\equiv \pm 1 \pmod{\ell}$, the remaining numerator values are odd, different from $\pm\ell$ and in $(-3\ell, 3\ell)$, and the remaining pole values are different from $0, \pm\ell$. For $k = 0$, $U_0(0) = 64 G_0(0)$ is the displayed product. For $k \in \{n-1, n\}$ use $G_{n-k}(\varepsilon) = -G_k(-\varepsilon)$ (Lemma 4.1(c)). Finally $[\varepsilon^\mu]Q(\pm\varepsilon) = (\pm 1)^\mu 2^{-6} \alpha_\mu \ell^{-\mu}$, where

$$\alpha_\mu := [x^\mu] \left(\frac{1+2x}{1+x} \right)^6 : \quad (\alpha_0, \dots, \alpha_6) = (1, 6, 9, -4, -6, 12, -10),$$

so $v_\ell([\varepsilon^\mu]G_k) \geq -\mu$, i.e. $v_\ell(r_{i,k}) \geq i-6$.

- (c) Sum (a) and (b) over k , and use (S5). □

For a power series G put

$$\Lambda[G] := \sum_{i=1}^6 (i)_4 [\varepsilon^{6-i}] G \cdot \left(-\frac{\ell}{2} \right)^{-(i+4)}.$$

Note that $\Lambda[G_k] = R_{n,k}^{(4)}(-k - \frac{\ell}{2})$, the value at $t + k = -\frac{\ell}{2}$.

Lemma 4.34 (Reduction to two terms). $\rho_0 \equiv X(n-1, l^*) + X(n, l^*) = -(\Lambda[G_0] + \Lambda[G_1]) \pmod{\ell^{-5}\mathbb{Z}_{(\ell)}}$.

Proof. Write $\rho_0 = \sum_{0 \leq l < k \leq n} X(k, l)$. If $l \neq l^*$, then $l + \frac{1}{2}$ is an ℓ -unit by (S4), and $v_\ell(X(k, l)) \geq \min_i v_\ell(r_{i,k}) \geq -5$ by Lemma 4.33. If $l = l^*$ and $\frac{n}{2} \leq k \leq n-2$, then $v_\ell(X(k, l^*)) \geq \min_i ((i-1) - (i+4)) = -5$. The two remaining terms satisfy exact identities: by (4.9) and (4.6),

$$\begin{aligned} X(n, l^*) &= R_{n,n}^{(4)}\left(\frac{\ell}{2} - n\right) = -R_{n,0}^{(4)}\left(-\frac{\ell}{2}\right) = -\Lambda[G_0], \\ X(n-1, l^*) &= R_{n,n-1}^{(4)}\left(-\frac{\ell}{2}\right) = -R_{n,1}^{(4)}\left(-1 - \frac{\ell}{2}\right) = -\Lambda[G_1]. \end{aligned}$$

□

The same congruence also follows from Lai's root trick [20, Lemma 4.6] (compare [21, Lemma 5.6]): the two central half-integers $-\frac{1}{2} - (\frac{n}{2} - 1)$ and $-\frac{1}{2} - \frac{n}{2}$ are zeros of R_n of order 6 (Lemma 4.1(b)), so $R_n^{(4)}$ vanishes there.

Lemma 4.35 (Universal local factor). *For $k \in \{0, 1\}$ put $\gamma_{k,\lambda} := [\varepsilon^\lambda]U_k$. Then*

$$\Lambda[G_k] = \sum_{\lambda=0}^5 \beta_\lambda \ell^{\lambda-10} \gamma_{k,\lambda}, \quad \beta_\lambda := 2^{-6} \sum_{i=1}^{6-\lambda} (i)_4 (-2)^{i+4} \alpha_{6-i-\lambda},$$

with $(\beta_0, \dots, \beta_5) = (0, -168, 168, -108, 48, -12)$. Hence

$$\Lambda[G_0] + \Lambda[G_1] \equiv -168 \ell^{-9} (\gamma_{0,1} + \gamma_{1,1}) \pmod{\ell^{-8} \mathbb{Z}_{(\ell)}}.$$

Proof. The functional Λ is linear and depends only on the coefficients of $\varepsilon^0, \dots, \varepsilon^5$, and $G_k = \sum_\lambda \gamma_{k,\lambda} \varepsilon^\lambda Q$. Since

$$\begin{aligned} [\varepsilon^{6-i}](\varepsilon^\lambda Q) &= 2^{-6} \alpha_{6-i-\lambda} \ell^{-(6-i-\lambda)}, \\ \left(-\frac{\ell}{2}\right)^{-(i+4)} &= (-2)^{i+4} \ell^{-(i+4)}, \end{aligned}$$

we get $\Lambda[\varepsilon^\lambda Q] = \beta_\lambda \ell^{\lambda-10}$. The values of β_λ are direct arithmetic; for instance

$$\begin{aligned} 64 \beta_1 &= 24 \cdot (-32) \cdot (-6) + 120 \cdot 64 \cdot (-4) + 360 \cdot (-128) \cdot 9 \\ &\quad + 840 \cdot 256 \cdot 6 + 1680 \cdot (-512) \cdot 1 = -10752. \end{aligned}$$

There is also a conceptual reason for $\beta_0 = 0$: the rational function $Q(\varepsilon)\varepsilon^{-6}$ has degree -6 , poles only at 0 and $-\ell$, is invariant under $\varepsilon \mapsto -\ell - \varepsilon$, and has a zero of order 6 at $-\frac{\ell}{2}$. Writing it as $\mathcal{P}(\varepsilon) + \mathcal{P}(-\ell - \varepsilon)$, with $\mathcal{P}$ its principal part at 0, and differentiating four times at $-\frac{\ell}{2}$ gives $0 = 2 \mathcal{P}^{(4)}(-\frac{\ell}{2}) = 2\Lambda[Q]$; here it matters that the order 4 of differentiation is even. The term $\lambda = 1$ gives $-168 \ell^{-9} \gamma_{k,1}$, and the terms with $\lambda \geq 2$ lie in $\ell^{-8} \mathbb{Z}_{(\ell)}$. $\square$

Lemma 4.36 (The two singular poles). *Let $\Theta_k := U'_k(0)/U_k(0) \in \mathbb{Z}_{(\ell)}$ for $k \in \{0, 1\}$. Then:*

- (a) $\frac{U_1(0)}{U_0(0)} = \frac{r_{6,1}}{r_{6,0}} = \frac{n-2}{n} n^6 \prod_{m=1}^6 \frac{-(2h_m+1)}{2n+2h_m-1} \equiv -1 \pmod{\ell};$
- (b) $\Theta_0 - \Theta_1 = \frac{2}{n} - \frac{2}{n-2} - 6\left(1 + \frac{1}{n}\right) + \sum_{m=1}^6 \left[\frac{2}{2h_m+1} + \frac{2}{2n+2h_m-1} \right];$
- (c) $\Theta_0 - \Theta_1 \equiv K_E \pmod{\ell}.$

Consequently $\gamma_{0,1} + \gamma_{1,1} = U_0(0)\Theta_0 + U_1(0)\Theta_1 \equiv U_0(0) K_E \pmod{\ell}.$

Proof. (a) By (4.16), $U_k(0) = 64 G_k(0) = 64 r_{6,k}$. Let $\text{Num}(t)$ be the product of the six blocks in (4.3). Then

$$\begin{aligned} \text{Num}(t) &= \prod_{m=1}^6 \prod_{c \in C_m} (t + \tfrac{1}{2} + c), & \frac{\text{Num}(-1)}{\text{Num}(0)} &= \prod_{m=1}^6 \frac{-h_m - \frac{1}{2}}{n + h_m - \frac{1}{2}}, \\ r_{6,0} &= \frac{n \text{Num}(0)}{(n!)^6}, & r_{6,1} &= \frac{(n-2) \text{Num}(-1)}{((n-1)!)^6}, \end{aligned}$$

where the product for $\text{Num}(-1)/\text{Num}(0)$ telescopes. Modulo ℓ we have $n \equiv 1$, $n-2 \equiv -1$, and $2n+2h_m-1 = (2h_m+1) + 2\ell \equiv 2h_m+1$, which is an ℓ -unit by the proof of (c).

(b) In (4.16) the six factors $\varepsilon + \frac{\ell}{2}$ and the factor $(\varepsilon + \ell)^{-6}$ are moved into Q ; they contribute exactly $(\log Q)'(0) = 6/\ell$ to $(\log G_k)'(0)$. Hence

$$\Theta_k = \frac{2}{n-2k} + \sum_m \sum_{\substack{c \in C_m \\ c \neq k+n/2-1}} \frac{1}{c-k+\frac{1}{2}} - 6 \sum_{\substack{0 \leq j \leq n \\ j \neq k, k+\ell}} \frac{1}{j-k}.$$

Shifting $c \mapsto c+1$ in the numerator sum of Θ_1 , the two index sets differ only in the end points $n+h_m-1$ and $-h_m-1$; the pole sums are $H_n - \frac{1}{n-1}$ and $-1 + H_{n-2}$, where $H_m = \sum_{j \leq m} 1/j$. This gives (b).

(c) Since $n \equiv 1 \pmod{\ell}$ and 40 is an ℓ -unit, $h_m = \frac{n}{40} \cdot 40\eta_m \equiv \eta_m$, so $2h_m + 1 \equiv 1 + 2\eta_m = \frac{20(1+2\eta_m)}{20}$, where $20(1+2\eta_m) \in \{3, 21, 22, 23, 25, 26\}$ are ℓ -units ($\ell \geq 79$). Hence $\Theta_0 - \Theta_1 \equiv 2 + 2 - 12 + 4 \sum_m (1 + 2\eta_m)^{-1} = K_E$. The last assertion follows from $\gamma_{k,1} = U_k(0)\Theta_k$ and (a). $\square$

Proof of Theorem 4.32. By Lemmas 4.34–4.36,

$$\begin{aligned} \rho_0 &\equiv -(\Lambda[G_0] + \Lambda[G_1]) \equiv 168 \ell^{-9} (\gamma_{0,1} + \gamma_{1,1}) \\ &\equiv 168 \ell^{-9} U_0(0) K_E \pmod{\ell^{-8} \mathbb{Z}(\ell)}. \end{aligned}$$

Here $168 = 2^3 \cdot 3 \cdot 7$ and $U_0(0)$ are ℓ -units, and so is K_E , because $\ell \geq 79$ and $\ell \neq 91079$. Hence $v_\ell(\rho_0) = -9$ and $\ell^9 \rho_0 \equiv 168 U_0 K_E \pmod{\ell}$. The bounds for Z_7, Z_9 are Lemma 4.33(c). $\square$

Remark 4.37. (1) *Exact checks.* For all 25 values $n \in \mathcal{N}$ with $n \leq 3000$, exact computation gives $v_\ell(\rho_0) = -9$, $v_\ell(c_3) = -2$, $v_\ell(c_5) = 0$ (one better than Lemma 4.33), and $\ell^9 \rho_0 \bmod \ell$ equals $168 U_0 K_E \bmod \ell$ (the residues are 43, 66, 167, 233, 24 for $n = 80, 200, 240, 360, 440$). An ℓ -adic computation confirms $v_\ell(\rho_0) = -9$ for all 243 values $n \in \mathcal{N}$ in $[3080, 40000]$ and for six values above 91080; at the excluded value $n = 91080$ it finds $v_\ell(\rho_0) = -8$, as predicted by $\ell \mid K_E$. (2) *Unconditional nonvanishing.* $S_n \neq 0$ for all sufficiently large $n \in \mathcal{N}$. Indeed, the $\mathbb{Q}$ -space V of relations $x_0 + x_1 \zeta_2(7) + x_2 \zeta_2(9) = 0$ contains no nonzero vector $(x_0, 0, 0)$, so there are $\alpha, \beta \in \mathbb{Q}$ with $x_0 = \alpha x_1 + \beta x_2$ on V . If $S_n = 0$ and ℓ does not divide the denominators of α, β , then $\rho_0 = \alpha Z_7 + \beta Z_9$ has $v_\ell(\rho_0) \geq -3$, a contradiction. This is ineffective, and the proof of Theorem A below obtains nonvanishing directly. (3) *Generality.* The argument uses neither the growth nor the denominator estimates. We have checked, but not written out, that the same proof applies to every even $a \geq 4$ and odd $j \leq a-3$, with rational η , $\sum_m \eta_m = 0$, $|\eta_m| < \frac{1}{2}$ and at least $j+2$ non-negative η_m : for all large n with $\ell = n-1$ prime, except for the finitely many primes dividing the numerators of $\beta_1(a, j)$, of the analogue of K_E and of the numbers $1 + 2\eta_m$, it gives $v_\ell(\rho_0) = -(a+j)$, while $v_\ell(c_i) \geq i-a$. The constant $\beta_1(a, j)$ is non-zero for all even $a \leq 30$ and odd $j \leq a-3$. For the undeformed family $(a, j) = (8, 3)$ of Theorem B this gives $v_{n-1}(\rho_0) = -11$ (confirmed numerically at $n = 20, 30, 60, 102, 138, 200$), and thus a possible second nonvanishing mechanism for Theorem B, along $n = \ell + 1$ instead of $n = 2^m - 1$. (4) *Formalization.* In Lean the nonvanishing statement is proved through the arithmetic condition **Stmt_LaiSprangCond**: for every bound $B \in \mathbb{N}$ there are infinitely many m with a prime $\ell > B$ that separates ρ_0 from Z_7, Z_9 (in the proof of Theorem A, B bounds the common denominator b of the assumed rational values). The witnesses are the

primes $\ell = 40m - 1$ with $\ell > \max(B, 10^9)$; the bound $\ell > 10^9$ replaces the conditions $\ell \geq 79$, $\ell \neq 91079$. The proof follows the one above, with the exact parity relation (4.6) in Lemma 4.34.

4.6. Proof of Theorem A.

Proof of Theorem A. Suppose, for a contradiction, that $\zeta_2(7) = u_7/b$ and $\zeta_2(9) = u_9/b$ with $u_7, u_9 \in \mathbb{Z}$ and $b \in \mathbb{Z}_{\geq 1}$. For $n \in \mathcal{N}$ let D_n be as in §4.3, choose $e_n \geq 0$ such that $a_{n,\cdot} := 2^{e_n} D_n \cdot (\rho_0, Z_7, Z_9) \in \mathbb{Z}^3$, put $L_n := a_{n,0} + a_{n,1}\zeta_2(7) + a_{n,2}\zeta_2(9) = 2^{e_n} D_n S_n$ and $M_n := \max(|\rho_0|, |Z_7|, |Z_9|)$.

Step 1: a nonzero integer. The number $bL_n = b a_{n,0} + u_7 a_{n,1} + u_9 a_{n,2}$ is an integer. Let $n \in \mathcal{N}$ with $\ell := n - 1 > b$. Since $\ell \nmid b$, $\ell \nmid 2^{e_n}$ and $D_n \in \mathbb{Z}$, Theorem 4.32 gives

$$\begin{aligned} v_\ell(b a_{n,0}) &= v_\ell(D_n) - 9, \\ v_\ell(u_7 a_{n,1}) &\geq v_\ell(D_n) - 3, \quad v_\ell(u_9 a_{n,2}) \geq v_\ell(D_n) - 1. \end{aligned}$$

By the ultrametric inequality $v_\ell(bL_n) = v_\ell(D_n) - 9 < \infty$; so $bL_n \neq 0$, and in particular $S_n \neq 0$.

Step 2: a Liouville-type inequality. A nonzero integer N satisfies $|N| \cdot |N|_2 \geq 1$. Here $|bL_n| \leq (b + |u_7| + |u_9|) \max_i |a_{n,i}| = (b + |u_7| + |u_9|) 2^{e_n} D_n M_n$, and, since D_n is odd, $|bL_n|_2 \leq 2^{-e_n} |S_n|_2 = 2^{-e_n} 2^{-v_2(S_n)}$. Hence, for all $n \in \mathcal{N}$ with $n > b + 1$,

$$(4.17) \quad 1 \leq (b + |u_7| + |u_9|) \cdot D_n \cdot M_n \cdot 2^{-v_2(S_n)}.$$

The power of 2 has cancelled exactly; only the odd part D_n matters.

Step 3: the right-hand side tends to 0. For $n \in \mathcal{N}$ with $n \geq 2000$:

- $\log D_n \leq (10 - R^{(0)})n + o(n)$ by Theorem 4.17, which holds for $n \rightarrow \infty$ through $40\mathbb{Z}$ and hence along $\mathcal{N}$;
- $\log M_n \leq -0.7812n - 4 \log n + \log(1.56 \cdot 10^{12})$ by Theorem 4.21(a);
- $v_2(S_n) \log 2 \geq 12n \log 2 - (1 + 10 \log_2(1.3n)) \log 2$ by Corollary 4.5.

So the logarithm of the right-hand side of (4.17) is at most

$$n(10 - R^{(0)} - 0.7812 - 12 \log 2) + o(n) = -0.18909n + o(n) \rightarrow -\infty$$

as $n \rightarrow \infty$ in $\mathcal{N}$. Since $\mathcal{N}$ is infinite, this contradicts (4.17). $\square$

Remark 4.38 (Logic and slack). Step 1 is condition (2) of Lemma 2.10 with $\ell(n) = n - 1$, and Steps 2 and 3 are Lemma 2.9; writing them out avoids any mismatch of hypotheses. The four ingredients are used only in weak forms, each with a large safety margin: with $R = R^{(0)}$ the growth exponent must be < -0.5921 and is ≤ -0.7812 ; the denominator savings must exceed $10 - 12 \log 2 + g$ ($= 0.9009638$ for $g = -0.78127$) and are $R^{(0)} = 1.0901$ ($R_E = 1.4517$ with the refinements claimed in Remark 4.19); the 2-adic bound must be $12n - o(n)$ and is $12n - O(\log n)$; and any strict ℓ -adic separation suffices, while Theorem 4.32 gives $-9 < -3$.

The total margin. Replacing $R^{(0)}$ by the sharper constants claimed in Remark 4.19 (proofs not included), and -0.7812 by the exponent $g = -0.78127$ of Theorem 4.21(b), the same computation gives, along $\mathcal{N}$,

$$\begin{aligned} \limsup_{n \in \mathcal{N}} \frac{1}{n} \log \left(\max_i |a_{n,i}| \cdot |L_n|_2 \right) &= \limsup_{n \in \mathcal{N}} \frac{1}{n} \log (D_n M_n 2^{-v_2(S_n)}) \\ &\leq 10 - 12 \log 2 + g - R, \end{aligned}$$

with $10 - 12 \log 2 = 1.682233833280657$. Table 12 lists the values. At $g = -0.78127$ the exponent is negative as long as $R > 0.9009638$, i.e. as long as $\log D_n \leq (9.0990362 - \epsilon)n$; at $R = R_E$ it is negative as long as $g < -0.2305288$.

TABLE 12. The exponent of the proof of Theorem A along $\mathcal{N}$. Only row (1) is needed: it uses the residue-level bound of Theorem 4.17; the proof in §4.6 uses the slightly weaker growth exponent -0.7812 of Theorem 4.21(a), which gives -0.18909 . Rows (2)–(4) use the refinements claimed in Remark 4.19, whose proofs are not included (second-order lemma and anti-diagonal switch; row cancellation on $[1, 11]$, respectively $[1, 41]$), and row (5) uses the numerically determined limit $g^* = \Psi(\tau^*) = -0.7812761397$ of the growth rate, which is not proved.

denominator bound	R	g	$10 - 12 \log 2 + g - R$
(1) residue level	1.0901241927613	-0.78127	-0.1891604
(2) + second order, switch	1.2901241927613	-0.78127	-0.3891604
(3) + rows on $[1, 11]$	1.4322937593059	-0.78127	-0.5313299
(4) + rows on $[1, 41]$	1.4517049996220	-0.78127	-0.5507412
(5) as (4), $g = g^*$	1.4517049996220	-0.7812761	-0.5507473

Remark 4.39 (Effectivity and finite n). The $o(n)$ of Theorem 4.17 comes from the prime number theorem, so the proof gives no explicit n_0 . At finite n the residue-level bound is far from the truth: at $n = 2000$ and $n = 2400$ (both in $\mathcal{N}$), Theorem 4.13 gives only $\frac{1}{n} \log D_n \leq 9.433$ and 9.386 , and the exponent would be $+0.356$ and $+0.306$. The crude small-prime bound of Theorem 4.13(b) costs 0.94 per n at $n = 2000$, against an asymptotic share of about 0.30 , and this cost decays only like $n^{-1/2}$ (0.14 at $n = 10^5$, 0.014 at $n = 10^7$). With the per-prime refinements of Remark 4.19, which are claimed to hold as inequalities at each fixed n (proofs not included here, checked only by AI agents), one gets $\frac{1}{n} \log(\max_i |a_{n,i}| \cdot |L_n|_2) \leq -0.7824 + 8.5730 - 8.2945 = -0.5039$ at $n = 2000$ and $\leq -0.7825 + 8.5580 - 8.2978 = -0.5223$ at $n = 2400$; with the true odd denominators ($\frac{1}{n} \log D_n = 8.5437$ and 8.5318) these become -0.533 and -0.548 . Table 13 gives the exact value of the quantity, computed with the true D_n , M_n and $v_2(S_n)$, at the first elements of $\mathcal{N}$; in each case $S_n \neq 0$.

TABLE 13. Exact values along $\mathcal{N}$ (true odd denominators, true sizes, true 2-adic valuations).

n	$\frac{1}{n} \log(D_n M_n 2^{-v_2(S_n)})$	$12n - v_2(S_n)$
80	-1.057	25
200	-0.646	35
240	-0.695	27
360	-0.697	37
440	-0.666	34
480	-0.658	29
600	-0.696	41
720	-0.656	39

Remark 4.40 (The formal proof). The Lean theorem

`Zeta2.Pair.zeta2_7_9_not_both_rational_unconditional`

follows the same route with weaker constants: $v_2(S_n) \geq 12n - 10\lceil \log_2 6n \rceil$ (Remark 4.10), odd denominators $\leq e^{(9+\epsilon)n}$ for large n (proved with the constant 8.97), coefficients $\leq e^{(-0.72+\epsilon)n}$ (Remark 4.31), and the ℓ -adic separation at $\ell = n - 1 > 10^9$. The margin condition is $g + \delta = -0.72 + 9 = 8.28 < 12 \log 2 = 8.3178$. The prime number theorem is proved in the formalization, and the only cited input is the identification of `zeta2` with the Kubota–Leopoldt value (Remark 2.5).

4.7. Comments on the construction. *Why configuration E.* The shift vector was found by numerical exploration. Configuration E has three features. First, $\sum_m h_m = 0$, which keeps $\deg R_n = -5$ and the 2-adic rate $12 \log 2$. Second, it has five strictly positive shifts $h_m \geq n/40$; they produce the critical zeros, which remove all primes $q > n$ from D_n and allow the contour of §4.4 to cross the real axis. Third, it has one strongly negative shift $h_1 = -17n/40$, which produces the prime-selective savings $R^{(0)} > 0$ (Table 9). A search over 8619 configurations on the grid $0.0125\mathbb{Z}$ found configurations with a better residue-level exponent (with estimated growth rates), for instance

$$\eta = (-0.5, 0.025, 0.0625, 0.1, 0.1375, 0.175),$$

with -0.260 against -0.190 for E. For that configuration an analogue of Theorem 4.21 (with the exponent -0.88 , for $80 \mid n$) was also derived in the working notes (not included here), but its third-order denominator savings are smaller, and with all the ingredients of the working notes E is the better of the two (-0.551 against about -0.33).

Generality. Theorem 4.4 holds for all admissible shift vectors and all (a, j, ι) . The denominator lemmas hold for every rational η with $\sum_m \eta_m = 0$, all $\eta_m \geq -\frac{1}{2}$, and at least five non-negative entries; only the numerical value of $R^{(0)}$ is specific to E. The growth argument applies to every configuration whose saddle-point and crossing values can be certified. For the ℓ -adic separation see Remark 4.37(3).

Relation with Theorem B. Theorem A implies Theorem B. The two proofs share the linear-form mechanism (Lemma 4.2) and the 2-adic Δ -calculus, but their other ingredients differ: the proof of Theorem B uses the Andrews transformation for the denominators, a direct bound for the coefficient growth, and a 2-adic dominant term along $n = 2^m - 1$ for the nonvanishing; the proof of Theorem A uses Legendre counting with critical zeros, a certified saddle-point bound, and the auxiliary prime $\ell = n - 1$.

5. FORMAL VERIFICATION

Both theorems are formalized in Lean 4 [27] with Mathlib [26]. The formalization of Theorem B [13] came first. The formalization of Theorem A started as a copy of it, and the two share their infrastructure, as follows.

- The six definitions of §5.2 are verbatim identical in both projects.
- In the directory `Zeta2Lean/Proofs/`, the five files `JConvergence`, `Translation`, `Criterion`, `DeltaCalculus` and `DeltaFunctions` (extension `.lean`; the convergence of J_s , the translation formula, Lai’s criterion and the Δ -calculus) are byte-identical, and so are the vendored files for the prime number theorem.

- The wrapper for the prime number theorem has identical statements and proofs, but is organized differently: for Theorem A it is the single file `Zeta2Lean/Cited/PNT.lean`, for Theorem B it consists of the files `Statements`, `Assembly` and `Main` of the directory `Zeta2Lean/Cited/`.
- The files `Zeta2Lean/Defs.lean` and `Zeta2Lean/Statements.lean` of the formalization of Theorem A are trimmed versions of those of Theorem B.

This section describes what exactly is proved, what has to be trusted, and how to check it. Lean names are quoted verbatim, and theorems are displayed with their full names (in the source files they are stated inside the namespaces `Zeta2` and `Zeta2.Pair`); in the displayed Lean code, the Unicode symbols of the source are typeset as mathematical symbols.

5.1. The statements. The main theorem of the formalization of Theorem A (file `Zeta2Lean/Pair/Unconditional.lean`) is

```
theorem Zeta2.Pair.zeta2_7_9_not_both_rational_unconditional :
  (∀ s : ℕ, HasVolkenborn (halfPow s) (J s)) ∧
  ¬ ((∃ q : ℚ, zeta2 7 = q) ∧ (∃ q : ℚ, zeta2 9 = q))
```

The formal statement of Theorem B, in the form registered with Palomar (file `Solution.lean`; the last line break is ours), is

```
theorem Zeta2.zeta2_7_9_11_not_all_rational_palomar :
  (∀ s : ℕ, HasVolkenborn (halfPow s) (J s)) ∧
  ¬ ((∃ q : ℚ, zeta2 7 = q) ∧ (∃ q : ℚ, zeta2 9 = q) ∧
    (∃ q : ℚ, zeta2 11 = q))
```

It is proved by the theorem

```
theorem Zeta2.zeta2_7_9_11_not_all_rational_unconditional : MainStatement
```

of the file `Zeta2Lean/Cited/Main.lean`, whose type `MainStatement` unfolds to the same proposition. Similarly, `Zeta2.Pair.zeta2_7_9_not_both_rational_palomar` restates Theorem A in the Palomar format.

In these statements `q` is coerced from `ℚ` to Mathlib’s field `ℚ_[2]` of 2-adic numbers, so “`∃ q : ℚ, zeta2 7 = q`” says that `zeta2 7` is rational in the sense of §1.2. The second conjunct is therefore Theorem A, respectively Theorem B, for the numbers `zeta2 s` defined below. The first conjunct says that for every `s` the Riemann sums of $(x + \frac{1}{2})^{-s}$ converge in $\mathbb{Q}_2$ to `J s`. It is there because `J s` is defined with Mathlib’s `limUnder`, which returns an unspecified value when the limit does not exist; thanks to the first conjunct, no such value enters the statements.

5.2. The trusted definitions. The statements use six definitions of the projects (namespace `Zeta2`, file `Zeta2Lean/Defs.lean`; these six definitions are verbatim identical in both projects, with the namespaces `Filter`, `Topology`, `Finset` and `PowerSeries` opened):

```
def volkenbornSum (f : ℕ → ℚ_[2]) (N : ℕ) : ℚ_[2] :=
  ((2 : ℚ_[2]) ^ N)-1 * ∑ x ∈ range (2 ^ N), f x

def HasVolkenborn (f : ℕ → ℚ_[2]) (I : ℚ_[2]) : Prop :=
  Tendsto (volkenbornSum f) atTop (ℕ I)

def volkenborn (f : ℕ → ℚ_[2]) : ℚ_[2] :=
  limUnder atTop (volkenbornSum f)
```

```

def halfPow (s x : ℕ) : ℚ_[2] :=
  ((x : ℚ_[2]) + 2-1)-1 ^ s

def J (s : ℕ) : ℚ_[2] :=
  volkenborn (halfPow s)

def zeta2 (s : ℕ) : ℚ_[2] :=
  J (s - 1) / (((s : ℚ_[2]) - 1) * 2 ^ s)

```

Here $\mathcal{N}I$ is Mathlib’s notation for the neighborhood filter of I (unrelated to the index set $\mathcal{N}$ of §4.5), so `HasVolkenborn f I` says that the Riemann sums (2.1) of f (for $p = 2$) converge to I . Only the values of f at natural numbers enter, and these are all that (2.1) uses. Hence `J s` is the integral J_s of (2.2), and `zeta2 s` = $J_{s-1}/((s-1)2^s)$; in particular `zeta2 7` = $J_6/768$, `zeta2 9` = $J_8/4096$ and `zeta2 11` = $J_{10}/20480$.

These are the only definitions of the projects that occur in the statements. A reader who wants to know what has been proved needs to check only them, the two statements, and the Mathlib notions they use. All other definitions (the rational functions, ρ_0 , Z_s , configuration E, the rates) are internal to the proofs: if one of them differed from the text, the formal theorems would still hold as stated. The file `Challenge.lean` of each project imports only Mathlib and contains verbatim copies of the six definitions and of the statement, with the proof replaced by `sorry`; `Comparator` checks that `Solution.lean` proves exactly this statement (§5.4).

5.3. What is cited. The identification `zeta2 s` = $\zeta_2(s) = L_2(s, \omega^{1-s})$ for $s \geq 2$ is Lemma 2.8 of [23] (Lemma 2.3 and Remark 2.5). That lemma is a short computation from the formula for $L_2(s, \omega^{1-s})$ in terms of the 2-adic Hurwitz zeta function [23, (2.1)–(2.2)], and it relies on the construction of the Kubota–Leopoldt L -functions through p -adic Hurwitz zeta functions [10, Ch. 11]. It is cited and not formalized, since Mathlib does not yet contain the Kubota–Leopoldt L -functions; a Lean 3 formalization of p -adic L -functions exists [28], but it is not part of Mathlib. It is needed only to read the formal theorems as statements about Kubota–Leopoldt values, not in their proofs. The factor $(s-1)2^s$ is a non-zero rational number, so, given this identification, the formal theorems are equivalent to Theorems A and B, and to the analogous statements for every normalization of $\zeta_2(s)$ that differs by a non-zero rational factor, such as those of Coleman and Furusho.

5.4. Trust base and checks.

- *Lean and Mathlib.* Both projects use Lean in version `v4.35.0-rc2` (the toolchain `leanprover/lean4:v4.35.0-rc2`) and Mathlib at the tag of the same name (commit `065356127b`).
- *Axioms.* `#print axioms` lists exactly `propext`, `Classical.choice` and `Quot.sound` for both main theorems and their Palomar restatements. There is no `sorry` (except the intended one in each `Challenge.lean`), no axiom declaration, and no `native_decide`, `implemented_by` or `extern`; the script `scripts/audit.sh` checks this.
- *Numerical certificates.* In the formalization of Theorem A, the 766 interval certificates for the residue function $\varpi(x)$ on $[1, 41]$ (§4.3) and the resulting constant $T = 8.9607\dots \leq 8.97$ are checked by `decide +kernel`, that is, by reduction in the kernel. The landscape inequalities of the growth bound are proved by explicit subdivisions with rational enclosures of log

and arctan, whose side conditions are checked by `norm_num` on exact rationals (Remarks 4.20 and 4.31). The Python programs that generated these certificates need not be trusted.

- *Vendored code.* The proof of the prime number theorem uses Lean files from open Mathlib pull requests by T. Tao, adapted from the PrimeNumberTheoremAnd project [4]. They are included verbatim, with porting notes and with their Apache-2.0 headers and authors. Being Lean proofs checked by the kernel, they add nothing to the trust base.
- *Kernel replay.* For Theorem A, Lean’s `leanchecker` replayed all 31 modules of the import closure of `Solution.lean`, one module at a time, without error. For Theorem B, continuous integration builds the project, checks the axioms and runs `leanchecker` on every module.
- *Comparator.* For both projects, Comparator [12], run with Lean’s kernel and the independent kernels NanoDa [29] and con-ron, accepted `Solution.lean` against `Challenge.lean`. These local runs were not sandboxed. As negative controls, altered copies of `Challenge.lean` were rejected; for Theorem A, one with `zeta2 11` in place of `zeta2 9` in the theorem and one with $2 \wedge (s + 1)$ in place of $2 \wedge s$ in the definition of `zeta2`. For Theorem B, the registry’s own checks passed before the registration [30].

5.5. Structure of the formal proofs. Both formalizations follow a blueprint. Each lemma of the informal proof is a Lean proposition `Stmt.X`, proved in its own file with the statements it depends on as hypotheses, and an assembly file derives the main theorem from the statements.

For Theorem B the correspondence with Section 3 is the table in Remark 3.19. The formal proof follows the text, including Andrews’ transformation (Theorem 3.6, proved via the $q = 1$ Bailey chain and a polynomial Pfaff–Saalschütz identity) and the prime number theorem; Lemma 3.10 is formalized with unspecified constants.

For Theorem A the correspondence is Table 14. The formal proof follows the residue-level route of Section 4, with the following differences (Remarks 4.3(3), 4.10(3), 4.20, 4.31, 4.37(4) and 4.40):

- (1) S_n is defined as $-\int_{\mathbb{Z}_2} R_n'''(t + \frac{1}{2}) dt$, which changes the signs of ρ_0 , Z_7 , Z_9 ;
- (2) the constants are weaker: $v_2(S_n) \geq 12n - 10\lfloor \log_2 6n \rfloor$ for every admissible shift vector (Remark 4.10(3)); the odd part of the denominators is at most $e^{(9+\epsilon)n}$ for large n (proved with the constant 8.97, the continuum integral being truncated at $x = 81$); and the coefficients are at most $e^{(-0.72+\epsilon)n}$ for large n ;
- (3) the growth bound uses the kernels $K_s(t) = \sum_{\nu \in \mathbb{Z}} (t - \nu - \frac{1}{2})^{-s}$ for $s = 2, 4, 5$, a contour moved to $\text{Re } t = 0.07n$ above height 1, and two landscape inequalities in place of the three of Lemma 4.28;
- (4) the auxiliary prime is $\ell = 40m - 1 > 10^9$, which replaces the conditions $\ell \geq 79$ and $\ell \neq 91079$;
- (5) Lai’s criterion is used in the form “ $|a_{n,j}| \leq B_n$ for all j , $B_n|L_n|_2 \rightarrow 0$ and $L_n \neq 0$ for infinitely many n ”, along all $n = 40m$, together with a nonvanishing statement that assumes the rationality of $\zeta_2(7)$ and $\zeta_2(9)$.

The margin condition of the formal proof is $g + \delta = -0.72 + 9 = 8.28 < 12 \log 2 = 8.3178$. The sharper constants of Section 4 (in particular Theorem 4.21(b) and

Remark 4.19) are not formalized; the theorem does not need them. The largest proof files are `Denominators.lean` (3370 lines), `Growth.lean` (3251 lines) and `Nonvanishing.lean` (1557 lines).

TABLE 14. The formalization of Theorem A. Unless stated otherwise, Lean names are in the namespace `Zeta2.Pair` and proof files in the directory `Zeta2Lean/Pair/Proofs/`.

this paper	Lean statement; proof files	formal version
Lemma 4.2	<code>Stmt.L1</code> ; <code>GenLinearForm</code> , <code>PartialFractions</code> , <code>CoeffVanish</code> , <code>LinearForm</code>	$S_n = \rho_0 + 60c_3J_6 + 210c_5J_8$ (sign convention (1))
Theorem 4.4, Corollary 4.5	<code>Stmt.Valuation</code> ; <code>Valuation</code>	$v_2(S_n) \geq 12n - 10\lfloor \log_2 6n \rfloor$
Theorems 4.13, 4.17	<code>Stmt.Denominators configE</code> <code>deltaE</code> ; <code>Denominators</code> , <code>CrudeIntegrality</code>	odd part $\leq e^{(9+\epsilon)n}$ for large n ; constant $8.9607 \dots \leq 8.97$
Theorem 4.21	<code>Stmt.Growth configE gE</code> ; <code>Growth</code> , <code>Landscape/*</code>	$ \rho_0 , Z_7 , Z_9 \leq e^{(-0.72+\epsilon)n}$ for large n
Theorem 4.32	<code>Stmt.LaiSprangCond</code> , <code>Stmt.Nonvanishing</code> ; <code>Nonvanishing</code> , <code>NonvanishingLS</code>	$v_\ell(\rho_0) = -9$, $v_\ell(Z_7) \geq -3$, $v_\ell(Z_9) \geq -1$ for primes $\ell = 40m - 1 > 10^9$
Lemmas 2.1, 2.7, 2.8, 2.9; (2.2)	<code>Stmt.Translation</code> , <code>Stmt.Delta</code> , <code>Stmt.DeltaFun</code> , <code>Stmt.Criterion</code> , <code>Stmt.JConv</code> (namespace <code>Zeta2</code>); <code>Zeta2Lean/Proofs/</code>	files identical to those of the formalization of Theorem B
(2.4)	<code>PNT.Stmt</code> ; <code>Zeta2Lean/Cited/PNT</code>	$\psi(x)/x \rightarrow 1$, via the Wiener–Ikehara theorem
§4.6	<code>main_of_stmts</code> , <code>marginE</code> ; <code>Zeta2Lean/Pair/Assembly</code>	$g + \delta = 8.28 < 12 \log 2$
Theorem A	<code>zeta2.7.9.not.both.rational.</code> <code>unconditional</code> ; <code>Zeta2Lean/Pair/Unconditional</code>	no hypotheses

5.6. How to check. The formalization of Theorem B is available at <https://github.com/gmDevi/zeta2-7-9-11-lean>; the registered version is the commit `2ae1bccb82f1f72ed7a7aeab3196aa1cdbbee165a`

The formalization of Theorem A is available at <https://github.com/gmDevi/zeta2-7-9-lean>; the registered version is the commit `3797cb6e4f8653744c0aa8e66ec8eb3645e75dcd`

Both repositories contain the informal proof (`docs/proof.md`), exact-arithmetic Python mirrors of the definitions with numerical checks of every statement (`python/`), and the files `Challenge.lean`, `Solution.lean`, `comparator.json` and `formalization.yaml` for Palomar. With `elan` installed, the following commands, run in the root directory of a repository, fetch the compiled Mathlib, build the project (including `Challenge` and `Solution`), print the axioms of the main theorem, run `Comparator` (which needs `bubblewrap`) and the census of `sorry` and forbidden constructs (the third line is run in the repository of Theorem B, the fourth in that of Theorem A):

```
lake exe cache get
lake build
lake env lean Zeta2Lean/Cited/Main.lean          # Theorem B
lake env lean Zeta2Lean/Pair/Unconditional.lean    # Theorem A
lake comparator
bash scripts/audit.sh
```

The kernel replay runs `leanchecker` on one module at a time (for Theorem A, `bash scripts/kernels.sh Solution` replays the 31 modules of the import closure of `Solution.lean`; without argument the script starts from `Zeta2Lean.Pair.Unconditional`, whose closure has 30 modules); each run needs between 6 and 7 GB of memory, mostly for loading Mathlib.

6. LIMITS OF THE METHOD

6.1. What the theorems do not say. Theorems A and B are lower bounds for the dimensions of $\mathbb{Q}$ -vector spaces: $\dim_{\mathbb{Q}}(\mathbb{Q} + \mathbb{Q}\zeta_2(7) + \mathbb{Q}\zeta_2(9)) \geq 2$, and similarly for $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$. They do not say which of the numbers is irrational, and they give no irrationality measure. A single sequence of linear forms combined with Lai’s criterion cannot give more. If, say, $\zeta_2(7)$ were rational, the forms of Theorem A would be linear forms in 1 and $\zeta_2(9)$ alone, and the argument would show that $\zeta_2(9)$ is irrational, in accordance with the theorem. To prove that one specific value is irrational, one needs linear forms in which the other values do not occur. In the families of rational functions examined in this work, no construction reaching $\zeta_2(7)$ alone was found.

6.2. Two of the three. “At least two of $\zeta_2(7)$, $\zeta_2(9)$, $\zeta_2(11)$ are irrational” means that each of the pairs $\{\zeta_2(7), \zeta_2(9)\}$, $\{\zeta_2(7), \zeta_2(11)\}$ and $\{\zeta_2(9), \zeta_2(11)\}$ contains an irrational number. Theorem A settles the first pair. For the other two pairs we found no suitable construction in the families considered here. The other route, a dimension bound $\dim_{\mathbb{Q}}(\mathbb{Q} + \mathbb{Q}\zeta_2(7) + \mathbb{Q}\zeta_2(9) + \mathbb{Q}\zeta_2(11)) \geq 3$, is out of reach of the linear forms of Section 3: the p -adic criterion of Nesterenko type [36, Theorem 1.4] could give at most the lower bound $\alpha/\beta = 1.004$ for this dimension, even granting its hypotheses, whereas 3 would need a ratio above 2 (Remark 3.18).

6.3. Thin margins and effectivity. The exponents are small: -0.0904 per n for Theorem B and -0.189 per n for Theorem A at the residue level (-0.551 with the refinements claimed in Remark 4.19, whose proofs are not included); the formal proof of Theorem A works with the margin $12 \log 2 - 8.28$, that is, about 0.038 per n . Both proofs use the prime number theorem in the form $\psi(x) = x + o(x)$, and the $o(n)$ terms are not made explicit, so neither proof gives an explicit n_0 from which on the linear forms are small enough (Remarks 3.16 and 4.39). For Theorem B, explicit versions of the prime number theorem would give one. For Theorem A, the crude bound for the small primes in Theorem 4.13(b) is too weak at moderate n ; the sharper per-prime bounds claimed in Remark 4.19 would give negative exponents at $n = 2000$ and $n = 2400$ (Remark 4.39).

6.4. Other primes and real values. Both constructions integrate $R_n'''(t + \frac{1}{2})$ over $\mathbb{Z}_2$ and use Lemma 2.3; they are specific to $p = 2$. For primes $p \geq 5$ see [21]. The method says nothing about the real numbers $\zeta(7)$ and $\zeta(9)$.

7. AI USE AND VERIFICATION

7.1. How this work was produced. Apart from the published results it cites and the vendored, human-written Lean files for the prime number theorem [4], everything in this paper was produced by AI agents, namely Anthropic Claude models running in Claude Code, under the direction of the author, between 23 and 25 September 2026. According to the project logs, the agents that produced and checked the results of this paper ran Claude Opus 5.5; runs of another Claude model in the same logs belong to a separate program of the project on the real value $\zeta(7)$ and contributed nothing to this paper. The author posed the research question (the irrationality of odd zeta values), directed and supervised the work, and is responsible for this paper. In detail:

- (1) *Research.* An orchestrating agent planned the research and launched sub-agents. Research agents explored families of rational functions with exact rational arithmetic, found the constructions of Sections 3 and 4 (configuration E was found by a numerical search) and wrote the informal proofs.
- (2) *Verification of the informal proofs.* Each of the four main steps of the proof of Theorem A (the 2-adic size, the denominators, the growth and the nonvanishing) was re-derived and checked by a separate adversarial verifier agent with its own code, and an assembly agent combined them and wrote an AI-written verification report on the assembled proof. The verifiers found errors of detail in the first versions (for instance a wrong concavity interval in the growth argument, a sign error in the denominator argument, and an unjustified clause dropping the pole $k = n/2$ in the first version of Theorem 4.13(c), which was repaired); all were corrected, and none affected the conclusions. Their existence means that the text may still contain further slips. The informal proof of Theorem B was also checked by adversarial verifier agents.
- (3) *Formalization.* An architect agent wrote the Lean definitions and the statements of all lemmas. Prover agents wrote the proofs against these frozen statements, and reported problems instead of changing them. Auditor agents checked the definitions and statements for faithfulness to the informal proofs and for soundness (no vacuous statements, no loopholes through junk values such as $x/0 = 0$ or truncated subtraction), with independent exact-arithmetic mirrors in Python. Integrator agents built the projects, ran the census of `sorry` and forbidden constructs, and replayed every module with `leanchecker`.
- (4) *Literature.* The literature search of §1.6 was carried out by AI agents.
- (5) *This paper* was written by AI agents from the informal proofs, the Lean repositories and the cited papers. The numerical values in it were copied from the exact computations of the project.

7.2. What has been verified, and how.

- *Machine-checked:* the formal statements of §5.1, in terms of the six definitions of §5.2, from the three standard axioms, by the Lean kernel, with a replay by `leanchecker` and a comparison with the independent kernels NanoDa and con-ron through Comparator.
- *Cited, not machine-checked:* the identification of the Lean definition `zeta2 s` with the Kubota–Leopoldt value $\zeta_2(s)$ (Lemma 2.3, from [23]).

- *Checked by AI agents only:* the informal proofs of Sections 3 and 4 as written, in particular the constants of Section 4 that the formal proof does not use (the growth rate -0.7812 , the value of $R^{(0)}$, and the refinements of Remark 4.19); the numerical remarks; and the three interval-arithmetic certifications of Lemma 4.28. The programs for these certifications were written by AI agents and have not been reviewed by a human. One of them uses no floating point and no external library, and is about 440 lines long. The same applies to the results claimed in Remarks 4.19 and 4.39, whose proofs are not included in this paper.
- *Not done:* no human expert has refereed the mathematics, the Lean statements or this paper.

7.3. Suggestions for a human referee. Since the formal proofs cover Theorems A and B completely, a referee who accepts Lean and Mathlib needs to check (i) that the six definitions of §5.2 define the Volkenborn integrals J_s and the numbers $J_{s-1}/((s-1)2^s)$; (ii) that the statements of §5.1 express Theorems A and B for these numbers; and (iii) Lemma 2.8 of [23] (see §5.3). For the informal proof of Theorem A, the AI-written verification report of the assembled proof suggests the following order: Proposition 4.23 and the proof of Theorem 4.21, together with an independent certification of the inequalities of Lemma 4.28 (a weak form, $\Psi \leq -0.75$, suffices); Theorem 4.13(a), (c) and the bound $R^{(0)} > 0.901$; Theorem 4.32 with the constants $\beta_1 = -168$ and K_E ; and Theorem 4.4.

7.4. Code and data availability. The programs behind the computer-assisted steps of the informal proof of Theorem A accompany this paper as ancillary files, in the directory `anc/` of its source distribution (`anc/README.md` lists them): the computation of $R^{(0)}$ (`rzero.py`, with its output); the three certification programs for Lemma 4.28 (`certify.py` and `certify_sharp.py`, based on `mpmath.iv`; `vcertify.py`, with exact rational boxes; `cert3.py` and `cert3b.py`, without floating point, with their logs), together with their validation scripts; the explicit constants of Table 11 (`explicit_constant.py`, `constants.py`); and the checks of Theorem 4.13 and Remark 4.39 (`resid_check.py`, `finite_n.py`). They will also be included in the public repository of the formalization of Theorem A. The formal proofs do not depend on any of these programs: all numerical inputs of the formal proofs are checked by the Lean kernel (§5.4).

Acknowledgements. The formal proofs rely on Mathlib, and on the Lean formalization of the Wiener–Ikehara theorem by T. Tao and the contributors of the PrimeNumberTheoremAnd project [4].

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- Theorem and lemma numbers refer to arXiv:2304.00816v1 for [20], arXiv:2306.10393v2 for [22], arXiv:2505.23088v1 for [21], and arXiv:2505.05005 (v1 and v2 have the same numbering) for [23].
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